Quantum Field Theory

Lecture notes (Sunil Mukhi)

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Lecture 1

QFT is considered difficult subject. It is basic underline formalism which describe laws of nature, we describe laws in terms of fields with each associated particles.

Q: Why does the subject play central role in particle physics?
A: for that let's ask ourselves, how do we describe most fundamental system, i.e. dynamics of particles. So our question is how shall we describe dynamics of the particles? We have so many particles like photons, electrons, quarks...

Before we answer let's ask what do we mean by dynamics of a particle, i.e. what will the particle do? If it's free, it will just propagate, if it is with other particles, it will interact and scattering will occur. So we can ask about the scattering. Suppose we put so many particles and the attraction makes them live in bound state. Ex. Proton is bound state of "UUD". In this case we would like to find energy levels, possible excitation of system... we have radiation and chemistry...

The phenomena of scattering is tested in accelerators, (accelerators can be naturally occurring also..., cosmic rays). We would like to have theory which provides calculation of rate at which scattering happens (we will calculate scattering amplitude which will provide probability of scattering or rate of scattering), which can be tested experimentally.

Strictly speaking quantum field theories are not fundamental theories, these are effective field theories which only included appropriate degree of freedom to describe physical phenomena occurring at chosen length and energy scale while ignoring the d.o.f. at shorter distances and substructure. This is why the answers to considering proton as fundamental particle agrees with experiment at energy scale where we can't see it's substructure ($O(\text{GeV})$). The breakdown of theory at some decimal place can indicate sign of substructure.

So how do we make the theory?

$$E \to i\hbar\frac{\partial}{\partial t}, \qquad \vec{p} = -i\hbar\nabla, \qquad E^2 = p^2 + m^2$$

$$\Rightarrow \quad E^2\psi = \left(p^2 + m^2\right)\psi$$

$$\left(E^2 - p^2 - m^2\right)\psi = 0 \qquad \psi = \text{wave function}$$

$$\Rightarrow \quad \hbar^2\left(\frac{\partial^2}{\partial t^2} - \nabla^2\right)\psi + m^2\psi = 0 \qquad \text{— ① K.G. eq}^{\text{n}}$$

General solution is $\psi(x) = e^{ik_\mu x^\mu}$ (trial sol$^{\text{n}}$). Using $\psi(x) = e^{ik\cdot x}$ in K.G. eq$^{\text{n}}$ we get

$$\boxed{\hbar^2 k^2 = m^2}$$

Q: What is the energy of a particle having some fixed momentum?

Using above eq$^{\text{n}}$

$$\begin{aligned}
\hbar^2 k^2 &= m^2 \\
\hbar^2\left(k_0^2 - \vec{k}^2\right) &= m^2 \\
E = \hbar k_0 &= \pm\sqrt{m^2 + \hbar^2 k^2} \\
E &= \pm\sqrt{m^2 + p^2} \quad \text{— ②} \quad (p = \hbar k)
\end{aligned}$$

↑ energy of particle with fixed momentum.

We can find eigen values of $\hat{E}$ & $\hat{P}$ operator for wave fun. $\psi = e^{ik\cdot x}$:

$$\hat{E}\psi = i\hbar\frac{\partial}{\partial t}\psi = i\hbar(ik_0)\psi = -\hbar k_0 \quad \Rightarrow \quad \boxed{\hat{E}\psi = -\hbar k_0\,\psi}$$

$$\begin{aligned}
\hat{P}\psi &= -i\hbar\nabla\psi = -i\hbar\left(\frac{\partial}{\partial x}\hat{i} + \frac{\partial}{\partial y}\hat{j} + \frac{\partial}{\partial z}\hat{k}\right)e^{i(k_0 t - \vec{k}\cdot\vec{r})} \\
&= -i\hbar\left(-i\vec{k}\right)\psi
\end{aligned}$$

$$\boxed{\hat{P}\psi = \hbar\vec{k}\,\psi}$$

<u>Note</u> — $\nabla_i = \dfrac{\partial}{\partial x^i}$, $\quad x^i$ — contravariant vector.

Using eq$^{\text{n}}$ ② we have $\hbar k_0 = \pm\sqrt{\hbar^2 k^2 + m^2}$.

We have possibility of having negative energy solutions which does not make any sense. In field theory if we consider ψ as field rather than wave fun. then this problem does not arise.

In quantum mechanics we can't throw away the negative energy solutions. As solution is superposition of all possible solutions.

The idea of fields

We do know the scalar fields (scalar potential $\phi(x,t)$) in theory of electrodynamics. But it comes together with vector field called the vector potential. This field is certainly very well observed experimentally. We have these two classical fields which obey dynamics of:

$$\left.\begin{aligned} &\nabla\cdot\vec{B} = 0 \qquad &&\nabla\times\vec{B} = +\frac{\partial\vec{E}}{\partial t} \\ &\nabla\times\vec{E} = -\frac{\partial\vec{B}}{\partial t} \qquad &&\nabla\cdot\vec{E} = 0 \end{aligned}\right\} \text{In absence of sources and sinks.}$$

where

$$\vec{E} = -\frac{\partial\vec{A}}{\partial t} - \nabla\varphi \qquad \text{and} \qquad \vec{B} = \nabla\times\vec{A}$$

<u>Relativistic form</u>:

$$\begin{gathered} A_\mu = (\varphi, -\vec{A}), \qquad A^\mu = (\varphi, \vec{A}) \\ F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu \quad \text{(anti-symmetric)} \end{gathered}$$

$F_{\mu\nu}$ has 6 independent parameters.

$$F_{0i} = -E_i\ ; \qquad F_{ij} = \epsilon_{ijk}B_k$$

Relativistic form is —

$$\left.\begin{aligned} \partial^\mu F_{\mu\nu} &= 0 \quad 4\ \text{eq}^{\text{n}} \\ \epsilon_{\mu\nu\rho\sigma}\,\partial^\nu F^{\rho\sigma} &= 0 \quad 4\ \text{eq}^{\text{n}}\text{s} \end{aligned}\right\} \text{Total 8 equations.}$$

(1) $\partial^\mu F_{\mu\nu} = 0 \ \Rightarrow$

$$\begin{aligned} \partial^\mu\left(\partial_\mu A_\nu - \partial_\nu A_\mu\right) &= 0 \\ \partial^2 A_\nu - \partial^\mu\left(\partial_\nu A_\mu\right) &= 0 \\ \partial^2 A_\nu - \partial_\nu\left(\partial^\mu A_\mu\right) &= 0 \\ \partial^2 A_\nu - \partial_\nu\left(\partial\cdot A\right) &= 0 \end{aligned}$$

For $\partial\cdot A = 0$ (Lorentz gauge) we have $\partial^2 A_\nu = 0$, which is similar to $\partial^\mu\partial_\mu\varphi = 0$ where φ is scalar field. But here K.G. eq$^{\text{n}}$ was wave eq$^{\text{n}}$, but we have similar eq$^{\text{n}}$ for fields.

We give up the idea of wave equations for relativistic particles, we are forced to do this by two problems. We replace it with the idea of field equations for the relativistic fields. We will get arbitrary number of particles. We also have to show that quantising fields gives us particles. Now $\phi(x,t)$ is no longer a wave function but rather is a field. ϕ and $|\phi\phi^*|$ is no longer probability but again some operator that creates particles. In next lecture we will see that this quantum field always creates particles of positive energy and never creates negative energy particles, secondly we can use this field again and again to create as many particles.

$$\text{classical particle} \xrightarrow{\text{QM}} \text{Quantum particle} \qquad \text{(inconsistent with relativity)}$$

$$\begin{tikzcd}
\text{Classical fields} \arrow[r, "\text{QM}"] & \text{Quantum fields} \arrow[r] & \text{Quantum particles.} \arrow[dl, bend right] \\
& \text{consistent with relativity} &
\end{tikzcd}$$

Also to note. field is an operator where as in QM $|\psi\rangle$ is a state vector or $\psi(x,t)$ is component in $\{|x\rangle\}$ basis.

Lecture 2

Lorentz transformations: all operations which keeps the space time interval $\delta S^2 = t^2 - x^2 - y^2 - z^2$ fixed are called as Lorentz transformations.

$$\begin{aligned} x_\mu x^\mu &= t^2 - x^2 - y^2 - z^2 \qquad \text{(invariant)} \\ &= t^2 - (x^2 + y^2 + z^2) \end{aligned}$$

We can have x^2 invariant also by keeping t fixed and have $(x^2 + y^2 + z^2)$ fixed by set of rotations. So Rotations are included in Lorentz transformation.

For $y = z = 0$:

$$x_\mu x^\mu = t^2 - x^2 = (t - x)(t + x) = x_+ x_-$$

$$\text{Define } x_\pm = t \pm x \quad \Rightarrow \quad t = \frac{x_+ + x_-}{2}, \quad x = \frac{x_+ - x_-}{2}$$

$x_\mu x^\mu = x_+ x_- =$ product of two independent coordinates. Lets ask what are the transformations on x_+ & x_- such that product $x_+ x_- = x^\mu x_\mu$ remain invariant/fixed.

$$x_\pm \longrightarrow e^{\pm\alpha} x_\pm \quad \text{will preserve } x_\mu x^\mu.$$

If

$$\left.\begin{aligned} x_+ &\longrightarrow e^{\alpha} x_+ \\ x_- &\longrightarrow e^{-\alpha} x_- \end{aligned}\right\} \Rightarrow t = \frac{x_+ + x_-}{2} \quad \text{under this transformation}$$

$$\begin{aligned} t &= \frac{e^\alpha x_+ + e^{-\alpha} x_-}{2} = \frac{1}{2}\left(e^\alpha (t + x) + e^{-\alpha}(t - x)\right) \\ &= \frac{1}{2}\left(\left(e^\alpha + e^{-\alpha}\right) t + \left(e^\alpha - e^{-\alpha}\right) x\right) = \left(\frac{e^\alpha + e^{-\alpha}}{2}\right) t + \left(\frac{e^\alpha - e^{-\alpha}}{2}\right) x \\ t &= (\cosh\alpha)\, t + (\sinh\alpha)\, x \end{aligned}$$

so under Lorentz transformation $t \longrightarrow t(\cosh\alpha) + x(\sinh\alpha)$

$$\Rightarrow \qquad \begin{aligned} t' &= (\cosh\alpha)\, t + (\sinh\alpha)\, x \\ x' &= (\sinh\alpha)\, t + (\cosh\alpha)\, x \end{aligned}$$

similarly

$$x = \frac{x_+ - x_-}{2}, \quad x' = \frac{x'_+ - x'_-}{2} = \frac{e^\alpha x_+ - e^{-\alpha} x_-}{2} = \frac{e^\alpha (t + x) - e^{-\alpha}(t - x)}{2} = \left(\frac{e^\alpha - e^{-\alpha}}{2}\right) t + \left(\frac{e^\alpha + e^{-\alpha}}{2}\right) x$$

$$\Rightarrow \quad \begin{pmatrix} t' \\ x' \end{pmatrix} = \underbrace{\begin{pmatrix} \cosh\alpha & \sinh\alpha \\ \sinh\alpha & \cosh\alpha \end{pmatrix}}_{\Lambda} \begin{pmatrix} t \\ x \end{pmatrix}$$

Lorentz transformation is mathematically similar to rotation. Since

$$\begin{vmatrix} \cosh\alpha & \sinh\alpha \\ \sinh\alpha & \cosh\alpha \end{vmatrix} = \cosh^2\alpha - \sinh^2\alpha = 1$$

Conceptually the boost & rotations are similar but we can't make any amount of rotation, since rotation is bounded by $[0, 2\pi]$. But $\cosh\alpha$ is not periodic which implies we can make any amount of boost.

We had,

$$t' = \frac{t - vx/c^2}{\sqrt{1 - v^2/c^2}} = \frac{t - vx}{\sqrt{1 - v^2}}, \qquad x' = \frac{x - vt}{\sqrt{1 - (v/c)^2}} = \frac{x - vt}{\sqrt{1 - v^2}}$$

The transformation matrix

$$\Lambda = \begin{pmatrix} \dfrac{1}{\sqrt{1 - v^2}} & \dfrac{-v}{\sqrt{1 - v^2}} \\ \dfrac{-v}{\sqrt{1 - v^2}} & \dfrac{1}{\sqrt{1 - v^2}} \end{pmatrix}$$

One nice thing about this formalism is that the Lorentz invariant quantities are having same number of plus and minus indices, for example we can take any two component vector A_μ, with just (A_t, A_x) then the quantity $A_- A_+$ is Lorentz invariant, and we can see that Lorentz Invariance is just scaling up and scaling down of A_+ and A_-, to make it invariant.

Field Equation

The field equation for the Klein gordon field is:

$$\hbar^2 \partial_\mu \partial^\mu \varphi + m^2 \varphi = 0 \quad —(1) \qquad \text{(or)} \quad \boxed{\left(\partial^2 + m^2\right) \varphi = 0}$$

treat $\varphi(x, t)$ as an operator in quantum theory rather than a wave function.

We will start with revision of quantum harmonic oscillator

$$L = \frac{1}{2} m \dot{q}^2 - \frac{1}{2} m \omega^2 q^2$$
$$H = \dot{q} p - L = \frac{p^2}{2m} + \frac{1}{2} m \omega^2 q^2$$

We do change of variables,

$$a = q + ip$$
$$a^\dagger = q - ip$$

We need to construct quantities of dimension length and momentum from $m, \omega, \hbar$:

$$q' \propto m^\alpha \omega^\beta \hbar^\gamma \qquad\qquad E = \hbar\omega \;\Rightarrow\; [\hbar] = [E][T]$$
$$\propto [M]^\alpha \, [T^{-1}]^\beta \, [ML^2T^{-2}]^\gamma \, [T]^\gamma$$
$$q' \propto [M]^{\alpha+\gamma} \, [L]^{2\gamma} \, [T]^{-\beta-\gamma}$$

For q' to be length we need $2\gamma = 1$, $\alpha + \gamma = 0$, $\beta = -\gamma$

$$\Rightarrow \; \gamma = 1/2, \;\; \alpha = -1/2, \;\; \beta = -1/2$$
$$\Rightarrow \; q' = \sqrt{\frac{\hbar}{\omega m}}$$

For q' to be momentum we require $\alpha + \gamma = 1$:

$$p \propto (\hbar)^\alpha (\omega)^\beta (m)^\gamma, \qquad MLT^{-1} \propto (ML^2T^{-1})^{\cdots}$$
$$2\gamma = 1 \;\Rightarrow\; \gamma = 1/2, \; \alpha = +1/2; \qquad -\beta - \gamma = -1 \;\Rightarrow\; \beta = 1 - 1/2 = +1/2$$
$$p' = \sqrt{m\omega\hbar}$$

So,

$$a = \left(\frac{q}{q'} + \frac{ip}{p'}\right) = q\sqrt{\frac{\omega m}{2\hbar}} + \frac{ip}{\sqrt{2m\omega\hbar}}$$

$$a^\dagger = q\sqrt{\frac{m\omega}{2\hbar}} - \frac{ip}{\sqrt{2m\omega\hbar}} \qquad \because \text{ where } C \text{ is some constant.}$$

Our Total Hamiltonian becomes—

$$H = \frac{p^2}{2m} + \frac{1}{2}m\omega^2 q^2$$
$$H = \hbar\omega\left(a^\dagger a + \frac{1}{2}\right) \qquad \nexists \text{ No state with negative energy.}$$

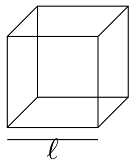
We note that $[a, a^\dagger] = 1$, assume there is state satisfying $a|0\rangle = 0$. This assumption ensures we don't end up with minus infinity as energy of lowest state, (also we don't see vacuum decay to photons).

The reason we had gone through this exercise is that because equation (1) gives us collection of simple harmonic oscillators, all decoupled from each other, so we can quantise each of those quantum harmonic oscillators and therefore in such way we have quantised the field φ.

Remember in quantum physics we put t and x on different footing but in quantum field theory we put time and space on the same footing.

So, object we are interested in is $\varphi(t, \vec{x})$, for the moment we will treat time and space on different footing but we will see that the result will be relativistically invariant.

Lets bound $\vec{x}$ to size of cube. Later we will take size of box 'L' to be infinity, then we can accommodate a system as large as we like to use. why?



We need boundary conditions/periodic boundary conditions.

$$\text{i.e.} \qquad \left.\begin{aligned} \varphi(t,\, x{+}L,\, y,\, z) &= \varphi(t, x, y, z) \\ \varphi(t,\, x,\, y{+}L,\, z) &= \varphi(t, x, y, z) \\ \varphi(t,\, x,\, y,\, z{+}L) &= \varphi(t, x, y, z) \end{aligned}\right\} \text{—}(2)$$

Till now we are treating φ as some function rather than an operator. We can write φ as some vector whose basis set are some exponential function (fourier series). Let's take $e^{i\vec{k}\cdot\vec{x}}$ as some general function, with periodic boundary conditions. Using $\varphi(x,t) = e^{i\vec{k}\cdot\vec{x}}$ in eq$^{\text{n}}$ (2) we get (RHS is missing time! anyway time is fixed!)

$$e^{i\vec{k}\cdot(\vec{x}+L)} = e^{i\vec{k}\cdot\vec{x}}$$
$$e^{i(k_x x + k_y y + k_z z)} \cdot e^{ik_x L} \text{ form: } \quad e^{i(k_x(x+L) + k_y y + k_z z)}$$
$$e^{ik_x x} = e^{ik_x x}\, e^{ik_x L}$$
$$\Rightarrow \quad e^{ik_x L} = \cos(k_x L) + i\sin(k_x L) = 1$$
$$\Rightarrow \quad k_x L = 2\pi m_1, \qquad k_x = \frac{2\pi m_1}{L}$$

Similarly $e^{i(k_x x + k_y(y+L) + k_z z)} = e^{i(k_x x + k_y y + k_z z)}$ gives us $k_y = \dfrac{2\pi m_2}{L}$. Similarly we get $k_z = \dfrac{2\pi m_3}{L}$

$$\text{(or)} \qquad \vec{k} = \frac{2\pi}{L}(m_1, m_2, m_3)$$

In fact we have taken all possible functions, as we can expand them in $e^{i\vec{k}\cdot\vec{x}}$ with some coefficients $(q_k(t))$. Therefore,

$$\varphi(t,x) = \underbrace{N}_{\substack{\text{Normalisation}\\ \text{factor}}} \sum_{\vec{k}} q_k(t)\, e^{i\vec{k}\cdot\vec{x}}$$

$\hookleftarrow$ expansion of most general configuration $\varphi(t,\vec{x})$. $\quad\hookrightarrow$ time dependence has to be taken by $q_k(t)$, and label k for 'q' notify the seperate coefficients for each $\vec{k}$.

We can use this expression and plug it into equation (1) $\left(\partial^2 + m^2\right)\varphi = 0$, and ask values of $q_k(t)$.

$$\Rightarrow \quad \left(\partial^2 + m^2\right)\sum_{\vec{k}} q_k(t)\, e^{i\vec{k}\cdot\vec{x}} = 0 \qquad \text{—}\,(3)$$

$$\Rightarrow \quad \partial^2 = \partial_\mu\partial^\mu : \left(\frac{\partial}{\partial t}, \nabla\right)\begin{pmatrix}\frac{\partial}{\partial t}\\ -\nabla\end{pmatrix} = \frac{\partial^2}{\partial t^2} - \nabla^2$$

$$\begin{aligned}
&\text{(i)} \quad \partial^2 q_k(t) = \frac{\partial^2}{\partial t^2}q_k(t) - \nabla^2 q_k(t) = \ddot{q}_k(t) - 0\\
&\text{(ii)} \quad \partial^2 e^{i\vec{k}\cdot\vec{x}} = \frac{\partial^2}{\partial t^2}e^{i\vec{k}\cdot\vec{x}} - \left(\frac{\partial^2}{\partial x^2}e^{i\vec{k}\cdot\vec{x}} + \frac{\partial^2}{\partial y^2}e^{i\vec{k}\cdot\vec{x}} + \frac{\partial^2}{\partial z^2}e^{i\vec{k}\cdot\vec{x}}\right)\\
&\qquad\qquad\quad = k^2 e^{i\vec{k}\cdot\vec{x}} \qquad\qquad \left(k^2 = k_x^2 + k_y^2 + k_z^2\right)
\end{aligned}$$

$\Rightarrow$ equation (3) reduces to —

$$\begin{aligned}
\sum_{\vec{k}}\left(\left(\partial^2 q_k(t)\right)e^{i\vec{k}\cdot\vec{x}} + q_k(t)\,\partial^2 e^{i\vec{k}\cdot\vec{x}} + m^2 q_k(t)\, e^{i\vec{k}\cdot\vec{x}}\right) &= 0\\
\sum_{\vec{k}}\left(\left(\ddot{q}_k(t) + q_k(t)\,k^2\right)e^{i\vec{k}\cdot\vec{x}} + m^2 q_k(t)\, e^{i\vec{k}\cdot\vec{x}}\right) &= 0\\
\sum_{\vec{k}}\left(\ddot{q}_k(t) + q_k(t)\,k^2 + m^2 q_k(t)\right)e^{i\vec{k}\cdot\vec{x}} &= 0
\end{aligned}$$

$$\underset{\substack{\text{Infinite set of}\\ \text{decoupled harmonic}\\ \text{oscillators}}}{\Longrightarrow} \qquad \boxed{\ddot{q}_k(t) + \left(k^2 + m^2\right)q_k(t) = 0} \qquad \text{—}\,(4)$$

Note that this is infinite set of differential equations one for each value of $\vec{k}$ satisfying $\vec{k} = \frac{2\pi}{L}(m_1, m_2, m_3)$.

Eq (4) looks like differential eq$^{\mathrm{n}}$ of SHO, with frequency of oscillation

$$\omega_k = \sqrt{k^2 + m^2} \qquad (\vec{k}\text{ dependent frequency})$$

| $\ddot{q} + \omega^2 q = 0$
| $\hookrightarrow$ SHO eq$^{\mathrm{n}}$ of motion.

We need to know if the $\varphi(t,x)$ is real or complex. In QM we were forced to live with complex wave functions, but here its upon us to decide if we want to deal with real fields or complex fields. For the moment we choose $\varphi(t,x)$ to real fields. Later we will extend the result to complex fields. Which implies—

$$\begin{aligned}
\varphi(t,x) &= \varphi^*(t,x)\\
\Rightarrow \quad \sum_{\vec{k}} q_k(t)\, e^{i\vec{k}\cdot\vec{x}} &= \sum_{\vec{k}} q_k^*(t)\, e^{-i\vec{k}\cdot\vec{x}}
\end{aligned}$$

$$\boxed{q_k^*(t) = q_{-k}(t)}$$

So, this is not a real harmonic oscillator, except for case $\vec{k} = \vec{0} = (0,0,0)$, only for that case $(q_0^* = q_0)$, else its a complex harmonic oscillator.

$$\ddot{q}_k(t) + \left(k^2 + m^2\right) q_k(t) = 0 \qquad \begin{cases} \text{Real H.O.} & \vec{k} = \vec{0} \\ \text{Complex H.O.} & \vec{k} \neq \vec{0} \end{cases}$$

q_k^* and q_{-k} has same frequency since frequency only depend upon $\vec{k}^2$ $\left(\omega_k^2 = k^2 + m^2\right)$.

We claim that lagrangian for $\left(\sum_{\vec{k}} \ddot{q}_k(t) + \underbrace{\left(k^2 + m^2\right)}_{\omega_k^2} q_k(t) = 0\right)$ is given by

$$\boxed{L = \underbrace{\frac{1}{2}\dot{q}_0^2 - \frac{1}{2}\omega_0^2 q_0^2}_{\vec{k}=\vec{0}} + \sum_{\vec{k}>0}\left(\dot{q}_{-\vec{k}}\,\dot{q}_{\vec{k}} - \omega_{\vec{k}}^2\, q_{-\vec{k}}\, q_{\vec{k}}\right)} \qquad \text{How?}$$

$\vec{k} > 0$ means $\left(\text{first non zero coefficient in } \vec{k} = \frac{2\pi}{L}(m_1, m_2, m_3)\right) > 0$. In general if $m_1 > 0$ we say $\vec{k} > 0$, if $m_1 = 0$, we look at sign of m_2, if $m_2 = 0$, we look at sign of m_3. If all $m_1 m_2 m_3$ are zero then $\vec{k} = \vec{0}$.

The quantisation requires working with hamiltonian rather than lagrangian. So we will find canonical momentum and then do legendre transform to find H.

$$H = \sum_i \dot{q}_i p_i - L \qquad \left\{ p_i = \frac{\partial L}{\partial \dot{q}_i} \right.$$

$$p_0 = \frac{\partial L}{\partial \dot{q}_0} = \dot{q}_0 \quad \text{for } \vec{k} = 0$$

$$p_k = \frac{\partial L}{\partial \dot{q}_k} = \dot{q}_{-k} \qquad \forall \vec{k}$$

So,

$$\boxed{p_k = \frac{\partial L}{\partial \dot{q}_k} = \dot{q}_{-k}} \qquad \text{also for } \vec{k} = \vec{0}$$

$$\begin{aligned}
H &= \sum p\dot{q} - L \\
&= p_0\dot{q}_0 + \sum_{\vec{k}>0} p_k\dot{q}_k + p_{-k}\dot{q}_{-k} - L \\
&= p_0 p_0 + \sum_{\vec{k}>0}\left(p_k p_k + p_k p_k\right) - \left(\frac{1}{2}\dot{q}_0^2 - \frac{1}{2}\omega^2 q_0^2 + \sum_{\vec{k}>0} \dot{q}_k\dot{q}_{-k} - \omega^2 q_{-k} q_k\right) \\
&= \underline{p_0^2} + \sum_{\vec{k}>0} p_k p_k + p_{-k} p_k - \left(\frac{1}{2}p_0^2 - \frac{1}{2}\omega^2 q_0^2 + \sum_{\vec{k}>0} p_k p_{-k} - \omega^2 q_{-k} q_k\right) \\
&= \frac{p_0^2}{2} + \sum_{\vec{k}>0}\left(\cancel{p_k p_{-k}} - \cancel{p_k p_{-k}} + p_{-k} p_k + \omega^2 q_{-k} q_k\right) + \frac{1}{2}\omega^2 q_0^2
\end{aligned}$$

$$\boxed{H = \frac{p_0^2}{2} + \frac{1}{2}\omega_0^2 q_0^2 + \sum_{\vec{k}>0} p_{\vec{k}}\, p_{-\vec{k}} + \omega_{\vec{k}}^2\, q_{-\vec{k}}\, q_{\vec{k}}}$$

This Hamiltonian is sum of independent hamiltonians of many harmonic oscillators each one's can be solved by writing them in terms of a and $a^\dagger$. That is what we are going to do now.

Let's write canonical commutation relations. Writing canonical commutation relations make canonical variables p_k & q_k Operators which is only possible if field itself is an operator. Therefore imposition of canonical commutation relations makes one field as operators.

$$[q_0, p_0] = i\hbar$$
$$[q_k, p_k] = i\hbar \qquad \text{or} \qquad [q_{\vec{k}}, p_{\vec{k}}] = i\hbar$$
$$\boxed{[q_{\vec{k}},\, p_{\vec{k}'}] = i\,\delta_{\vec{k},\vec{k}'}}$$

$$\checkmark \quad \begin{aligned} a_{\vec{k}} &= \frac{1}{\sqrt{2\omega_k}}\left(ip_{\vec{k}} + \omega_{\vec{k}}\, q_{\vec{k}}\right) \\ a_{\vec{k}}^{\dagger} &= \frac{1}{\sqrt{2\omega_k}}\left(-ip_{\vec{k}}^{*} + \omega_{\vec{k}}\, q_{\vec{k}}^{*}\right) = \frac{1}{\sqrt{2\omega_k}}\left(-ip_{-\vec{k}} + \omega_{\vec{k}}\, q_{-\vec{k}}\right) \end{aligned}$$

using $q_k^* = q_{-k}$
and $p_k^\dagger = p_{-k}$
$\vec{k} \to -\vec{k}$

Let us check commutation relation —

$$\begin{aligned} [a_{\vec{k}},\, a_{-\vec{k}}^{\dagger}] &= \frac{1}{2\omega_k}\Big(- i^2 \underbrace{[p_{\vec{k}},\, q_{\vec{k}}]}_{\nearrow 0} i\omega_k [p_{\vec{k}},\, q_{\vec{k}}] + \omega_k(-i)[q_{\vec{k}},\, p_{\vec{k}}] + \omega_{\vec{k}}^2 \underbrace{[q_{\vec{k}},\, q_{\vec{k}}]}_{\nearrow 0} \Big) \\ &= \frac{1}{2\omega_k}\left(-i^2\omega_k - i^2\omega_k\right) = 1 \end{aligned}$$

in general,

$$\Rightarrow \quad \boxed{[a_{\vec{k}},\, a_{-\vec{k}'}^{\dagger}] = \delta_{\vec{k},\vec{k}'}}$$

Our hamiltonian of infinite decoupled harmonic oscillators

$$H = \frac{p_0^2}{2} + \frac{1}{2}\omega_0^2 q_0^2 + \sum_{\vec{k}>0}\left(p_{\vec{k}}\, p_{-\vec{k}} + \omega_{\vec{k}}^2\, q_{-\vec{k}}\, q_{\vec{k}}\right)$$

becomes,

$$\begin{aligned} a_{\vec{k}} &= \frac{1}{\sqrt{2\omega_k}}\left(ip_{\vec{k}} + \omega_{\vec{k}}\, q_{\vec{k}}\right) \\ a_{\vec{k}}^{\dagger} = \frac{1}{\sqrt{2\omega_k}}\left(-ip_{\vec{k}}^{*} + \omega_{\vec{k}}\, q_{\vec{k}}^{*}\right) \;\Rightarrow\; a_{-\vec{k}}^{\dagger} &= \frac{1}{\sqrt{2\omega_{-k}}}\left(-ip_{\vec{k}} + \omega_{-\vec{k}}\, q_{\vec{k}}\right) \qquad \left\{\omega_k = \omega_{-k} = \sqrt{k^2+m^2}\right. \\ &= \frac{1}{\sqrt{2\omega_k}}\left(-ip_{-\vec{k}} + \omega_{\vec{k}}\, q_{-\vec{k}}\right) \end{aligned}$$

$$\begin{aligned} a_{\vec{k}} + a_{-\vec{k}}^{\dagger} &= \frac{1}{\sqrt{2\omega_k}}\left(0 + 2\omega_k q_k\right) \;\Rightarrow\; q_k = \frac{\left(a_{\vec{k}} + a_{-\vec{k}}^{\dagger}\right)}{\sqrt{2\omega_k}} \\ a_{\vec{k}} - a_{-\vec{k}}^{\dagger} &= \frac{1}{\sqrt{2\omega_k}}\left(2ip_k + 0\right) \;\Rightarrow\; p_k = \frac{\sqrt{2\omega_k}}{2i}\left(a_k - a_{-k}^{\dagger}\right) \end{aligned}$$

Using set (4) in original hamiltonian we get

$$\begin{aligned} H = \frac{1}{2}\frac{\left[\sqrt{2\omega_0}\right]^2}{(2i)^2}\left(a_0 - a_0^{\dagger}\right)^2 &+ \frac{1}{2}\omega_0^2\frac{1}{2\omega_0}\left(a_0 + a_0^{\dagger}\right)^2 \\ &+ \sum_{\vec{k}>0}\frac{1}{(2i)^2}\left(\sqrt{2\omega_k}\right)^2\left(a_{-k} - a_k^{\dagger}\right)\left(a_k - a_{-k}^{\dagger}\right) \\ &\quad + \frac{\omega_k^2}{2\omega_k}\left(a_{-k} + a_k^{\dagger}\right)\left(a_k + a_{-k}^{\dagger}\right) \end{aligned}$$

$$H = -\frac{\omega_0}{4}\left(\cancel{a_0^2} - a_0 a_0^\dagger - a_0^\dagger a_0 + \cancel{a_0^\dagger a_0^\dagger}\right) + \frac{1}{2}\omega_0^2 \frac{1}{2\omega_0}\left(\cancel{a_0 a_0} + a_0 a_0^\dagger + a_0^\dagger a_0 + \cancel{a_0^\dagger a_0^\dagger}\right)$$
$$+ \sum_{\vec{k}>0} (-)\frac{\omega_k}{2}\left(\cancel{a_{-k} a_k} - a_{-k} a_{-k}^\dagger - a_k^\dagger a_k + \cancel{a_k^\dagger a_{-k}^\dagger}\right)$$
$$+ \frac{1}{2}\omega_k\left(\cancel{a_{-k} a_k} + a_{-k} a_{-k}^\dagger + a_k^\dagger a_k + \cancel{a_k^\dagger a_{-k}^\dagger}\right)$$

$$\boxed{H = \frac{\omega_0}{2}\left(a_0 a_0^\dagger + a_0^\dagger a_0\right) + \sum_{\vec{k}>0} \omega_k \left(a_{-k}\, a_{-k}^\dagger + a_k^\dagger\, a_k\right)}$$

Using $[a_k, a_k^\dagger] = 1 \;\Rightarrow\; a_k a_{-k}^\dagger - a_{-k}^\dagger a_k = 1$ we get

$$a_0 a_0^\dagger - a_0^\dagger a_0 = 1 \;\Rightarrow\; a_0 a_0^\dagger = 1 + a_0^\dagger a_0$$

but it should be $\left[a_k\, a_{-k}^\dagger\right] = 1$:

$$\boxed{H = \sum_{\text{all } \vec{k}} \omega_{\vec{k}}\, a_k^\dagger a_k + \frac{1}{2}\sum_{\substack{\text{all}\\ \vec{k}}} \omega_{\vec{k}}}$$

For all $\vec{k}$, $\sum_{\vec{k}} \omega_k \longrightarrow \infty \Rightarrow$ ultraviolet divergence!

$$\text{as } \omega_k = \sqrt{k^2 + m^2}, \qquad \sum_k \omega_k = \sum_k \sqrt{k^2+m^2} \longrightarrow \infty.$$

For a field we always subtract the zero-point energy, because all measurements are made with respect to it.

Free Klein-Gordan field—

$$H = \sum_k \omega_k\, a_k^\dagger\, a_{-k}$$

Lets ask what are states of system, energies and quantum numbers.

(1) States—
Lowest energy state, satisfies

$$a_{\vec{k}}|0\rangle = 0 \qquad \forall\, \vec{k}$$
$$\text{So,} \quad H|0\rangle = \sum_{\vec{k}} \omega_k\, a_k^\dagger\, a_{-k}|0\rangle = 0$$

(2) Excited states—
Most general excited state is

$$a_{k_n}^\dagger\, a_{k_{n-1}}^\dagger \ldots a_{k_2}^\dagger\, a_{k_1}^\dagger |0\rangle$$

All these creation operators are independent and can be repeated.

(3) What is reasonable interpretation of system?
We first look for momentum operator,

$$\vec{P} = \sum_{\vec{k}} \vec{k}\, a_{-k}^\dagger\, a_k$$

$$\vec{P}\left(a_{k_n}^\dagger \ldots a_{k_2}^\dagger a_{k_1}^\dagger |0\rangle\right) = \left(k_n\, a_{-k_n}^\dagger\, a_{k_n} + \ldots + k_2\, a_{-k_2}^\dagger a_{k_2} + k_1\, a_{-k_1}^\dagger\, a_{k_1}\right)\left(a_{k_n}^\dagger \ldots a_{k_2}^\dagger a_{k_1}^\dagger |0\rangle\right)$$
$$= (k_n + k_{n-1} + \cdots + k_2 + k_1)\left(a_{k_n}^\dagger \ldots a_{k_2}^\dagger a_{k_1}^\dagger |0\rangle\right)$$

Similarly

$$\hat{H}\left(a^\dagger_{k_n}\dots a^\dagger_{k_2}a^\dagger_{k_1}|0\rangle\right) = (\omega_{k_1}+\omega_{k_2}+\dots+\omega_{k_n})\left(a^\dagger_{k_n}\dots a^\dagger_{k_2}a^\dagger_{k_1}|0\rangle\right)$$

where $\omega_{k_i} = \sqrt{k_i^2+m^2} \quad (E^2 = p^2+m^2)$.

This implies that the state behaves like n-particle state with momenta $(\vec{k}_1+\vec{k}_2\cdots+\vec{k}_n)$ and energy $(\omega_1+\omega_2+\cdots+\omega_n)$.

Vacuum— and state $|0\rangle$ has no momentum and no energy, so we should associate it with vacuum. That is change of terminology, in harmonic oscillator case (in QM), Harmonic oscillator can be in ground state and can be in excited state. In field theory nothing is there, so we call it Vacuum.

Field operators are then simply linear combination of $e^{i\vec{k}\cdot\vec{x}}$

$$\varphi = \sum q_{\vec{k}}\, e^{i\vec{k}\cdot\vec{x}}$$

So as we quantise the field, we get all possible no. of particles as states in the theory. This is not achievable by Schrödinger equation.

$$\left.\begin{array}{l} a\begin{cases}\text{Lowering operator — QM}\\ \text{Annihilation operator — QFT}\end{cases}\\ a^\dagger\begin{cases}\text{Raising operator — QM}\\ \text{Creation operator — QFT}\end{cases}\end{array}\right\}\ \text{Particle interpretation implies.}$$

Consider two states

$$a^\dagger_{k_2}\, a^\dagger_{k_1}|0\rangle\,, \qquad a^\dagger_{k_1}\, a^\dagger_{k_2}|0\rangle$$

These states are same. Since $[a^\dagger_{k_1}, a^\dagger_{k_2}] = 0$

$$\Rightarrow \quad \left(a^\dagger_{k_1}a^\dagger_{k_2} - a^\dagger_{k_2}a^\dagger_{k_1}\right)|0\rangle = 0$$
$$\Rightarrow \quad a^\dagger_{k_1}a^\dagger_{k_2}|0\rangle = a^\dagger_{k_2}a^\dagger_{k_1}|0\rangle$$

Thus, if we interchange two particles, the states are same, such particles are called bosons and follow Bose statistics.

Quantisation of K.G. field gives us particles obeying Bose statistics.

Spin of such n-particle state — We could construct the angular momentum in field theory (but that is complex way to go), there is simpler way to get spin of state. From statistics those are integer spin particles (since the particles are bosons).

Scalar field does not change under L.T., spin is something that should change under L.T., so spin should be zero.

There is even simpler way to tell spin of those bosons. Since particle with spin forms multiplet, but we don't see any label for those multiplets in general n-particle states. So these particles are spinless.

$$\text{2 particle state} = a^\dagger_{k_2}\ \underbrace{a^\dagger_{k_1}|0\rangle} \qquad \text{— No label for spin}$$

Every n-particle state is singlet. $\Rightarrow$ spinless particles. (Spin $= 0$)

If we want to have interactions, we need to put $\varphi^2 \sim \varphi^3 \sim \varphi^4 \dots$ terms in EOM.

Non linear equations $\Rightarrow$ Interactions.

Also note that, we don't have negative energy:

$$E_T = \sum_k \omega_k > 0, \quad \text{since all } \omega_k = \sqrt{k^2 + m^2} > 0 \ \forall\, k.$$

So the problem of negative energy only arise when we take wavefn. interpretation of $\varphi(t, x)$. In field rep$^{\text{n}}$ we don't have negative energy.

"There is room for wavefn. interpretation in field space"
— QFT by Suvrat.!

Lecture 3 (Sunil Mukhi)

Yesterday what we did, was to try and convert Klein gordon field into its mode q_k.

$$\varphi(x,t) = N \sum_{\vec{k}} q_{\vec{k}}(t)\, e^{i\vec{k}\cdot\vec{x}}$$

— $q_k(t)$ denote Infinitely many decoupled harmonic oscillators, and these oscillators have $a_k^\dagger$ and a_k which increase and decrease the energy respectively.

— Lowest energy state $|0\rangle$ is called vacuum. In fact vacuum is only empty for non interacting system (like we did in earlier lectures), For interactions vacuum is not empty.

— States are

$$a^\dagger_{k_1}\, a^\dagger_{k_2}\, a^\dagger_{k_3} \cdots a^\dagger_{k_n} |0\rangle \qquad \text{— multi particle state.} \quad \text{(must be spinless bosons)}$$

From the fact that $\left[a_k^\dagger,\, a_{k'}^\dagger\right] = 0 \Rightarrow$ these n particle states are bosons and since no other label describe states, these are spinless particles.

— The hamiltonian

$$H = \sum_{\text{all } \vec{k}} \omega_{\vec{k}}\, a^\dagger_{\vec{k}}\, a_{\vec{k}}$$

The values of $\vec{k}$ take discrete values $\frac{2\pi}{L}(m_1, m_2, m_3)$, because we took the system in a cubical box of length L. Now let us take limit $L \to \infty$, then $\vec{k}$ will take continuous values

$$H \xrightarrow{L\to\infty} \int \frac{d^3k}{(2\pi)^3}\, \omega_{\vec{k}}\, a^\dagger_{-\vec{k}}\, a_{\vec{k}}$$

Commutation relation becomes $\longrightarrow$

$$\left[a_k,\, a^\dagger_{-k'}\right] = (2\pi)^3\, \delta^3(\vec{k} - \vec{k}') \qquad \hookleftarrow \text{kronecker delta becomes dirac delta}$$

From here we will only work in this limit ($L \to \infty$).

— Recall that $\varphi(t,\vec{x}) = \sum_{\text{all } \vec{k}} q_k(t)\, e^{i\vec{k}\cdot\vec{x}}$ will become

$$\varphi(t,\vec{x}) = \int \frac{d^3k}{(2\pi)^3}\, q_k(t)\, e^{i\vec{k}\cdot\vec{x}}$$

$$\left.\begin{aligned}
a_{\vec{k}} &= \frac{1}{\sqrt{2\omega_k}}\left(i\bar{p}_{\vec{k}} + \omega_{\vec{k}}\, q_{\vec{k}}\right) \quad \text{— (a)} \\
a^\dagger_{\vec{k}} &= \frac{1}{\sqrt{2\omega_k}}\left(-i\bar{p}^*_{\vec{k}} + \omega_{\vec{k}}\, q^*_{\vec{k}}\right) \\
&= \frac{1}{\sqrt{2\omega_k}}\left(-ip_{-\vec{k}} + \omega_{\vec{k}}\, q_{-\vec{k}}\right) \\
a^\dagger_{\vec{k}} &= \frac{1}{\sqrt{2\omega_k}}\left(-ip_{\vec{k}} + \omega_k\, q_{\vec{k}}\right) \quad \text{— (b)}
\end{aligned}\right\} \quad
\begin{aligned}
&\text{From (a) \& (b) we get:} \\
&\frac{a_k + a^\dagger_{-k}}{2} = \sqrt{\frac{\omega_k}{2}}\, q_k \\
&q_k = \frac{1}{\sqrt{2\omega_k}}\left(a_k + a^\dagger_{-k}\right)
\end{aligned}$$

Note that $a_k = a_k(t)$; $a^\dagger_{-k} = a^\dagger_{-k}(t)$.

We know,

$$q_k(t) = \left(a_k(t) + a^\dagger_{-k}(t)\right) \frac{1}{\sqrt{2\omega_k}}$$

Using above,

$$\varphi(t,x) = \int \frac{d^3k}{(2\pi)^3} \left(a_k + a^\dagger_{-k}\right) \frac{1}{\sqrt{2\omega_k}}\, e^{i\vec{k}\cdot\vec{x}}$$

We can also see that this $\varphi(t,x)$ is real:

$$\varphi^\dagger(t,x) = \int \frac{d^3k}{(2\pi)^3} \frac{\left(a^\dagger_k + a_{-k}\right)}{\sqrt{2\omega_k}}\, e^{-i\vec{k}\cdot\vec{x}}$$

replacing $k \to -\vec{k}$:
$$\varphi^\dagger(t,x) = \int \frac{d^3k}{(2\pi)^3} \frac{\left(a^\dagger_{-k} + a_k\right)}{\sqrt{2\omega_k}}\, e^{i\vec{k}\cdot\vec{x}} \qquad \left(\omega_k = \omega_{-k} = \frac{\omega_k}{\sqrt{k^2+m^2}}\right)$$
$$= \varphi(t,x) \;\Rightarrow\; \varphi(t,x) \text{ is real}$$

We can write $\varphi(t,x)$ as:

$$\varphi(t,\vec{x}) = \int \frac{d^3k}{(2\pi)^3} \frac{1}{\sqrt{2\omega_k}} \left(a_k\, e^{i\vec{k}\cdot\vec{x}} + a^\dagger_k\, e^{-i\vec{k}\cdot\vec{x}}\right) \text{ — (1)}$$

$\hookleftarrow$ (replaced $\vec{k} \to -\vec{k}$ since integral remains same.)

We also had $p_k(t)$ as canonical momentum (from eq (a)−(b))—

$$p_k = \frac{-i}{\sqrt{2\omega_k}} \left(a_k(t) - a^\dagger_{-k}(t)\right)$$

So,

$$\begin{aligned}
\pi(t,\vec{x}) &= \int \frac{d^3k}{(2\pi)^3}\, p_k\, e^{i\vec{k}\cdot\vec{x}} \\
&= \int \frac{d^3k}{(2\pi)^3} \frac{(-i)}{\sqrt{2\omega_k}} \left(a_k - a^\dagger_{-k}\right) e^{i\vec{k}\cdot\vec{x}} \\
\underset{\substack{\text{canonical conjugate} \\ \text{field momentum to } \varphi(t,\vec{x})}}{\underset{\downarrow}{\pi(t,\vec{x})}} &= (-i) \int \frac{d^3k}{(2\pi)^3} \left(a_k\, e^{i\vec{k}\cdot\vec{x}} - a^\dagger_k\, e^{-i\vec{k}\cdot\vec{x}}\right) \text{ — (2)}
\end{aligned}$$

$\hookleftarrow$ again we flipped $\vec{k} \to -\vec{k}$ since integral remains

Lets find

$$\boxed{\left[\varphi(t,x),\, \pi(t,x')\right] = i\hbar\, \delta^3(\vec{x} - \vec{x}')} \qquad \text{(Equal time commutation relation)}$$

for $\hbar = 1$:

$$\underbrace{\boxed{\left[\varphi(t,x),\, \pi(t,\vec{x}')\right] = i\, \delta^3(x - x')}}_{\text{quantised}}$$

— This relation is sufficient for telling φ, π are quantised.

But to see energy spectrum, we need equation (1) & (2).

Comment about Normalisation —
Let us normalise

$$\begin{aligned}
a^\dagger_k|0\rangle &= c_k|k\rangle \\
\langle 0|a_k &= \langle k|c^*_k \\
\langle 0|a_k\, a^\dagger_k|0\rangle &= \langle k|c^*_k c_k|k\rangle
\end{aligned}$$

For

$$\langle 0|a_{\vec{k}}\, a^\dagger_{\vec{k}'}|0\rangle = (2\pi)^3\, \delta^3(\vec{k} - \vec{k}')$$

$\hookleftarrow$ not Lorentz invariant!
{(as $\vec{k}$ is only appearing with sides. $\vec{k}$ should mix with k_0
to show some Lorentz Invariance.)

We change definition, so that we have Lorentz invariant quantity. Suppose we made boost along $x^1 \longrightarrow$

$$\text{Then} \quad \begin{pmatrix} k_0' \\ k_1' \end{pmatrix} = \Lambda \begin{pmatrix} k_0 \\ k_1 \end{pmatrix} \quad ; \qquad \Lambda = \begin{bmatrix} \cosh\alpha & \sinh\alpha \\ \sinh\alpha & \cosh\alpha \end{bmatrix}$$

Since $\delta^3(\vec{k}-\vec{l}) = \delta(k_1-l_1)\,\delta(k_2-l_2)\,\delta(k_3-l_3)$ is not invariant. $\Big($i.e. $\delta^3(\vec{k}'-\vec{l}') \neq \delta^3(\vec{k}-\vec{l})\Big)$

$$k_1'-l_1' = \sinh\alpha\,(k_0-l_0) + \cosh\alpha\,(k_1-l_1)$$

$$\text{we can see,} \quad \delta(k_1'-l_1') \neq \delta(k_1-l_1) \ \Rightarrow \ \delta^3(\vec{k}-\vec{l}) \text{ is not Lorentz Invariant.}$$

We rather have a beautiful relation —

$$\delta\left(k_1'-l_1'\right) = \delta\left(\sinh\alpha\,(k_0-l_0) + \cosh\alpha\,(k_1-l_1)\right)$$

$$\delta\left(k_1'-l_1'\right) = \frac{\omega_{\vec{k}}}{\omega_{\vec{k}'}}\,\delta(k_1-l_1) \text{ — } (7)$$

$$\Rightarrow \quad \delta(k_1'-l_1') \neq \delta(k_1-l_1) \ \text{ because of the factor } \frac{\omega_{\vec{k}}}{\omega_{\vec{k}'}} \text{ being not equal to one.}$$

Therefore this tells us —

$$\boxed{\omega_{\vec{k}}\,\delta(\vec{k}-\vec{l}) \ \text{ has to be Lorentz invariant}}$$

Using eq (7):

$$\begin{aligned} \omega_{\vec{k}'}\,\delta(\vec{k}'-\vec{l}') &= \cancel{\omega_{\vec{k}'}}\left(\frac{\omega_{\vec{k}}}{\cancel{\omega_{\vec{k}'}}}\,\delta(\vec{k}-\vec{l})\right) \\ &= \omega_{\vec{k}}\,\delta(\vec{k}-\vec{l}) \\ \Rightarrow \quad & \omega_{\vec{k}}\,\delta(\vec{k}-\vec{l}) \text{ is Lorentz invariant!} \end{aligned}$$

Therefore we choose the normalisation such that,

$$\langle\vec{k}|\vec{k}'\rangle = (2\pi)^3 \qquad \underset{\substack{\uparrow \\ \text{depends on us, we could} \\ \text{have just taken } \omega_k}}{2\omega_{\vec{k}}} \qquad \delta^3(\vec{k}-\vec{k}')$$

This is true if

$$\boxed{|\vec{k}\rangle = \sqrt{2\omega_k}\;a^\dagger_{\vec{k}}\,|0\rangle}$$

Note, we have written $\varphi(t,x)$ as mode expansion which treats $t, \vec{x}$ on different footings:

$$\varphi(t,x) = \int \frac{d^3k}{(2\pi)^3}\, q_k(t)\, e^{i\vec{k}\cdot\vec{x}}$$

From QM (heisenberg picture) we can write,

$$\varphi(t,x) = e^{iHt}\,\varphi(0,x)\,e^{-iHt} \text{ — } (5)$$

$$\text{and} \quad \left.\begin{aligned} a_k(t) &= e^{iHt}\,a_k(0)\,e^{-iHt} \\ a_k^\dagger(t) &= e^{iHt}\,a_k^\dagger(0)\,e^{-iHt} \end{aligned}\right\} \text{ — } (6)$$

To further solve eq (6), we require $[H, a_k^\dagger]$ & $[H, a_k]$, where

$$H = \int \frac{d^3k}{(2\pi)^3}\,\omega_k\,a_k^\dagger\,a_k \qquad \left|\quad \begin{aligned} \text{use: } [A,BC] &= [A,B]C + B[A,C] \\ [AB,C] &= A[B,C] + [A,C]B \\ \text{and } [a_k\,a_{k'}^\dagger] &= (2\pi)^3\,\delta(k-k') \end{aligned}\right.$$

So,

$$[H,\, a_k] = -\omega_k\, a_k$$
$$[H,\, a_k^\dagger] = +\omega_k\, a_k^\dagger$$

Eq$^{\text{n}}$ (6) becomes,

$$a_k(t) = e^{iHt}\, a_k(0)\, e^{-iHt} = e^{-i\omega_k t}\, a_k$$
$$a_k^\dagger(t) = e^{iHt}\, a_k^\dagger(0)\, e^{-iHt} = e^{i\omega_k t}\, a_k^\dagger$$

Using

$$\varphi(0,x) = \int \frac{d^3k}{(2\pi)^3}\, \frac{1}{\sqrt{2\omega_k}} \left(a_k\, e^{i\vec{k}\cdot\vec{x}} + a_k^\dagger\, e^{-i\vec{k}\cdot\vec{x}} \right)$$

$$\begin{aligned}
\varphi(t,x) &= e^{iHt}\, \varphi(0,\vec{x})\, e^{-iHt} \\
&= \int \frac{d^3k}{(2\pi)^3}\, \frac{1}{\sqrt{2\omega_k}} \left(e^{-i\omega_k t}\, a_k\, e^{i\vec{k}\cdot\vec{x}} + e^{i\omega_k t}\, a_k^\dagger\, e^{-i\vec{k}\cdot\vec{x}} \right) \\
&\qquad x^\mu = (t,\vec{x}), \quad k^\mu = (k^0,\vec{k}), \quad k_\mu = (k_0, -\vec{k}) \\
&= \int \frac{d^3k}{(2\pi)^3}\, \frac{1}{\sqrt{2\omega_k}} \left(a_k\, e^{-i(k_\mu x^\mu)} + a_k^\dagger\, e^{i\, k_\mu x^\mu} \right)
\end{aligned}$$

$$\boxed{\varphi(t,x) = \int \frac{d^3k}{(2\pi)^3}\, \frac{1}{\sqrt{2\omega_k}} \left(a_k\, e^{-ik\cdot x} + a_k^\dagger\, e^{ik\cdot x} \right)}$$

Here a_k, $a_k^\dagger$ are time independent!

The important part is $e^{-ik\cdot x}$ & $e^{ik\cdot x}$ as these are Lorentz Invariant.

We can regard $e^{-ik\cdot x}$ as +ve frequency mode as we get $(+\omega_k)$ for $i\frac{\partial}{\partial t}\left(e^{-ik\cdot x}\right)$ & we call $e^{+ik\cdot x}$ as −ve frequency mode

$$\text{Since} \quad i\frac{\partial}{\partial t}\left(e^{ik\cdot x}\right) = i\frac{\partial}{\partial t}\, e^{i(k^0x^0 - \vec{k}\cdot\vec{x})}, \qquad i(ik^0) = -k^0 = -\omega_k.$$

It does not mean that these corresponds to +ve & −ve energies, instead of that annihilation operator a_k multiplies +ve frequency mode and creation operator $a_k^\dagger$ multiplies −ve frequency mode.

So, the field operator either creates state with one less particle or creates state with one more particle.

$$\text{Field} \longrightarrow \text{change no of particles}$$

— There is no definite no of particles in field. (as one part creates and one part destroys the particle).

— Convert φ to a complex field.

$$\text{i.e.} \qquad \varphi^\dagger \neq \varphi$$

In same space it is done:

$$\varphi(t,\vec{x}) = \int \frac{d^3k}{(2\pi)^3\sqrt{2\omega_k}} \left(b_k\, e^{-i\vec{k}\cdot\vec{x}} + a_k^\dagger\, e^{i\vec{k}\cdot\vec{x}} \right)$$
$$\varphi^*(t,\vec{x}) = \int \frac{d^3k}{(2\pi)^3}\, \frac{1}{\sqrt{2\omega_k}} \left(a_k\, e^{-i\vec{k}\cdot\vec{x}} + b_k^\dagger\, e^{i\vec{k}\cdot\vec{x}} \right)$$

Now we have here this sets of oscillators $(a, a^\dagger)$ and $(b, b^\dagger)$ with equal time commutation relations —

$$[a_k,\, a_{k'}^\dagger] = (2\pi)^3\, \delta^3(\vec{k} - \vec{k}')$$

$$[b_k,\, b_{k'}^\dagger] = (2\pi)^3\, \delta^3(k-k')$$
$$[a_k,\, b_k] = [a_k,\, b_k^\dagger] = [a_k^\dagger,\, b_k^\dagger] = 0$$

We will see that charge can be associated with complex fields φ & φ^*.

$$\text{Say} \quad \begin{cases} \varphi \longrightarrow +1 \ \text{ charge} \\ \varphi^* \longrightarrow -1 \ \text{ charge} \end{cases}$$

Then we can see that, if $a_k^\dagger$ creates particle of charge $+1$, and b_k destroys particle of charge -1, then in both cases there is increment of charge by $+1$ (i.e. both terms create charge that adds charge of $+1$).

Similarly, φ^* induces charge change of -1.

$$\begin{array}{ll} \text{Action of } \varphi \text{ induces} & \Delta q = +1 \\ \qquad\qquad \varphi^* & \Delta q = -1 \end{array} \Bigg\} \ \begin{array}{l} \text{if } a_k^\dagger \text{ produces particle \& } b_k^\dagger \\ \text{produces anti-particle} \end{array}$$

If we say $a, a^\dagger$ are associated with particles of some charge, then b & $b^\dagger$ will be associated with antiparticles of opposite charge. It turns out everything is same except for charge.

(Antiparticles have same spin, mass, other quantum nos as of particles.) They don't travel backward in time! — have negative mass, spin, energy...

In physics we call particles to those which are more in number, if antiparticles were more than we might call them by term "particles".

Propagators — (Real fields)

Purpose — (1) Particles can propagate from one time to future time. (2) It tests our ability to compute expectation values.

(A) Vacuum expectation values are not that interesting:

$$\begin{aligned} \langle 0|\varphi|0\rangle &= \langle 0|\varphi(x,t)|0\rangle \\ &= \Big\langle 0\Big| \int \frac{d^3k}{(2\pi)^3\sqrt{2\omega_k}} \left(a_k\, e^{-ik\cdot x} + a_k^\dagger\, e^{ik\cdot x}\right) \Big|0\Big\rangle \\ &= \int \frac{d^3k}{(2\pi)^3} \frac{1}{\sqrt{2\omega_k}} \Big(\underbrace{\langle 0|a_k\, e^{-ik\cdot x}|0\rangle}_{\to 0} + \underbrace{\langle 0|a_k^\dagger\, e^{ik\cdot x}|0\rangle}_{\to 0} \Big) \end{aligned}$$

$$\boxed{\langle 0|\varphi(t,x)|0\rangle = 0} \qquad \text{(In free field theory)}$$

$\langle 0|\varphi|0\rangle$ could be interesting in interacting theory.

(B) $\quad \langle 0|\varphi(x)\, \varphi(y)|0\rangle \qquad$ — 2 point function.

$$\langle 0|\varphi(x^0,\vec{x})\ \varphi(y^0,\vec{y})|0\rangle$$

Note that $\varphi(y^0,\vec{y})|0\rangle$ creates particle at $y = (y^0,\vec{y})$; $\langle 0|\varphi(x^0,\vec{x})$ annihilates particle at $x = (x^0,\vec{x})$.

i.e. $\langle 0|\varphi(x)\varphi(y)|0\rangle$ gives us probability amplitude for particle to be created at Y and destroyed at X (here $x^0 > y^0$). That is why we call it a propagator. Something (particle) pops out of vacuum at $y\,(y^0,\vec{y})$, propagates to $\vec{x}$ and gets destroyed at time x^0.

For $y^0 > x^0$ it would be more sensible/natural to calculate $\langle 0|\varphi(y)\,\varphi(x)|0\rangle$, which describe amplitude/probability amplitude for particle to be created at $\vec{x}$ at time x^0 and gets destroyed at time y^0 at position $\vec{y}$.

$$
\begin{aligned}
&\langle 0|\varphi(x)\varphi(y)|0\rangle \\
&\quad = \int \frac{d^3k}{(2\pi)^3}\frac{1}{\sqrt{2\omega_k}}\Big(\underbrace{\langle 0|a_k\,a_{k'}|0\rangle}_{\to 0}\,e^{-ikx}e^{-ik'y} + \langle 0|a_k\,a^\dagger_{k'}|0\rangle\,e^{-ikx}e^{ik'y} \\
&\qquad\qquad + \underbrace{\langle 0|a^\dagger_k\,a_{k'}|0\rangle}_{\to 0}\,e^{ikx}e^{-ik'y} + \underbrace{\langle 0|a^\dagger_k\,a^\dagger_{k'}|0\rangle}_{\to 0}\,e^{ikx}e^{ik'y}\Big) \\
&\quad = \int \frac{d^3k}{(2\pi)^3}\int \frac{d^3k'}{(2\pi)^3}\,\frac{1}{2\omega_k}\,\langle 0|a_k\,a^\dagger_{k'}|0\rangle\,e^{-ik\cdot x}\,e^{+ik'\cdot y}
\end{aligned}
$$

$$
a^\dagger_{k'}|0\rangle = \frac{1}{\sqrt{2\omega_{k'}}}\,|k'\rangle
$$

$$
\langle 0|a_k = \frac{1}{\sqrt{2\omega_k}}\,\langle k|
$$

$$
\begin{aligned}
&= \int \frac{d^3k}{(2\pi)^3}\,\frac{d^3k'}{(2\pi)^3}\,\frac{1}{2\omega_k}\,\frac{1}{\left(2\sqrt{\omega_k\,\omega_{k'}}\right)}\,\underline{\langle k|k'\rangle}\,e^{-ik\cdot x}\,e^{ik'\cdot y} \\
&= \int \frac{d^3k}{(2\pi)^3}\,\frac{d^3k'}{(2\pi)^3}\,\frac{1}{2\omega_k}\,\frac{1}{\sqrt{\omega_k\,\omega_{k'}}}\,(2\pi)^3\,\delta^3(k-k')\,\cancel{2\omega_k}\,e^{-ik\cdot x}\,e^{ik'\cdot y} \\
&= \int \frac{d^3k}{(2\pi)^3}\,\frac{1}{2\omega_k}\,e^{ik\cdot(y-x)} \qquad\qquad (\text{different } x, y)
\end{aligned}
$$

This motivates us to define

$$
\begin{aligned}
D_0(x,y) &= \theta(x^0-y^0)\,\langle 0|\varphi(x)\,\varphi(y)|0\rangle \\
&\quad + \theta(y^0-x^0)\,\langle 0|\varphi(y)\,\varphi(x)|0\rangle \\
&= \langle 0|\,T\{\varphi(x)\,\varphi(y)\}\,|0\rangle
\end{aligned}
$$

This object shall be studied for particle propagation rather than just $\langle 0|\varphi(x)\varphi(y)|0\rangle$ or $\langle 0|\varphi(y)\varphi(x)|0\rangle$, where

$$
\theta(x-y) = \begin{cases} 1 & x > y \\ 0 & x < y \end{cases}
$$

Here $T\{\varphi(x)\,\varphi(y)\} = T\{\varphi(y)\,\varphi(x)\}$ is called time ordered product.

It turns out that $D_0(x,y)$ (for these variables only) depends upon just the difference '$x-y$'.

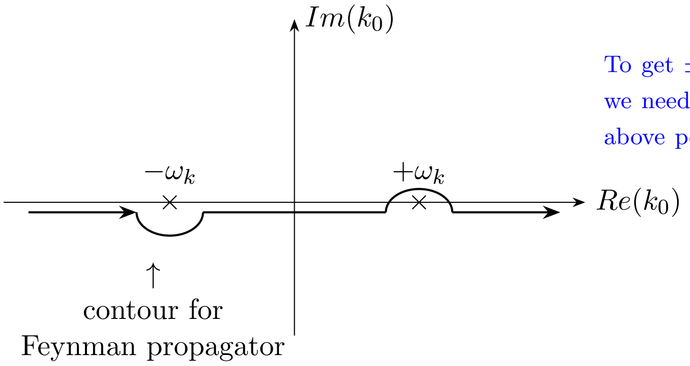
$$
\text{i.e.} \qquad D_0(x,y) = D_0(x-y) = D_0(x^\mu - y^\mu)
$$

Because universe has geometry of translational invariance, we can translate whole system to another place. Therefore $D_0(x,y)$ shall only depend upon relative position $x^\mu - y^\mu$ or $x-y$.

$$
D_0(x-y) = \int \frac{d^4k}{(2\pi)^4}\,\frac{i}{k^2-m^2}\,e^{-i\,k\cdot(x-y)}
$$

$$
\begin{gathered}
\downarrow \\
\text{pole at } (k^0)^2 - \vec{k}^2 - m^2 = 0 \\
k^0 = \pm\sqrt{\vec{k}^2 + m^2} \\
k_0 = k^0 = \pm\,\omega_{\vec{k}}
\end{gathered}
$$

We'll see how to go around the pole and calculable $\mathrm{D_F}$:



To get $\pm$ve energy waves we need to go pole below & above pole respectively

How do we calculate contour integral — by closing contour and making(?) the arc at infinity does not contribute, then,

$$\int_{C_F \text{ open}} = \oint_{\text{closed}} \qquad \text{(so that we can use residue theorem)}$$

$$D_F(x-y) = \lim_{\epsilon \to 0} \int \frac{d^4k}{(2\pi)^4} \frac{i}{k^2 - m^2 + i\epsilon} \qquad \underset{\substack{e^{-ik^0(x^0-y^0)}\, e^{-i\vec{k}\cdot(\vec{x}-\vec{y})} \\ k^0 = -iR \text{ (LHP)}}}{e^{-i\vec{k}\cdot(x-y)} \atop \downarrow}$$

We can close the contour in lower half plane (LHP) so that k^0 takes $-$ve imaginary values $\left(k^0 = Re^{i\theta}\right)$: the factor $e^{-i(k^0)(x^0-y^0)}$ converges to '0' as $R \to \infty$. For $(x^0 - y^0 > 0)$

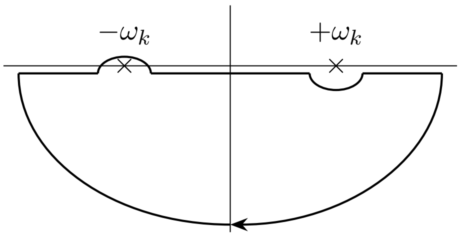
i.e. C_F can be closed in LHP if $x^0 > y^0$; ditto UHP if $x^0 < y^0$.

For $x^0 > y^0$ —

$$\frac{i}{k^2 - m^2} = \frac{i}{k_0^2 - \vec{k}^2 - m^2} = \frac{i}{k_0^2 - \omega_k^2} = \frac{i}{(k_0 - \omega_k)(k_0 + \omega_k)}$$

$$= \left(\frac{i}{k_0 - \omega_k} - \frac{i}{k_0 + \omega_k}\right) \frac{1}{2\omega_{\vec{k}}}$$

$\hookleftarrow$ Not useful pole since we are closing in LHP.

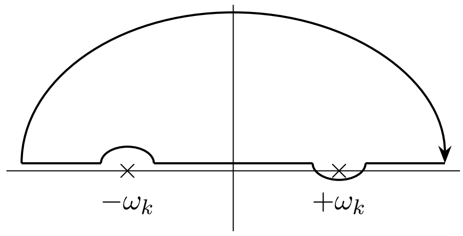


$$\begin{aligned}
D_F(x-y) &= \int \frac{d^4k}{(2\pi)^4} \frac{i}{k^2 - m^2} e^{-i\vec{k}\cdot(x-y)} \\
&= \int \frac{dk_0}{2\pi} \int \frac{d^3k}{(2\pi)^3} \left(\frac{i}{k_0 - \omega_k} - \frac{i}{k_0 + \omega_k}\right) \frac{1}{2\omega_k} e^{-ik_0(x^0-y^0) + i\vec{k}\cdot(\vec{x}-\vec{y})} \\
&\quad \text{using } \left\{k_\mu (x-y)^\mu = \vec{k}\cdot(x-y)\right\} = k^0(x^0 - y^0) - \vec{k}\cdot(\vec{x}-\vec{y}) \\
&= \int \underbrace{\frac{(-2\pi i \operatorname{Res}(k_0 = \omega_k))}{(2\pi)} \frac{1}{2\omega_k}}_{\text{clockwise (LHP)}} \frac{d^3k}{(2\pi)^3} e^{-ik(x-y)} \Big|_{k^0 = \omega_k}
\end{aligned}$$

$$= \int \frac{-2\pi i\,(i)}{2\pi\; 2\omega_k} \frac{d^3k}{(2\pi)^3}\; e^{-ik(x-y)}\bigg|_{k^0=\omega_k}$$

$$D_F(x-y) = \int \frac{d^3k}{(2\pi)^3} \frac{1}{2\omega_k}\; e^{-ik(x-y)}\bigg|_{k^0=\omega_k} \qquad \left(\text{For } x^0 > y^0\right)$$

For $x^0 < y^0$ we close the contour in UHP.



$$\begin{aligned}
D_F(x-y) &= \int \frac{d^4k}{(2\pi)^4} \frac{i}{k^2-m^2}\; e^{-i\vec{k}\cdot(x-y)} \\
&= \int \frac{d^3k}{(2\pi)^3} \int \frac{dk^0}{2\pi} \left(\underbrace{\frac{i}{k_0-\omega_k}}_{\text{Not included}} - \frac{i}{k_0+\omega_k} \right) \frac{1}{2\omega_{\vec{k}}}\; e^{-ik^0(x^0-y^0)+i\vec{k}\cdot(\vec{x}-\vec{y})} \\
&= \int \frac{2\pi i\,(-i)}{(2\pi)\,2\omega_k} \frac{d^3k}{(2\pi)^3}\; e^{-i\,k\cdot(x-y)}\bigg|_{k_0=-\omega_k}
\end{aligned}$$

$$D_F(x-y) = \int \frac{d^3k}{(2\pi)^3} \frac{1}{2\omega_k}\; e^{-i\,k\cdot(x-y)}\bigg|_{k^0=-\omega_k} \qquad \left(x^0 < y^0\right)$$

The reason why feynman propagator is important is that, it is basic ingredient to understand interactions.

Suppose, a theory is weakly interacting, then on average particles with interact with each other once in a while, the more time they interact with each other, lower will be the probability.

Particle interact then propagates freely . . . and then they interact and again interact freely.

Process of Interaction $\approx$ Free propagation + point interactions once in a while.

In between any two interactions particle propagate freely and that is job of feynman propagator ($D_F(x-y)$ to find amplitude for free theory propagation).

Lecture 4 (Sunil Mukhi)

Propagators — The basic idea of propagator is that, it is vacuum expectation of product of two fields. There are many such vacuum expectation values, we will start with most basic two fields.

(1)

$$\underset{\substack{\downarrow \\ \text{comes from} \\ \text{translational invariance of } \Delta(x,y)}}{\Delta(x-y)} = \langle 0|\varphi(x)\,\varphi(y)|0\rangle \quad \begin{cases} \text{Just a propagator, not useful \& does not have any name.} \\ \text{Physically useful one's will be linear combination} \\ \text{of these in some way.} \end{cases}$$

$$\varphi(x) = \underbrace{\int \frac{d^3k}{(2\pi)^3}\frac{1}{\sqrt{2\omega_k}}\,a_k\,e^{-ik\cdot x}}_{\substack{\varphi_+(x) \\ \text{Removes particle} \\ \text{of momentum } \vec{k}}} + \underbrace{\int \frac{d^3k}{(2\pi)^3}\frac{1}{\sqrt{2\omega_k}}\,a_k^\dagger\,e^{ik\cdot x}}_{\substack{\varphi_-(x) \\ \text{Adds particle} \\ \text{of momentum } \vec{k}}}$$

$$\varphi(x) = \varphi_+(x) + \varphi_-(x)$$

here, $e^{ik\cdot x} = e^{i(k^0x^0 - \vec{k}\cdot\vec{x})}$ $\quad \begin{pmatrix} k^0=\omega \\ x^0=t \end{pmatrix}$

→ Since a_k is destruction operator, so $\varphi_+(x)$ which is linear combination of a_k is also a destruction/annihilation operator.

→ Similarly $\varphi_-(x)$ is a creation operator.

$$\begin{aligned}
\langle 0|\varphi(x)\varphi(y)|0\rangle &= \langle 0|\left(\varphi_+(x)+\varphi_-(x)\right)\left(\varphi_+(y)+\varphi_-(y)\right)|0\rangle \\
&= \langle 0|\varphi_+(x)\,\varphi_-(y)|0\rangle + 0 + 0 + 0 \\
&= \left\langle 0\right|\int \frac{d^3k}{(2\pi)^3}\frac{1}{\sqrt{2\omega_k}}\,a_k\,e^{-ik\cdot x}\int \frac{d^3k'}{(2\pi)^3}\frac{1}{\sqrt{2\omega_{k'}}}\,a_{k'}^\dagger\,e^{ik'\cdot y}\left|0\right\rangle \\
&= \int \frac{d^3k}{(2\pi)^3}\frac{d^3k'}{(2\pi)^3}\frac{1}{\sqrt{2\omega_k}}\frac{1}{\sqrt{2\omega_{k'}}}\,\langle 0|a_k\,a_{k'}^\dagger|0\rangle\,e^{-ik\cdot x}\,e^{ik'\cdot y} \\
&\qquad\qquad\qquad \downarrow \\
&\qquad\qquad [a_k,\,a_{k'}^\dagger] = a_k a_{k'}^\dagger - a_{k'}^\dagger a_k = (2\pi)^3\,\delta^3(k-k') \\
&= \int \frac{d^3k}{(2\pi)^3}\frac{d^3k'}{(2\pi)^3}\frac{1}{\sqrt{2\omega_k}}\frac{1}{\sqrt{2\omega_{k'}}}\,e^{-ik\cdot x}\,e^{ik'\cdot y}\Big(\underbrace{\langle 0|a_{k'}^\dagger a_k|0\rangle}_{\to 0} + \langle 0|[a_k,a_{k'}^\dagger]|0\rangle\Big) \\
&= \int \frac{d^3k}{(2\pi)^3}\frac{d^3k'}{(2\pi)^3}\frac{1}{\sqrt{2\omega_k}}\frac{1}{\sqrt{2\omega_{k'}}}\,e^{-ik\cdot x}\,e^{ik'\cdot y}\,(2\pi)^3\,\delta^3(k-k') \\
&= \int \frac{d^3k}{(2\pi)^3}\frac{1}{2\omega_k}\,e^{-i\,k\cdot(x-y)} \qquad \text{which we found in lec 3}
\end{aligned}$$

$$\boxed{\Delta(x-y) = \langle 0|\varphi(x)\,\varphi(y)|0\rangle = \int \frac{d^3k}{(2\pi)^3}\frac{1}{2\omega_k}\,e^{-i\,k\cdot(x-y)}}$$

Feynman Propagator

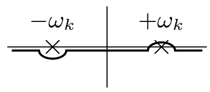
$$D_F(x-y) = \langle 0|T\left(\varphi(x)\varphi(y)\right)|0\rangle = \langle 0|T\left(\varphi(y)\varphi(x)\right)|0\rangle$$

$$= \theta(x^0 - y^0)\, \langle 0|\varphi(x)\varphi(y)|0\rangle + \theta(y^0 - x^0)\, \langle 0|\varphi(y)\varphi(x)|0\rangle$$
$$= \theta(x^0 - y^0)\, \Delta(x-y) + \theta(y^0 - x^0)\, \Delta(y-x) \quad —(1)$$

Therefore we can see that, feynman propagator is linear combination of two ordinary propagators.

Now we can easily see that the formula we wrote in previous lecture was correct:

$$D_F(x-y) = \int_{C_F} \frac{d^4k}{(2\pi)^4}\, \frac{i}{k^2 - m^2}\, e^{-ik(x-y)}$$

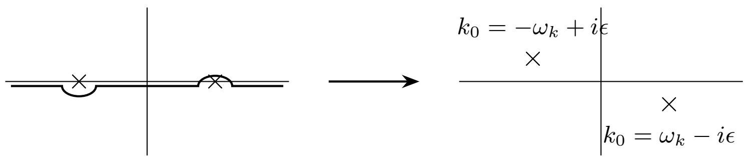


To show that after integrating over k^0:

$$D_F(x-y) = \begin{cases} \Delta(x-y) & x^0 > y^0 \\ \Delta(y-x) & x^0 < y^0 \end{cases}$$

If we go above/below both poles we end up getting retarded and advanced propagator.

Another way is to move poles bit up & down:



Done by replacing
$k^2 - m^2 \to k^2 - m^2 + i\epsilon$

$$\left\{ \begin{array}{ll} \text{Poles become:} & k^2 - m^2 + i\epsilon = 0 \\ & k_0 = \pm\sqrt{\vec{k}^2 + m^2 - i\epsilon} \\ & k_0 = \pm\,(\omega_k - i\epsilon) \\ \Rightarrow & k_{0_1} = \omega_k - i\epsilon \\ & k_{0_2} = -\omega_k + i\epsilon \end{array} \right.$$

Therefore we write

$$\boxed{D_F(x-y) = \lim_{\epsilon\to 0} \int \frac{d^4k}{(2\pi)^4}\, \frac{i}{k^2 - m^2 + i\epsilon}\, e^{-ik(x-y)}}$$

Eq (1) becomes —

$$D_F(x-y) = \theta(x^0 - y^0)\, \langle 0|\varphi(x)\,\varphi(y)|0\rangle + \theta(y^0 - x^0)\, \langle 0|\varphi(y)\,\varphi(x)|0\rangle$$
$$= \theta(x^0 - y^0)\, \langle 0|\, \underbrace{[\varphi_+(x),\, \varphi_-(y)]}_{\hookrightarrow \text{ complex number}}\, |0\rangle + \theta(y^0 - x^0)\, \langle 0|\, \underbrace{[\varphi_+(y),\, \varphi_-(x)]}_{\hookrightarrow \text{ complex number}}\, |0\rangle$$
$$= \theta(x^0 - y^0)\, [\varphi_+(x), \varphi_-(y)]\, \langle 0|0\rangle + \theta(y^0 - x^0)\, [\varphi_+(y), \varphi_-(x)]\, \langle 0|0\rangle$$

$$D_F(x-y) = \theta(x^0 - y^0)\, \underbrace{[\varphi_+(x),\, \varphi_-(y)]}_{\text{A complex number}} + \theta(y^0 - x^0)\, [\varphi_+(y),\, \varphi_-(x)]$$ (representing an operator)!

Lagrangian density & Action

Motivation — (1) Communicate important information about field equations, make symmetries manifest. (2) Hamiltonian required Lagrangian which require $\mathcal{L}$ and Action. (we need it for quantisation) (3) $\mathcal{L}$ itself is starting point for path integral quantisation.

Above are reasons to work with $\mathcal{L}$ and Action as opposed to just EOM.

EOM of interest was $\rightarrow \left(\partial_\mu \partial^\mu + m^2\right)\varphi(x) = 0$

$$\& \qquad \mathcal{L}(\varphi, \partial_\mu \varphi) = \frac{1}{2}\left(\partial_\mu \varphi\right)\left(\partial^\mu \varphi\right) - \frac{1}{2}m^2\varphi^2$$

$$\& \qquad L = \int d^3x\, \mathcal{L} \qquad (\mathcal{L} \text{ for value of } \varphi \text{ at each point } \vec{x})$$

$$\& \qquad A = \int dt\, L = \int d^4x\, \mathcal{L}$$

Expanding

$$\mathcal{L} = \frac{1}{2}\partial_\mu \varphi\, \partial^\mu \varphi - \frac{1}{2}m^2\varphi^2 = \underbrace{\frac{1}{2}\dot{\varphi}^2}_{T} - \underbrace{\frac{1}{2}(\nabla\varphi)^2 - \frac{1}{2}m^2\varphi^2}_{V}$$

<u>EOM</u> —

$$S = \int d^4x\, \mathcal{L} = \int d^4x \left(\frac{1}{2}\partial_\mu\varphi\, \partial^\mu\varphi - \frac{1}{2}m^2\varphi^2\right)$$

$$\frac{\delta S}{\delta(\varphi(y))} = 0 \qquad \text{from} \left\{ \frac{\delta\varphi(x)}{\delta\varphi(y)} = \delta^4(x-y) \right. \quad \hookrightarrow \text{variation at } \varphi(x) \text{ w.r.t. variation at } y \text{ for } \varphi.$$

$$\frac{\delta S_{KG}}{\delta\varphi(y)} = \int d^4x \left[\frac{1}{2}\, 2\, \underbrace{\frac{\delta\varphi(x)}{\delta\varphi(y)}}_{\delta_{xy}} \underbrace{\partial^\mu\varphi}_{A^\mu} + \frac{1}{2}\, 2\, \underbrace{\partial_\mu\varphi}_{A_\mu}\, \partial^\mu \underbrace{\frac{\delta\varphi(x)}{\delta\varphi(y)}}_{\delta_{xy}} - \frac{1}{2}m^2\, 2\, \varphi(x)\frac{\delta\varphi(x)}{\delta\varphi(y)}\right]$$

$$= \int d^4x \left[\underline{\partial^\mu\varphi\, \partial_\mu\left(\delta^4(x-y)\right)} - m^2\varphi(x)\,\delta^4(x-y)\right] \qquad \begin{cases} \partial_x^\mu = \dfrac{\partial}{\partial x_\mu} \\ \partial_x = \dfrac{\partial}{\partial x} \end{cases} \qquad (A^\mu B_\mu = A_\mu B^\mu)$$

Integral by parts —

$$= \int d^4x \left(\partial_x^\mu\left(\partial_\mu\varphi\, \delta^4(x-y)\right) - \partial^\mu\left(\partial_\mu\varphi(x)\right)\delta^4(x-y) - m^2\varphi(x)\,\delta^4(x-y)\right)$$

$$= \int d^4x \left(\cancel{\partial^\mu\left(\partial_\mu\varphi\, \delta^4(x-y)\right)}^{\,0\text{ (boundary)}} - \left(\partial^2 + m^2\right)\varphi(x)\,\delta^4(x-y)\right)$$

$$= -\left(\partial_0^2 - \nabla^2 + m^2\right)\varphi(y)$$

For S_{KG}:

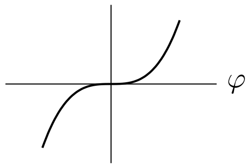
$$\frac{\delta S}{\delta\varphi(y)} = 0 \;\Rightarrow\; \boxed{\left(\partial^2 + m^2\right)\varphi(y) = 0}$$

Interacting theory (Non linear EOM)

$$S_{\text{int}} = \int d^4x \left(\frac{1}{2}\,\partial_\mu\varphi\, \partial^\mu\varphi - V(\varphi)\right) \qquad \begin{array}{l} \nearrow \text{ field potential} \\ \hookrightarrow \text{ potential in field space} \end{array}$$

<u>Possibilities of $V(\varphi)$</u> —

$$V(\varphi) = \begin{cases} 1) \text{ Constant } (\times) & \{\text{does not contribute to EOM} \\ 2) \text{ Linear in } \varphi\ (\times) & \{\text{as shifting by } \varphi \text{ will remove this term.} \\ 3) \text{ quadratic } \left(\frac{1}{2}m^2\varphi^2\right) & \{\text{already there in free theory} \\ 4) \text{ Cubic} & (\text{have problem of negative energy}) \\ 5) \ \varphi^4 & \end{cases}$$



System does not
have stable behaviour
for $V(\varphi) = \varphi^3$.

$$\text{So,} \qquad V(\varphi) = \varphi^{2n} \qquad \text{where } \left\{ n \in \mathbb{Z}^+ \;\&\; n > 1 \right.$$

So the general field potential we take is

$$V(\varphi) = \frac{1}{2}m^2\varphi^2 + \frac{\lambda}{4!}\varphi^4 \longrightarrow \text{interaction term}$$

This is known as 'φ^4 theory'. (we can have mix of φ^3 & φ^4 terms.)

This theory will describe particles which not only propagate but also interact.

Here, λ — strength of interaction/coupling constant.

<u>Dimension</u> — 'S' is dimensionless $\left(e^{iS/\hbar},\ \hbar = (\!(\ )\!)\right)$

$$c = 1 \;\Rightarrow\; [L] = [T]$$

$$S = \int d^4x \left(\frac{1}{2}\partial^\mu\varphi\,\partial_\mu\varphi - \frac{1}{2}m^2\varphi^2 - \frac{\lambda\varphi^4}{4!}\right)$$

$$[S] = 0 = \Big(4 - 2 + 2[\varphi]\Big) \;\Rightarrow\; [\varphi] = -1 \text{ in length unit } / \; +1 \text{ in energy/mass unit}$$

$$\text{mass dimension of } L, T = -1$$

$$0 = -4 - [\lambda] + 4[\varphi]$$

$$4 = [\lambda] + 4 \;\Rightarrow\; \boxed{[\lambda] = 0} \qquad \text{mass dimension of } \lambda \text{ is } 0.$$

If we use $\varphi^5, \varphi^6, \ldots \varphi^n$ interaction terms we get coupling constant λ with negative mass dimension & theories with $-$ve mass dimension of λ are not renormalizable. So we only study φ^4.

Corresponding to lagrangian density, we have notion of $\mathcal{H}$ (Hamiltonian density):

$$\pi(x) = \frac{\delta\mathcal{L}}{\delta(\dot\varphi(x))}$$

$$\begin{aligned} \mathcal{L} &= \frac{1}{2}\partial^\mu\varphi\,\partial_\mu\varphi - V(\varphi) \\ &= \frac{1}{2}\left(\partial^0\varphi\,\partial_0\varphi - \partial^i\varphi\,\partial_i\varphi\right) - V(\varphi) \\ &= \frac{1}{2}\dot\varphi^2 - \frac{1}{2}(\nabla\varphi)^2 - V(\varphi) \end{aligned}$$

We then find $\mathcal{H} = \dot\varphi(x)\,\pi - \mathcal{L}$, for $\mathcal{L} = \frac{1}{2}\dot\varphi^2 - \frac{1}{2}(\nabla\varphi)^2 - V(\varphi)$:

$$\pi = \frac{\partial\mathcal{L}}{\partial\dot\varphi} = \dot\varphi$$

$$\text{So} \quad \mathcal{H} = \dot\varphi\,\pi - \mathcal{L} = \pi^2 - \left(\frac{1}{2}\pi^2 - \frac{1}{2}(\nabla\varphi)^2 - V(\varphi)\right)$$

$$\boxed{\mathcal{H} = \frac{\pi^2}{2} + \frac{1}{2}(\nabla\varphi)^2 + V(\varphi)}$$

$\downarrow$
energy density in space

$$\text{i.e.} \quad H = \int d^3x\, \mathcal{H}$$

For

$$\mathcal{H} = \frac{\pi^2}{2} + \frac{1}{2}(\nabla\varphi)^2 + \frac{1}{2}m^2\varphi^2 + \frac{\lambda}{4!}\varphi^4$$

Our goal shall be to find eigen values and eigen functions of this hamiltonian. We would like to know how interactions will occur mathematically. We shall diagonalise the $\mathcal{H}$. We have effectively diagonalised first part in last lecture, we were able to rewrite it in terms of the $a, a^\dagger$. But it turns out we can't diagonalise the $\mathcal{H}$.

Using another approach we write

$$\mathcal{H} = H_0 + H_{\text{int}}$$
$$H_0 = \frac{\pi^2}{2} + \frac{1}{2}(\nabla\varphi)^2 + \frac{1}{2}m^2\varphi^2$$
$$H_{\text{int}} = \frac{\lambda}{4!}\varphi^4$$

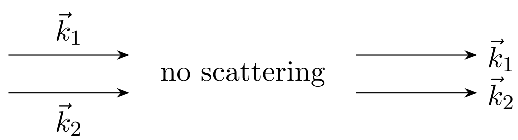
We assume ($\lambda \ll 1$) & use perturbation theory in powers of λ.

What is meaning of perturbation theory in QFT?

It means for any physical quantity, there will be contribution from part where $\lambda = 0$, then first correction is proportional to λ, & second correction is proportional to λ^2 & so on.

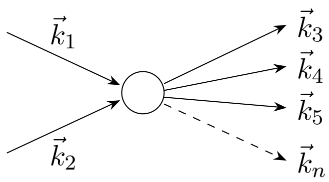
Zeroth order of λ was being discussed by us in last 3 lectures (i.e. free theory). In that theory there are free particles of arbitrary four momentum k_i $\left(\forall (3 \text{ momentum } \vec{k}, \text{ energy } \omega_k)\right)$ as many as we like, i.e. there are states with many particle, no particle, few particles.

In free theory there won't be any scattering, suppose we have 2 particle state with momentum $\vec{k}_1$ and $\vec{k}_2$, as long as $\lambda = 0$:



Since there is no interaction particles will keep propagating with initial 3 momentum $\vec{k}_i$.

To first order in 'λ', we expect particles to scatter exactly once! Then we have



$\Big\}$ final no of particles is not fixed!

We shall do a power series in λ & ask if the series converge. In fact all field theories have divergent power series, they satisfy weaker cond. i.e. they are asymptotic, which means that, to high enough orders we get closer & closer to some answer & after some more higher orders the series diverge.

Recall that in free theory, vacuum state $|0\rangle$ satisfies

$$H_0|0\rangle = 0 \quad \text{—(k)} \quad \text{\& all other states have positive energy w.r.t. the ground state.}$$

For case of interacting theory — The vacuum is denoted by $|\Omega\rangle$ and is defined by

$$H|\Omega\rangle = \underbrace{(H_0 + H_{\text{int}})\,|\Omega\rangle}_{\text{some minimum energy}} \quad \text{—(1)}$$

Free vacuum $|0\rangle$ is defined by H_0, where as interacting vacuum is defined by $(H_0 + H_{\text{int}})$.

— We can imagine that we have a system with no interactions $\lambda = 0$, and the system is in the vacuum. & imagine slowly turning on λ, the nature of vacuum will change. The vacuum will evolve from $|0\rangle$ to $|\Omega\rangle$.

We can see that $|\Omega\rangle \neq |0\rangle$, by looking at H_{int}:

$$H_{\text{int}} \propto \varphi^4 = (\varphi_+ + \varphi_-)^4$$

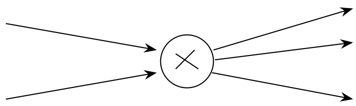
$$= \varphi_-^4 + \varphi_-^3\varphi_+ + \cdots$$

↪ create 4 particles from vacuum

contains 4 $\left(a_{k_1}^\dagger a_{k_2}^\dagger a_{k_3}^\dagger a_{k_4}^\dagger\right)$ in free theory

That means $|\Omega\rangle$ can not be $|0\rangle$, because that term creates particles from $|0\rangle$. So we must change the state in some way to compensate such that finally there are no particles in the interacting theory in new vacuum.

- When we turn 'on' the interaction, the vacuum state $|0\rangle$ even at fixed time, changes to different vacuum state $|\Omega\rangle$. We should not think in terms of time evolution. Its one state turning into another state.
- Note that $|\Omega\rangle$ contains particles in free theory but is empty in interacting theory.
- In truly free theory particles just propagate without interacting. For 1st order in λ they only interact once, which means for most of their life they remain free.



↪ from $-\infty$ to x, particles must propagate freely. If interacts on the way then we get another power of λ, then its not the lowest order term.

For first order/lowest order interaction, they are described by H_0 (most of time).

→ Suppose at some time $t = 0$, we have a field configuration $\varphi(0, \vec{x})$ & we want to study its time evolution (in interacting theory).

We can define,

$$\varphi_0(t, \vec{x}) = e^{iH_0 t}\, \varphi(0, \vec{x})\, e^{-iH_0 t} \qquad \begin{cases}\text{free Hamiltonian evolution of} \\ \text{free field.}\end{cases}$$

We can also define

$$\varphi(t, \vec{x}) = e^{iHt}\, \varphi(0, \vec{x})\, e^{-iHt} \qquad \begin{cases}\text{interacting field evolution} \\ \text{according to full Hamiltonian} \\ H = H_0 + H_{\text{int}}.\end{cases}$$

We can agree that we don't have free field in interacting theory, we only have field $\varphi(t, x)$ which evolves with full hamiltonian H. However field might not be some observable, its needed for our calculation.

The goal here is to express the interacting theory in terms of variables of free theory.

The simplest physical quantity of interacting theory/any field theory is the propagator. Lets find the interacting propagator. It will be amplitude of particle propagation from x to y:

$$(t,\vec{x}) \rightleftarrows (t',\vec{y}), \quad \text{or} \quad y \to x \text{ depending on time ordering.}$$

Interacting Propagator —

$$\begin{aligned} \langle\Omega|\, T\big(\overset{\substack{\text{interacting fields}\\ \downarrow}}{\varphi(x)\,\varphi(y)}\big)\, |\Omega\rangle \qquad &\Longleftarrow \text{ our goal!} \\ = \langle 0|\, f\,(\varphi_0,\cdots)\, |0\rangle \end{aligned}$$

Once we achieve this goal of writing down $\langle\Omega|T(\varphi(x)\varphi(y))|\Omega\rangle$ in terms of free fields i.e. $\langle 0|f(\varphi_0,\cdots)|0\rangle$, then note that it reduced to calculations in free theory (i.e. expectation value of product of free fields in free vacuum.)

The first term is easy to write, since interactions are turned off (leading term):

$$\langle\Omega|T\,(\varphi(x)\varphi(y))\,|\Omega\rangle = \langle 0|T\,(\varphi_0(x)\,\varphi_0(y))\,|0\rangle + \lambda\,\langle 0|T(\cdots)|0\rangle + \lambda^2\,\langle 0|T(\cdots)|0\rangle + \cdots$$
$$(\lambda \to \text{turn})$$

here Ω, φ depends upon λ:

$$\left.\begin{aligned} \Omega &= \Omega(\lambda) \\ \varphi &= \varphi(\lambda) \\ \varphi_0 &= \varphi(\lambda=0) \\ |\Omega(\lambda=0)\rangle &= |0\rangle \end{aligned}\right]$$

As we saw

$$\varphi(t,\vec{x}) = e^{iHt}\,\varphi(0,\vec{x})\,e^{-iHt} \qquad \begin{cases} \varphi_0(t,\vec{x}) = e^{iH_0t}\,\varphi(0,\vec{x})\,e^{-iH_0t} \\ \Rightarrow\ \varphi(0,\vec{x}) = e^{-iH_0t}\,\varphi_0(t,\vec{x})\,e^{iH_0t} \end{cases}$$

Relation between $\varphi_0(x)$ & $\varphi(x)$

$$\varphi(t,\vec{x}) = \underline{e^{iHt}\,e^{-iH_0t}}\,\varphi_0(t,\vec{x})\,\underline{e^{iH_0t}\,e^{-iHt}}$$

$$\neq e^{i(H-H_0)t}\,\varphi_0(t,\vec{x})\,e^{-i(H-H_0)t} \qquad \begin{cases} \text{Since } [H,H_0]\neq 0. \\ \text{✘ } [H_{\text{int}},H_0]\neq 0; \\ \Rightarrow [H_0,H]+[H_{\text{int}},H_0]\neq 0 \\ \text{1) } [H_{\text{int}},H_0]\neq 0 \\ \Rightarrow \left[(a^\dagger)^4,\, a^\dagger a\right]\neq 0 \\ \text{indeed its} \neq 0. \end{cases}$$

$$\varphi(t,x) \neq e^{iH_{\text{int}}t}\,\varphi_0(t,\vec{x})\,e^{-iH_{\text{int}}t}$$

We define,

$$\boxed{\begin{aligned} \varphi(t,x) &= U^{-1}\,\varphi_0(t,x)\,U \\ \text{then}\quad U &= e^{iH_0t}\,e^{-iHt} \end{aligned}} \qquad \begin{cases} \text{where `}U\text{' is some unitary operator} \\ U^\dagger U = UU^\dagger = 1 \end{cases}$$

Lecture 5 (Sunil Mukhi) — Part 1

We could write $\mathcal{H} = H_0 + H_{\text{int}}$, where

$$H_{\text{int}} = \int d^3x\, \mathcal{H}_{\text{int}} = \frac{\lambda}{4!}\int d^3x\, \varphi^4$$

$\varphi_0(x) =$ interaction picture field (or) Free field.
$\varphi(x)$: Interacting Field.

In earlier lecture we saw that, we can write

$$\begin{aligned}\varphi(x,t) &= e^{iHt}\,\varphi(x,0)\,e^{-iHt}\\ &= e^{iHt}\,e^{-iH_0t}\,\varphi_0(x,t)\,\underbrace{e^{iH_0t}\,e^{-iHt}}_{U}\end{aligned}$$

$$\boxed{\varphi(\vec{x},t) = U^\dagger\,\varphi_0(x,t)\,U}$$

Physical interpretation of above equation is that, we can express

$$\varphi_0(x,t) = U\,\varphi(x,t)\,U^\dagger \qquad \longrightarrow \text{ interacting field.}$$

$H_0|0\rangle = 0 \;\longrightarrow$ free vacuum
$(H_0 + H_{\text{int}})\,|\Omega\rangle = E_0|\Omega\rangle$
$\hookrightarrow$ interacting vacuum

This means we can get $\varphi(\vec{x},t)$ from $\varphi_0(x,t)$ by operating some conjugate operators $U^\dagger$ & U.

$E_0 \neq 0$ due to perturbation, ground state energy has shifted.

We will also write $|\Omega\rangle$ in terms of $|0\rangle$.
Now, strategy to solve for U, is to look for differential equation in U:

$$U = e^{iH_0t}\,e^{-iHt} \qquad \begin{cases} U \neq e^{iH_0t - iHt} \\ \text{as } [H_0, H] \neq 0\end{cases}$$

$$\begin{aligned}\frac{dU}{dt} &= iH_0U - iUH\\ &= iH_0\,e^{iH_0t}\,e^{-iHt} - i\,e^{+iH_0t}\,e^{-iHt}\,H\\ &= i\left(e^{iH_0t}H_0\,e^{-iHt} - e^{iH_0t}\,H\,e^{-iHt}\right)\\ &= i\left(e^{iH_0t}\,(H_0 - H)\,e^{-iHt}\right)\\ &= -i\,e^{iH_0t}\,H_{\text{int}}\,e^{-iHt}\\ &= -i\,\underbrace{e^{iH_0t}\,H_{\text{int}}\,e^{-iH_0t}}_{H_I(t)}\,\underbrace{e^{iH_0t}\,e^{-iHt}}_{U(t)}\end{aligned}$$

We define,

$$H_I(t) = e^{iH_0t}\,H_{\text{int}}\,e^{-iH_0t} \qquad \hookrightarrow \text{ free } H_0 \text{ evolution of } H_{\text{int}}.$$

then,

$$\boxed{i\,\frac{dU(t)}{dt} = H_I(t)\,U(t)}$$

where $H_I(t)$ = Interaction picture hamiltonian.

Suggestion of students —

$$U(t) = e^{-i\int_0^t H_I(t')\,dt'}$$

It is only true if $[H_I(t), H_I(t')] = 0 \quad \forall\, t, t' \in (-\infty, \infty)$.
But

$$\left[H_I(t) = \left[e^{iH_0t}\,H_{\text{int}}\,e^{-iH_0t},\; e^{iH_0t'}\,H_{\text{int}}\,e^{-iH_0t'}\right] = H_I(t')\right] \neq 0$$

Note that here $(0 < t' < t)$, here t' is dummy variable which is integrated over 0 to t.

What if we want $U(t'')$ such that $(0 < t'' < t)$:

$$U(t'') = e^{-i\int_0^{t''} H_I(t')\,dt'} \qquad \{\ (0 < t' < t'')$$

Then

$$\begin{aligned}
U(t) &= U(t)\,U(t'') \\
&= \underbrace{e^{-i\int_{t''}^{t} H_I(t')\,dt'}}_{\text{`}H'\text{ at }(t''<t'<t)} \cdot \underbrace{e^{-i\int_0^{t''} H_I(t')\,dt'}}_{H\text{ at }(0<t'<t'')} \\
&\neq e^{-i\int_0^t H_I(t')\,dt'} \qquad \text{Since } [H_I(t_1),\, H_I(t_2)] \neq 0.
\end{aligned}$$

Another way to see that <u>how</u> suggestion is wrong —
Lets find

$$\begin{aligned}
\frac{dU}{dt} = \frac{d}{dt}(U) = \frac{d}{dt}\Bigg(1 - i\int_0^t H_I(t')\,dt' + \frac{(-i)^2}{2}\int_0^t H_I(t')dt' \int_0^t H_I(t'')dt'' \\
+ \frac{(-i)^3}{3!}\int_0^t H_I(t')\,dt' \int_0^t H_I(t'')\,dt'' \int_0^t H_I(t''')\,dt''' + \cdots\Bigg) \\
= 0 - i\,H_I(t) + \frac{(-i)^2}{2!}\left(H_I(t)\int_0^t H_I(t')\,dt' + \int_0^t H_I(t')\,dt'\ H_I(t)\right) \\
+ \frac{(-i)^3}{3!}\Bigg(H_I(t)\int_0^t H_I(t'')\,dt'' \int_0^t H_I(t''')\,dt''' + \int_0^t H_I(t')\,dt'\ H_I(t)\int_0^t H_I(t'')\,dt'' \\
+ \int_0^t H_I(t')\,dt' \int_0^t H_I(t'')\,dt''\ H_I(t)\Bigg) + \cdots
\end{aligned}$$

We can see that

$$\int_0^t H_I(t')\,dt'\ H_I(t) \neq H_I(t)\int_0^t H_I(t')\,dt'$$

as $[H_I(t),\, H_I(t')] \neq 0$ for t' $(0 < t' < t)$; also $[H_I(t),\, H_I(t'')] \neq 0$ for t'' $(0 < t'' < t)$.

$$\Rightarrow \quad i\frac{dU}{dt} \neq H_I(t)\,U(t)$$

So our suggestion does not give correct answer. $\hookrightarrow$ <u>$U(t,0)$</u>

<u>Solution</u> —

$$U(t) = T\left(e^{-i\int_0^t H_I(t')\,dt'}\right)$$

$\downarrow$
Time ordering symbol.

Let us explain it —
We can write

$$\int_0^t H_I(t')\,dt' = \lim_{\epsilon\to 0}\ \epsilon\,[H_I(0) + H_I(\epsilon) + H_I(2\epsilon) + \cdots + H_I(t-\epsilon)] \qquad \text{where `}\epsilon\text{' is time step.}$$

First consider ordinary exponential of above quantity & then we will look into Time ordered exponential of this quantity.

The ordinary exponential:

$$T\left(e^{-i\int_0^t H_I(t')\,dt'}\right) = \lim_{\epsilon\to 0} T\left(e^{-i\epsilon\left(H^0 + H^\epsilon + H^{2\epsilon} + \cdots + H^{t-\epsilon}\right)}\right) — ①$$

Whereas Time ordered exponential means

$$T\left(\exp\left(-i\int_0^t H_I(t')\,dt'\right)\right) = \lim_{\epsilon\to 0} e^{-i\epsilon H^{t-\epsilon}} \cdots e^{-i\epsilon H^{2\epsilon}}\, e^{-i\epsilon H^{\epsilon}}\quad \underset{\substack{\uparrow\\ \text{earliest time}\\ \text{operate on}\\ \text{state first.}}}{e^{-i\epsilon H^{0}}} \quad \text{— (2)}$$

In limit $\epsilon \to 0$: eq (1) = eq (2).

If we write right Riemann sum we have

$$U = T\left(e^{-i\int_0^t H_I(t')\,dt'}\right) = \lim_{\epsilon\to 0} e^{-i\epsilon H_I^t}\, e^{-i\epsilon H_I^{t-\epsilon}} \cdots e^{-i\epsilon H^0}$$

such that

$$\frac{dU}{dt} = -i\,H_I(t)\left(e^{-i\epsilon H_I^t}\, e^{-i\epsilon H_I^{t-\epsilon}} \cdots e^{-i\epsilon H^0}\right)$$

$$\frac{dU}{dt} = H_I(t)\,U(t) \qquad \text{which recovers the eq}^{\text{n}}.$$

Now lets find $U^\dagger \to$

$$U^\dagger(t) = \text{It should be Anti time ordering of } \left(e^{+i\int_0^t H_I(t')\,dt'}\right): \qquad U^\dagger(t) = \bar{T}\left(e^{+i\int_0^t H_I(t')\,dt'}\right)$$

We can easily see that $UU^\dagger = U^\dagger U = 1$:

$$UU^\dagger = \left(\lim_{\epsilon\to 0} e^{-i\epsilon H^t}\cdots e^{-i\epsilon H^{2\epsilon}}\, e^{-i\epsilon H^{\epsilon}}\right) \lim_{\epsilon\to 0}\left(e^{i\epsilon H^{\epsilon}}\, e^{i\epsilon H^{2\epsilon}}\cdots e^{i\epsilon H^t}\right)$$

$$= \text{All terms cancel out and give } 1$$

$$\boxed{UU^\dagger = 1}$$

<u>Identity</u>:

$$U(t_1)\,U^\dagger(t_2) = U(t_1,t_2) = T\left(e^{-i\int_{t_1}^{t_2} H_I(t'')\,dt''}\right) \qquad \forall\ (t_1 > t_2).$$

$$\Rightarrow \quad T\left(\exp\left(-i\int_0^{t_1} H_I(t')\,dt'\right)\right)\bar{T}\left(\exp\left(+i\int_0^{t_2} H_I(t'')\,dt''\right)\right)$$

Since, $t_1 > t_2$ we can write LHS —

$$T\left(e^{-i\int_{t_2}^{t_1} H_I(t'')\,dt''}\right)\underline{T\left(e^{-i\int_0^{t_2} H_I(t')\,dt'}\right)\bar{T}\left(e^{+i\int_0^{t_2} H_I(t'')\,dt''}\right)}$$

$$= T\left(e^{-i\int_{t_2}^{t_1} H_I(t'')\,dt''}\right)\cdot 1$$

$$= U(t_1,t_2)$$

$$\text{i.e.} \qquad U(t_1)\,U^\dagger(t_2) = U(t_1,t_2) \qquad \text{for } t_1 > t_2 > 0$$

Remember we want to calculate —

$\langle\Omega|\,T\left(\varphi(x_1)\,\varphi(x_2)\right)|\Omega\rangle$ — called 2 point function or Feynman propagator. (Full propagator).

We can define multiple point functions. Now we have two Jobs —

1) Express $|\Omega\rangle$ in terms of $|0\rangle$
2) Express $\varphi(x)$ in terms of $\varphi_0(x)$

Job (2) is already done:

$$\varphi(x,t) = U^\dagger\,\varphi_0(x,t)\,U \qquad \text{where} \quad U(t) = T\left(e^{-i\int_0^t H_I(t')\,dt'}\right)$$

Lecture 5 — Part 2

(1) $|\Omega\rangle$ in terms of $|0\rangle$ —
To do that let us find

$$e^{-iHt}\,|0\rangle \qquad \hookrightarrow \text{ How free vacuum evolve in time with full hamiltonian of the system}$$

Remember that this hamiltonian 'H' has ground state $|\Omega\rangle$. Let $|n\rangle$ be the eigen state of $\hat{H}$, such that

$$H|n\rangle = E_n|n\rangle \qquad \hookrightarrow \text{ state with } n \text{ particles}$$
$$H|\Omega\rangle = \underset{\text{ground state of } H.}{\underset{\downarrow}{E_0}} \quad |\Omega\rangle \qquad\qquad E_0 < E_n \ \ (n \neq 0)$$

Then we can write

$$\begin{aligned}
e^{-iHt}\,|0\rangle &= e^{-iHt}\left(\sum_n |n\rangle\langle n|\right)|0\rangle \\
&= \sum_n e^{-iHt}\,|n\rangle\langle n|0\rangle \\
&= \sum_n e^{-iE_nt}\,|n\rangle\langle n|0\rangle \\
&= e^{-iE_0t}\,|\Omega\rangle\langle\Omega|0\rangle + \sum_{n\neq 0} e^{-iE_nt}\,|n\rangle\langle n|0\rangle
\end{aligned}$$

For large times —

$$= e^{-iE_0t}\left[\,|\Omega\rangle\langle\Omega|0\rangle + \underbrace{\sum_{n\neq 0} e^{-i(E_n-E_0)t}\,|n\rangle\langle n|0\rangle}_{\longrightarrow 0 \text{ as } T\to\infty(1-i\epsilon)}\right] \qquad \begin{array}{l}\downarrow\\ \text{Mathematical Prescription to}\\ \text{get rid of } \sum_{n\neq 0} e^{-i(E_n-E_0)t}|n\rangle\langle n|0\rangle\end{array}$$

$$\lim_{\lambda\to\infty}\sum_{n\neq 0} e^{-i(E_n-E_0)\,\lambda(1-i\epsilon)}\,\langle n|0\rangle\,|n\rangle$$

Because we need to find relation between $|\Omega\rangle$ & $|0\rangle$. We will see that 'ϵ' does not play much role.

$$\lim_{\lambda\to\infty}\sum_{n\neq 0}\underbrace{e^{-i(E_n-E_0)\lambda}}_{\text{oscillatory}}\ \underbrace{e^{-\lambda\epsilon(E_n-E_0)}}_{\substack{\text{vanishes}\to 0\\ \text{For } (E_n>E_0)}} \longrightarrow 0$$

So, we can write RHS as $\quad$ ($t \Leftrightarrow T \Rightarrow$ large times with "$-i\epsilon$" prescription)

$$\lim_{T\to\infty(1-i\epsilon)} e^{-iE_0T}\,|\Omega\rangle\langle\Omega|0\rangle$$

i.e.

$$\Rightarrow \lim_{T\to\infty(1-i\epsilon)} e^{-iHT}\,|0\rangle = \lim_{T\to\infty(1-i\epsilon)} e^{-iE_0T}\,|\Omega\rangle\langle\Omega|0\rangle \qquad \left\{ \text{ or,} \quad |\Omega\rangle \propto \lim_{T\to\infty(1-i\epsilon)} e^{-iHT}\,|0\rangle \right.$$

{If we start with free vacuum state $|0\rangle$ and evolve it using full hamiltonian for very long time, we get interacting vacuum in the end.}

$$|\Omega\rangle = \lim_{T\to\infty(1-i\epsilon)} \frac{1}{\underbrace{\langle\Omega|0\rangle\,e^{-iE_0T}}_{\hookrightarrow \text{ Normalisation}}}\ e^{-iHT}\,|0\rangle$$

Normalisation is just a proportionality factor, we are looking for quantum state. ($\langle\Omega|\Omega\rangle = 1 \Rightarrow$ Normalisation facto

So,

$$\boxed{|\Omega\rangle = \lim_{T\to\infty(1-i\epsilon)} e^{-iHT}\,|0\rangle} \qquad \text{Job (1) is also complete.}$$

Is $|\Omega\rangle$ normalizable?
Yes. In perturbation theory we expect it to be **normalizable**.

Interacting vacuum is what free vacuum will reach, if we wait long enough, when we let it evolve under the interacting theory.

Since $T \to \infty$ is not possible, $|\Omega\rangle$ we get is close enough approximation in reality.

We can further write

$$|\Omega\rangle = \lim_{T\to\infty(1-i\epsilon)} e^{-iHT}\cdot 1\cdot|0\rangle = \lim_{T\to\infty(1-i\epsilon)} e^{-iHT}\,e^{iH_0T}\,|0\rangle$$
$$\left(\text{as} \quad e^{iH_0T}|0\rangle = e^0\,|0\rangle = |0\rangle\right)$$

So

$$|\Omega\rangle = \lim_{T\to\infty(1-i\epsilon)} U^\dagger(-T)\,|0\rangle \qquad \begin{cases} \text{Since} \quad U(t) = e^{iH_0t}\,e^{-iHt} \\ \qquad U^\dagger(t) = e^{+iHt}\,e^{-iH_0t} \\ \quad U^\dagger(-T) = e^{-iHT}\,e^{iH_0T} \end{cases}$$

$$|v\rangle = AB\,|r\rangle$$
$$\langle v| = \langle r|(AB)^\dagger = \langle r|\,B^\dagger A^\dagger$$

We have following result —

$$1) \quad \begin{aligned} |\Omega\rangle &= \lim_{T\to\infty(1-i\epsilon)} U^\dagger(-T)\,|0\rangle \\ \langle\Omega| &= \lim_{T\to\infty(1-i\epsilon)} \langle 0|\,U(T) \end{aligned} \qquad \Bigg| \qquad \begin{aligned} U(t) &= e^{iH_0t}e^{-iHt} \\ |\Omega\rangle &= \lim_{T\to\infty(1-i\epsilon)} e^{-iHT}e^{iH_0T}|0\rangle \\ \langle\Omega| &\equiv \lim_{T\to\infty(1-i\epsilon)} \langle 0|\,e^{iH_0T}\,e^{-iHT} \\ &= \lim_{T\to\infty(1-i\epsilon)} \langle 0|\,U(T) \end{aligned}$$

Now we can write $\langle\Omega|T\{\varphi(x_1)\varphi(x_2)\}|\Omega\rangle$ in term of free field quantities:

$$\begin{aligned} \langle\Omega|T\left(\varphi(x)\varphi(y)\right)|\Omega\rangle = {} & \theta(x^0-y^0)\,\langle\Omega|\varphi(x)\,\varphi(y)|\Omega\rangle \\ & + \theta(y^0-x^0)\,\langle\Omega|\varphi(y)\,\varphi(x)|\Omega\rangle \end{aligned}$$

Let us choose 1$^{\text{st}}$ case $x^0 - y^0 > 0$; $\quad$ i.e. $x^0 > y^0$:

$$\begin{aligned} \langle\Omega|T\left(\varphi(x)\varphi(y)\right)|\Omega\rangle &= \langle\Omega|\,\varphi(x)\,\varphi(y)\,|\Omega\rangle \\ &= \lim_{T\to\infty(1-i\epsilon)} \langle 0|\,U(T)\left[U^\dagger(x^0)\,\varphi_0(x)\,U(x^0)\right]\left[U^\dagger(y^0)\,\varphi_0(y)\,U(y^0)\right]U^\dagger(-T)\,|0\rangle \\ &\qquad \left\{ \text{where } \varphi_0(x), \varphi_0(y) \text{ are free field operators; } x = (x^0, \vec{x}),\; y = (y^0, \vec{y}) \right) \\ &= \lim_{T\to\infty(1-i\epsilon)} \langle 0|\;U(T,x^0)\;\varphi_0(x)\;U(x^0,y^0)\;\varphi_0(y)\;U(y^0,-T)\;|0\rangle \end{aligned}$$

We can see that every operator is in correct time ordering:

$$= \lim_{T\to\infty(1-i\epsilon)} \langle 0|\;T\left[\,U(T,x^0)\;\varphi_0(x)\;U(x^0,y^0)\;\varphi_0(y)\;U(y^0,-T)\,\right]|0\rangle$$

We can interchange quantities inside Time ordering $T\left(\phi(t_1)\phi(t_2)\phi(t_3)\right) = T\left(\phi(t_3)\phi(t_2)\phi(t_1)\right)$:

$$= \lim_{T\to\infty(1-i\epsilon)} \langle 0|\;T\Big[\varphi_0(x)\,\varphi_0(y)\;\underbrace{U(T,x^0)}\;\underbrace{U(x^0,y^0)}\;\underbrace{U(y^0,-T)}\Big]|0\rangle$$

$$= \lim_{T\to\infty(1-i\epsilon)} \langle 0|\, T\Big[\varphi_0(x)\,\varphi_0(y)\ \overbrace{U(T)U^\dagger(x^0)\ U(x^0)U^\dagger(y^0)\ U(y^0)U^\dagger(-T)}\Big]|0\rangle$$

$$= \lim_{T\to\infty(1-i\epsilon)} \langle 0|\, T\Big[\varphi_0(x)\,\varphi_0(y)\ \underline{U(T)\,U^\dagger(-T)}\ \cancelto{1}{U^\dagger(x^0)U(x^0)}\ \cancelto{1}{U^\dagger(y^0)U(y^0)}\Big]|0\rangle$$

$$= \lim_{T\to\infty(1-i\epsilon)} \langle 0|\, T\Big[\varphi_0(x)\ \varphi_0(y)\ U(T,-T)\Big]|0\rangle$$

$$= \lim_{T\to\infty(1-i\epsilon)} \langle 0|\, T\Big[\varphi_0(x)\ \varphi_0(y)\ e^{-i\int_{-T}^{T} H_I(t)\,dt}\Big]|0\rangle$$

Normalisation of $|\Omega\rangle$ is taken care of by dividing the above expression by

$$\langle 0|\, T\Big[e^{-i\int_{-T}^{T} H_I(t)\,dt}\Big]|0\rangle$$

So, n-point function reduces to —

$$\boxed{\langle\Omega|\, T\left[\varphi(x_1)\,\varphi(x_2)\ldots\varphi(x_n)\right]|\Omega\rangle = \lim_{T\to\infty(1-i\epsilon)} \frac{\langle 0|\, T\Big[\varphi_0(x_1)\,\varphi_0(x_2)\ldots\varphi_0(x_n)\ e^{-i\int_{-T}^{T} H_I(t)\,dt}\Big]|0\rangle}{\langle 0|\, T\Big[\exp\Big(-i\int_{-T}^{T} H_I(t)\,dt\Big)\Big]|0\rangle}}$$

It says that vacuum expectation value of time ordered n-fields is equal to same quantity (vacuum expectation) in free field theory modified by exponential of the interaction, & divided by vacuum expectation value of time ordered exponential of interaction.

For $y^0 > x^0$, we also get same answer.

Note that,

$$H_I(t) = e^{iH_0 t}\ H_{\text{int}}\ e^{-iH_0 t} \qquad \text{(free } H_0 \text{ evolution of interaction)}$$

For φ^4 theory,

$$H_{\text{int}} = \int \frac{\lambda}{4!}\ \varphi^4(\vec{x},0)\ d^3x$$

$$H_I(t) = e^{iH_0 t}\left(\int \frac{\lambda}{4!}\,\varphi^4(\vec{x},0)\,d^3x\right) e^{-iH_0 t}$$

$$H_I(t) = \int \frac{\lambda}{4!}\ \underline{e^{iH_0 t}\ \varphi^4(\vec{x},0)\ e^{-iH_0 t}}\ d^3x \qquad \{\,\varphi(x,0) = \varphi_0(x,0)$$

$$= \int \frac{\lambda}{4!}\ \varphi_0^4(\vec{x},t)\ d^3x$$

$$\boxed{H_I(t) = \frac{\lambda}{4!}\int \varphi_0^4(x)\ d^3x}$$

Note

$$H_I(t) \neq H_{\text{int}}$$
$$\underline{\text{as}\quad \varphi(\vec{x},t) = \varphi(x) \neq \varphi(\vec{x},0)}$$

Lecture 6

Result of the conceptually difficult talk is (Things will be technically complex (not conceptually difficult) from here onwards)

$$\langle\Omega|\, T[\varphi(x_1)\,\varphi(x_2)\dots\varphi(x_n)]\,|\Omega\rangle = \lim_{T\to\infty(1-i\epsilon)} \frac{\langle 0|\, T\left(\varphi_0(x_1)\,\varphi_0(x_2)\cdots\varphi_0(x_n)\, e^{-i\int_{-T}^{T} H_I(t')\,dt'}\right)|0\rangle}{\langle 0|\, T\left[\exp\left(-i\int_{-T}^{T} H_I(t')\,dt'\right)\right]|0\rangle}$$

Note that this is completely Normalisation independent. (i.e. true even if $|\Omega\rangle$ is not normalized. (Since expression takes care of it via dividing by $\langle\Omega|\Omega\rangle$.)

The RHS can be computed using computers as all $\varphi_0(x_i)$ are oscillators, but the exponential gives us so many terms rapidly that even computer finds it hard to do. (As we expand exponential to some order $\left(1+x+\frac{x^2}{2!}+\cdots\right)$)

Technique to calculate $\langle 0|\, T[\varphi_0(x_1)\,\varphi_0(x_2)\cdots\varphi_0(x_n)]\,|0\rangle$:

Since RHS reduces to above expression after expanding exponential.

$$\begin{aligned}
\text{Ex}\quad & \langle 0|\, T\Big[\varphi_0(x_1)\,\varphi_0(x_2)\dots\varphi_0(x_n)\Big(1-i\int\varphi^4\,d^4y+\cdots\Big)\Big]|0\rangle \\
&= \langle 0|\, T\,[\varphi_0(x_1)\,\varphi_0(x_2)\dots\varphi_0(x_n)]\,|0\rangle - \int d^4y\,\langle 0|T\left[\varphi_0(x_1)\varphi_0(x_2)\dots\varphi_0(x_n)\,\varphi^4(y)\right]|0\rangle
\end{aligned}$$

(1) <u>2 point correlator</u> $\quad\langle 0|\, T[\varphi_0(x_1)\,\varphi_0(x_2)]\,|0\rangle$

$$\varphi_0(x) = \varphi_+ + \varphi_- \qquad \begin{matrix} \downarrow & & \searrow \\ \text{contains } \hat{a} & & \text{contains } \hat{a}^\dagger \\ \displaystyle\int \frac{d^3k}{(2\pi)^3}\frac{1}{\sqrt{2\omega_k}}\, a_k\, e^{-ik\cdot x} & : & \displaystyle\int \frac{d^3k}{(2\pi)^3}\frac{1}{\sqrt{2\omega_k}}\, a_k^\dagger\, e^{ik\cdot x} \end{matrix}$$

We have following fact —

$$T\left(\varphi_0(x_1)\,\varphi_0(x_2)\right) = \theta(t_1-t_2)\,\varphi_0(x_1)\,\varphi_0(x_2) + \theta(t_2-t_1)\,\varphi_0(x_2)\,\varphi_0(x_1)$$

Each $\varphi_0(x_i)$ has two terms $\varphi_+ + \varphi_-$:

$$\begin{aligned}
&= \theta(t_1-t_2)\Big[\varphi_+(x_1)\varphi_+(x_2) + \varphi_+(x_1)\varphi_-(x_2) + \varphi_-(x_1)\varphi_+(x_2) + \varphi_-(x_1)\varphi_-(x_2)\Big] \\
&\quad + \theta(t_2-t_1)\Big[\varphi_+(x_2)\varphi_+(x_1) + \varphi_+(x_2)\varphi_-(x_1) + \varphi_-(x_2)\varphi_+(x_1) + \varphi_-(x_2)\varphi_-(x_1)\Big]
\end{aligned}$$

$\varphi_+(x_1)\,\varphi_+(x_2) = \varphi_+(x_2)\,\varphi_+(x_1)\quad$ as $[a_k, a_{k'}] = 0$; $\qquad$ Similarly $\;\varphi_-(x_1)\,\varphi_-(x_2) = \varphi_-(x_2)\,\varphi_-(x_1)\quad$ as $[a_k^\dagger, a_{k'}^\dagger] =$

$$\begin{aligned}
&= \Big[\cancelto{1}{\theta(t_1-t_2)+\theta(t_2-t_1)}\Big]\Big(\varphi_+(x_1)\varphi_+(x_2) + \varphi_-(x_1)\varphi_-(x_2)\Big) \\
&\quad + \theta(t_1-t_2)\Big[\varphi_+(x_1)\,\varphi_-(x_2) + \varphi_-(x_1)\varphi_+(x_2)\Big] + \theta(t_2-t_1)\Big[\varphi_+(x_2)\,\varphi_-(x_1) + \varphi_-(x_2)\varphi_+(x_1)\Big]
\end{aligned}$$

We would like to compare it with normal ordering (which is important to define as will be of use later).

<u>Normal Ordering</u> —
All destruction operators move to right and all creation operators move to left, so that $\langle 0|\, a_k^\dagger\, a_k\,|0\rangle =$

0.

Symbol for normal ordering is

$$:\varphi_0(x_1)\,\varphi_0(x_2): \,=\, :\varphi_+(x_1)\varphi_+(x_2) + \varphi_+(x_1)\varphi_-(x_2) + \varphi_-(x_1)\varphi_+(x_2) + \varphi_-(x_1)\varphi_-(x_2):$$

$$\begin{aligned}
\langle 0|:\varphi_0(x_1)\,\varphi_0(x_2):|0\rangle &= \cancelto{0}{\langle 0|:\varphi_+(x_1)\varphi_+(x_2):|0\rangle} + \langle 0|:\overset{a}{\varphi_+}(x_1)\overset{a^\dagger}{\varphi_-}(x_2):|0\rangle \\
&\quad + \cancelto{0}{\langle 0|:\varphi_-(x_1)\varphi_+(x_2):|0\rangle} + \cancelto{0}{\langle 0|:\varphi_-(x_1)\varphi_-(x_2):|0\rangle} \\
&= \langle 0|\,\overset{a^\dagger}{\varphi_-}(x_2)\;\overset{a}{\varphi_+}(x_1)\,|0\rangle \\
&= 0
\end{aligned}$$

Let us find $T\left[\varphi_0(x_1)\,\varphi_0(x_2)\right] \;-\; :\varphi_0(x_1)\,\varphi_0(x_2):$

$$\begin{aligned}
&\Big[\varphi_+(x_1)\cancel{\varphi_+(x_2)} + \varphi_-(x_1)\cancel{\varphi_-(x_2)}\Big] + \theta(t_1-t_2)\Big[\varphi_+(x_1)\varphi_-(x_2) + \varphi_-(x_1)\varphi_+(x_2)\Big] \\
&\quad + \theta(t_2-t_1)\Big(\varphi_+(x_2)\varphi_-(x_1) + \varphi_-(x_2)\varphi_+(x_1)\Big) \\
&\quad - \underbrace{1}_{\to\, 1=\theta(t_1-t_2)+\theta(t_2-t_1)} \Big(\varphi_+(x_1)\cancel{\varphi_+(x_2)} + \varphi_-(x_2)\varphi_+(x_1) + \varphi_-(x_1)\varphi_+(x_2) + \varphi_+(x_1)\cancel{\varphi_-(x_2)}\Big) \\
&= \theta(t_1-t_2)\Big[\underline{\varphi_+(x_1)\varphi_-(x_2)} + \cancel{\varphi_-(x_1)\varphi_+(x_2)} - \underline{\varphi_-(x_2)\varphi_+(x_1)} - \cancel{\varphi_-(x_1)\varphi_+(x_2)}\Big] \\
&\quad + \theta(t_2-t_1)\Big[\underline{\varphi_+(x_2)\varphi_-(x_1)} + \cancel{\varphi_-(x_2)\varphi_+(x_1)} - \underline{\varphi_-(x_1)\varphi_+(x_2)} - \cancel{\varphi_-(x_2)\varphi_+(x_1)}\Big] \\
&= \theta(t_1-t_2)\,\Big[\varphi_+(x_1),\,\varphi_-(x_2)\Big] + \theta(t_2-t_1)\,\Big[\varphi_+(x_2),\,\varphi_-(x_1)\Big] \\
&= \langle 0|\,T\left(\varphi_0(x_1)\,\varphi_0(x_2)\right)|0\rangle \qquad \text{(difference of this operator turns out to be a complex number)}
\end{aligned}$$

$$T\left(\varphi_0(x_1)\,\varphi_0(x_2)\right) = \,:\varphi_0(x_1)\,\varphi_0(x_2): + \langle 0|\,T\left[\varphi_0(x_1)\,\varphi_0(x_2)\right]|0\rangle$$

Taking vacuum expectation value makes it identity:

$$\begin{aligned}
\langle 0|T\left(\varphi_0(x_1)\varphi_0(x_2)\right)|0\rangle &= \cancelto{0}{\langle 0|:\varphi_0(x_1)\varphi_0(x_2):|0\rangle} + \langle 0|T\left(\varphi_0(x_1)\varphi_0(x_2)\right)|0\rangle\,\langle 0|0\rangle \\
\langle 0|T\left[\varphi_0(x_1)\,\varphi_0(x_2)\right]|0\rangle &= \langle 0|T\left[\varphi_0(x_1)\,\varphi_0(x_2)\right]|0\rangle.
\end{aligned}$$

Let us consider time ordered product of 4 point fields —

$$\begin{aligned}
T\left(\varphi_0(x_1)\varphi_0(x_2)\varphi_0(x_3)\varphi_0(x_4)\right) = \,&:\varphi_0(x_1)\,\varphi_0(x_2)\,\varphi_0(x_3)\,\varphi_0(x_4): \\
&\begin{cases}
+\; :\varphi_0(x_1)\,\varphi_0(x_2):\;\langle 0|T[\varphi_0(x_3)\,\varphi_0(x_4)]|0\rangle \\
+\; :\varphi_0(x_1)\,\varphi_0(x_3):\;\langle 0|T[\varphi_0(x_2)\,\varphi_0(x_4)]|0\rangle \\
+\; :\varphi_0(x_1)\,\varphi_0(x_4):\;\langle 0|T[\varphi_0(x_2)\,\varphi_0(x_3)]|0\rangle \\
+\;\cdots
\end{cases} \\
&\qquad\qquad\qquad \text{6 such terms.} \\
&\begin{cases}
+\; :\varphi_0(x_1):\;\cancelto{0}{\langle 0|T[\varphi_0(x_2)\,\varphi_0(x_3)\,\varphi_0(x_4)]|0\rangle} \\
+\; :\varphi_0(x_2):\;\cancelto{0}{\langle 0|T[\varphi_0(x_1)\,\varphi_0(x_3)\,\varphi_0(x_4)]|0\rangle} \\
+\;\cdots
\end{cases} \\
&\qquad\qquad \langle 0|T[\text{odd \# of free fields}]|0\rangle = 0
\end{aligned}$$

$$
\begin{cases}
+ \ :\varphi_0(x_1)\varphi_0(x_2)\varphi_0(x_3): \ \cancelto{0}{\langle 0|T[\varphi_0(x_4)]|0\rangle} \\
+ \ :\varphi_0(x_2)\varphi_0(x_3)\varphi_0(x_4): \ \cancelto{0}{\langle 0|T[\varphi_0(x_1)]|0\rangle} \\
+ \ \cdots
\end{cases}
$$

3 such terms

$$
\begin{aligned}
&+ \langle 0|T[\varphi_0(x_1)\varphi_0(x_2)]|0\rangle \, \langle 0|T[\varphi_0(x_3)\varphi_0(x_4)]|0\rangle \\
&+ \langle 0|T[\varphi_0(x_1)\varphi_0(x_3)]|0\rangle \, \langle 0|T[\varphi_0(x_2)\varphi_0(x_4)]|0\rangle \\
&+ \langle 0|T[\varphi_0(x_1)\varphi_0(x_4)]|0\rangle \, \langle 0|T[\varphi_0(x_2)\varphi_0(x_3)]|0\rangle
\end{aligned}
$$

So the

$$
\begin{aligned}
\langle 0|T\left[\varphi_0(x_1)\varphi_0(x_2)\varphi_0(x_3)\varphi_0(x_4)\right]|0\rangle = {} & \cancelto{0}{\langle 0| :\varphi_0(x_1)\varphi_0(x_2)\varphi_0(x_3)\varphi_0(x_4): |0\rangle} \\
& + \cancelto{0}{\langle 0| :\varphi_0(x_1)\varphi_0(x_2): |0\rangle} \, \langle 0|T[\varphi_0(x_3)\varphi_0(x_4)]|0\rangle \ + \ \text{All such combinations} \\
& + \langle 0|T[\varphi_0(x_1)\,\varphi_0(x_2)]|0\rangle \, \langle 0|T[\varphi_0(x_3)\,\varphi_0(x_4)]|0\rangle \\
& + \langle 0|T[\varphi_0(x_1)\,\varphi_0(x_3)]|0\rangle \, \langle 0|T[\varphi_0(x_2)\,\varphi_0(x_4)]|0\rangle \\
& + \langle 0|T[\varphi_0(x_1)\,\varphi_0(x_4)]|0\rangle \, \langle 0|T[\varphi_0(x_2)\,\varphi_0(x_3)]|0\rangle
\end{aligned}
$$

So,

$$
\boxed{
\begin{aligned}
\langle 0|T[\varphi_0(x_1)\varphi_0(x_2)\varphi_0(x_3)\varphi_0(x_4)]|0\rangle = {} & \langle 0|T[\varphi_0(x_1)\varphi_0(x_2)]|0\rangle \, \langle 0|T[\varphi_0(x_3)\varphi_0(x_4)]|0\rangle \\
& + \langle 0|T[\varphi_0(x_1)\varphi_0(x_3)]|0\rangle \, \langle 0|T[\varphi_0(x_2)\varphi_0(x_4)]|0\rangle \\
& + \langle 0|T[\varphi_0(x_1)\varphi_0(x_4)]|0\rangle \, \langle 0|T[\varphi_0(x_2)\varphi_0(x_3)]|0\rangle
\end{aligned}
}
$$

<u>Wicks theorem</u> —

In free field theory $\quad \langle 0|T[\varphi_1\varphi_2\varphi_3\cdots\varphi_{2n}]|0\rangle \quad$ ($\nearrow$ even no. of fields)

$$
= \sum \langle 0|T[\varphi_1\varphi_2]|0\rangle \, \langle 0|T[\varphi_3\varphi_4]|0\rangle \cdots \text{ —(2)} \qquad + \text{ all distinct permutations.}
$$

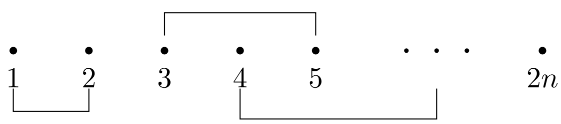
Also,

$$
\langle 0|T\left[\varphi_0(x_1)\,\varphi_0(x_2)\cdots\varphi_0(x_{2n+1})\right]|0\rangle = 0 \qquad \hookrightarrow \text{ odd no. of fields.}
$$

Since one field will always be outside

$$
\sim \ \cancelto{0}{\langle 0| :\varphi_0(x_i): |0\rangle} \left\langle 0 \middle| T\Big[\prod_{j\neq i}^{2n} \varphi_0(x_j)\Big] \middle| 0 \right\rangle \qquad \hookrightarrow \text{ Non zero.}
$$

To understand equation (2) we draw:



We make pairs out of these $2n$ points. Total no. of such combination are —

$$
(2n-1)(2n-3)\cdots 1 = (2n-1)!!
$$

Lets write interacting theory propagator —

$$
\frac{\langle \Omega|\,T[\varphi(x_1)\,\varphi(x_2)]\,|\Omega\rangle}{\langle \Omega|\Omega\rangle} = \frac{\langle 0|\,T\Big[\varphi_0(x_1)\,\varphi_0(x_2)\,e^{-i\int H_I(y)\,dy}\Big]|\Omega\rangle}{\langle 0|T\Big[\exp\left(-i\int H_I(y)\,dy\right)\Big]|0\rangle}
$$

where,

$$H_I(y) = \frac{\lambda}{4!}\,\varphi_0^4(y) \qquad \Bigg| \qquad \begin{aligned} &H_{\text{int}} = \varphi_0^4(0) \\ &H_I = e^{iH_0 y^0}\,\varphi_0^4(0)\,e^{-iH_0 y^0}\end{aligned}$$

So,

$$\frac{\langle\Omega|\,T[\varphi(x_1)\,\varphi(x_2)]\,|\Omega\rangle}{\langle\Omega|\Omega\rangle} = \frac{\langle 0|\,T\Big[\varphi_0(x_1)\varphi_0(x_2)\Big(1-\frac{i\lambda}{4!}\int d^4y\,\varphi_0^4(y)+\cdots\Big)\Big]|0\rangle}{\langle 0|T\Big[1-\frac{i\lambda}{4!}\int d^4y\,\varphi_0^4(y)+\cdots\Big]|0\rangle}$$

$$= \frac{\langle 0|T[\varphi_0(x_1)\,\varphi_0(x_2)]|0\rangle - \frac{i\lambda}{4!}\,\langle 0|T\left[\varphi_0(x_1)\varphi_0(x_2)\int d^4y\,\varphi_0^4(y)\right]|0\rangle+\cdots}{\langle 0|0\rangle - \frac{i\lambda}{4!}\,\langle 0|T\left[\int d^4y\,\varphi_0^4(y)\right]|0\rangle+\cdots} \qquad \hookleftarrow \text{ all 4 fields are}$$

$$= \frac{\langle 0|T[\varphi_0(x_1)\,\varphi_0(x_2)]|0\rangle - i\frac{\lambda}{4!}\int d^4y\,\langle 0|T[\varphi_0(x_1)\varphi_0(x_2)\,\varphi_0^4(y)]|0\rangle + \left(\frac{i\lambda}{4!}\right)^2\frac{1}{2!}\int d^4y\,\langle 0|T[\varphi_0(x_1)\varphi_0(x_2)\,\varphi_0^4(y)\,\varphi_0^4(y')]|0\rangle + \text{higher order}}{1-\frac{i\lambda}{4!}\int d^4y\,\varphi_0^4(y)+\left(\frac{i\lambda}{4!}\right)^2\frac{1}{2!}\int d^4y\,\varphi_0^4(y)\int d^4y'\,\varphi_0^4(y')+\cdots}$$

For upto 1$^{\text{st}}$ order

$$= \frac{D_F(x_1-x_2) - \frac{i\lambda}{4!}\int d^4y\,\langle 0|T[\varphi_0(x_1)\varphi_0(x_2)\,\varphi_0^4(y)]|0\rangle}{\left(1-\frac{i\lambda}{4!}\int d^4y\,\langle 0|T[\varphi_0^4(y)]|0\rangle\right)}$$

$$= \left(D_F(x_1-x_2) - \frac{i\lambda}{4!}\int d^4y\,\langle 0|T[\varphi_0(x_1)\varphi_0(x_2)\varphi_0^4(y)]|0\rangle\right)\left(1+\frac{i\lambda}{4!}\int d^4y\,\langle 0|T[\varphi_0^4(y)]|0\rangle\right) \qquad \left\{\frac{1}{1-x} = 1+x+\right.$$

$$= D_F(x_1-x_2)\left(1+\frac{i\lambda}{4!}\int d^4y\,\langle 0|T[\varphi_0^4(y)]|0\rangle\right) - \frac{i\lambda}{4!}\int d^4y\,\langle 0|T[\varphi_0(x_1)\,\varphi_0(x_2)\,\varphi_0^4(y)]|0\rangle \qquad (\text{dropping } \lambda^2 \text{ term})$$

Since $\langle 0|T[\varphi_0^4(y)]|0\rangle \Rightarrow$ loops $\to \infty$.
Assume all UV divergences are "regularized" through a cut off. The cut off can be following —

— Either the cut off says no two points are allowed to get infinitely closed. So we split the hamiltonian and instead of $\varphi^4(y)$ we use $\varphi(y)\,\varphi(y+\epsilon)\,\varphi(y+2\epsilon)\,\varphi(y+3\epsilon)$.

So, we assume that $\langle 0|T\left[\varphi_0(y)\,\varphi_0(y+\epsilon)\,\varphi_0(y+2\epsilon)\,\varphi_0(y+3\epsilon)\right]|0\rangle$ can be handled later, but first we do the perturbation series in λ. (So it is question of non-commuting limits).

In high energy — short distance and high momentum are same, so short distance can be regularized by splitting the points or high momentum can be regularized by putting a finite cut off.

Naturally, it turns out that it is the only way to proceed, and it leads to sensible answers.

$$\langle\Omega|\,T[\varphi(x_1)\,\varphi(x_2)]\,|\Omega\rangle = D_F(x_1-x_2) + \frac{i\lambda}{4!}\int D_F(x_1-x_2)\,\langle 0|T[\varphi_0^4(y)]|0\rangle \qquad \nearrow\ 3\text{ ways}$$

$$- \frac{i\lambda}{4!}\int d^4y\,\langle 0|T[\varphi_0(x_1)\,\varphi_0(x_2)\,\varphi_0^4(y)]|0\rangle \qquad \left\{\begin{aligned}&3\,D_{x_1x_2}\,D_{yy}\,D_{yy}+\\ &12\,D_{x_1y}\,D_{x_2y}\,D_{yy}\end{aligned}\right.$$

Since we have 6 fields, Total no. of combinations $(2n-1)!! = 5!! = 5\cdot 3\cdot 1 = 15$; ways to select y' for x_1: $4\times 3 = 12$ ways, $\to$ ways to select y for x_2

$$= D_F(x_1-x_2) + \frac{i\lambda}{4!}\int \cancel{D_F(x_1-x_2)\,3\,D_F(y-y)\,D_F(y-y)}$$

$$- \frac{i\lambda}{4!}\int d^4y\;\cancel{3\,D_F(x_1-x_2)\,D_F(y-y)\,D_F(y\cdot y)} \qquad \to \text{ These completely disconnected diagram cancel from de}$$

$$- \frac{i\lambda}{4!}\int d^4y\,(12)\,D_F(x_1-y)\,D_F(x_2-y)\,D_F(y-y)$$

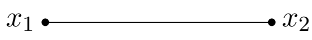
Assuming all expressions are regularized/finite we must cancel terms which are cancelling each other.

$$= D_F(x_1 - x_2) - \frac{i\lambda}{2} \int d^4y \; D_F(x_1 - y) \; D_F(x_2 - y) \; D_F(0)$$

Pictorial Way (Feynman diagrams)

Suppose we want to calculate 2 point interacting theory propagator (b/w x_1 & x_2) with $H_I = \frac{1}{4!}\varphi_0^4(x)$.

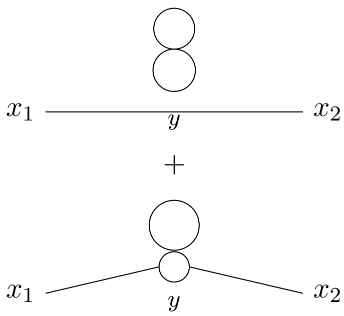
We draw two points x_1 & x_2 in space. (For 0^{th} order)



we join them via line & it is just $D_F(x_1 - x_2)$

For 1^{st} order:

3 completely disconnected diagrams cancels out from denominator $\rightarrow$



$-\frac{i\lambda}{4!}$

$-\frac{i\lambda}{4!}$

Using jump & land rule we have such 3 diagrams; we have such 4×3 diagrams $\{$ particle starts at either x_1 or x_2, produce a virtual particle at y & reach at x_2 or x_1.

Now we add all

$$\langle \Omega | \, T[\varphi(x_1)\varphi(x_2)] \, |\Omega\rangle = D_F(x_1 - x_2) - \frac{i\lambda}{4} \int d^4y \; D_F(x_1 - y) \; D_F(x_2 - y) \; D_F(y - y)$$

Note that denominator cancels out completely disconnected diagrams (vacuum fluctuation) to all orders in λ.

Note that the denominator only contains $\varphi_0^4(y)$ fields which means denominator only tells us the vacuum fluctuations & vacuum fluctuation do not affect particle of concern to us. (So normalizing the theory $\langle\Omega|\Omega\rangle$ gave us the denominator which also cancels all completely disconnected diagrams).

So,

$$D_F(y - y) = D_F(0) = \int \frac{d^4k}{(2\pi)^4} \, \frac{i}{k^2 - m^2} \, e^0 \quad \longrightarrow \text{ divergent} \qquad \left\{ \sim \frac{d^4k}{k^2} \sim \int k^2 \, dk \to \infty \right.$$

We will deal with this divergence later.

All above procedure to find 2-point correlator in φ^4 theory is valid for dirac theory, abelian/non-abelian gauge theory i.e. all quantum field theories. As long as we are doing perturbation theory, the relation b/w free fields and n-point function is true.

Vertex may not be 4 point always/it may be 3 point vertex. Ex. in electrodynamics we have 3 point vertex connecting (e^-, e^+, γ). In scalar φ^4 theory, since we have just one kind of field φ, so we have one kind of particle representing all lines. But when we have multiple fields one type of line will represent just one field/particle, so for feynman rules, we have D_F (for specific particle) $\hookrightarrow$ came from that line.

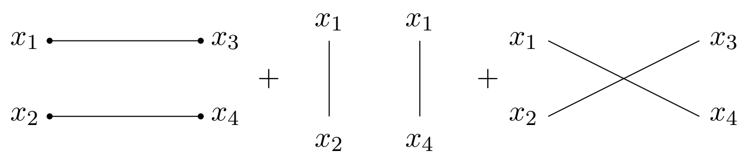
4-point Function

Also we will try to get what kind of physical process we can get.

$\langle\Omega|\, T\left[\varphi_0(x_1)\,\varphi_0(x_2)\,\varphi_0(x_3)\,\varphi_0(x_4)\right]|\Omega\rangle$ shall describe two particles scattering with interaction.

Using Feynman diagrams we can write the 4 point correlator (without writing the formula). We have to fix the labels.

For 0^{th} order:

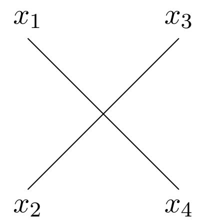


(Free level diagrams) as both particles reaches final state without interaction

Not completely disconnected as Def$^{\text{n}}$ of completely disconnected is that virtual particles should not be connected to any external vertex.

For <u>1^{st} order</u>:

(i)



$-\frac{i\lambda}{4!}$

} here two particles interact at vertex and again was found to be at two another positions.

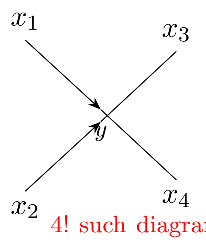
Tree level (No loops)

We can choose any direction of time   This arrow represents dir$^{\text{n}}$ of time

i.e. $\leftarrow x_1^0, x_2^0 > x_3^0\, x_4^0$ or $x_1^0, x_3^0 < x_2^0, x_4^0 \downarrow$ or $\uparrow x_2^0, x_4^0 < x_1^0\, x_3^0$ or $x_1^0\, x_2^0 < x_3^0\, x_4^0 \rightarrow$

Assuming $x_1^0, x_2^0 < x_3^0, x_4^0$:

$\left(-\frac{i\lambda}{4!}\right)$



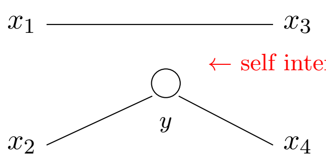
4! such diagrams possible.

$\Rightarrow$ particle at 1,2 interacted at vertex and were found at 3,4 with some probability amplitude given by

$$-\frac{i\lambda}{4!}\int d^4y\; D_F(x_1-y)\; D_F(x_2-y)\; D_F(x_3-y)\; D_F(x_4-y)\times 4!$$

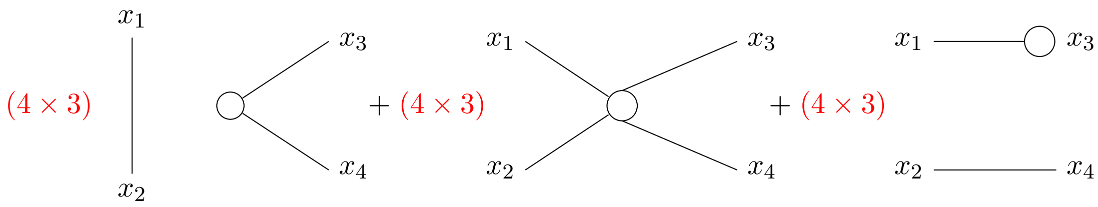
$\left\{\begin{array}{l}\text{If we start at } x_1 \text{ we can land at } y \text{ in} \\ \text{4 ways. then } x_2 \text{ has 3 ways} \\ x_3 \text{ has 2 ways \& } x_4 \text{ has 1 way.} \\ 4\times 3\times 2\times 1 = 4! \end{array}\right.$

(ii) (4×3)



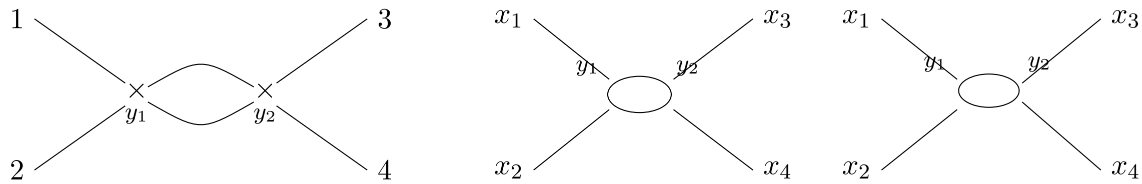
$\leftarrow$ self interaction

1 particle moves freely, another particle undergoes self interaction and reach another point.



In total we have $4 \times 4 \times 3 = 48$ such diagrams.

For <u>2nd order</u> $\big(O(\lambda^2)\big)$:



We can write combinatorial factors along with the diagrams to evaluate propagator in momentum space.

We have not discussed physical interpretation of 4-point function. There is one step to go from 4-point function to $\mathcal{M}$, couple of steps to find scattering matrix which will take us to Crossection.

So using this quantity we can calculate scattering crossection of 2 particles $\longrightarrow$ 2 particles.

We can give physical interpretation to these diagrams, but originally we are expanding interacting fields in language of free fields, (nature does not do that, nature has some interacting fields which interact) its the technique we are using to understand the field theory. If it helps us we should give its physical interpretation. & if physical interpretation is confusing we should not make it. Physical interpretation is something we give to understand better. It is not there to be in theory. There is no obligation that it has to have some physical interpretation.

Physical Interpretation are just ways of thinking about these diagrams. Look around us, there are no interaction vertices in space. But if we calculate something to order (λ^2) which is equal to 1 millionth of measured answer, then we can say that all higher order $O(\lambda^4)$, $O(\lambda^3)$... don't contribute.

We concentrate on lowest order/tree level processes. This is the spirit of taking physical interpretation.

We know that

$$D_F(x-y) = \int \frac{d^4k}{(2\pi)^4} \, \frac{i}{k^2 - m^2} \, e^{-i\vec{k}\cdot(x-y)}$$

$\swarrow$ Position space Feynman propagator

$\{\, x, y, k$ four vectors.

$\downarrow$ Fourier transform of $\dfrac{i}{k^2-m^2}$. So,

$$\widetilde{D}_F(k) = \frac{i}{k^2 - m^2}$$

$\{$is called momentum space Feynman propagator.

Suppose $G(x_1, x_2 \cdots x_{2n}) = \langle \Omega | \, T \left[\varphi(x_1) \, \varphi(x_2) \ldots \varphi(x_{2n}) \right] | \Omega \rangle$: we have a $2n$-point function. Then we can transform it to momentum space as follows.

$$G(x_1 \, x_2 \ldots x_{2n}) = \langle \Omega | \, T \left[\varphi(x_1) \, \varphi(x_2) \cdots \varphi(x_{2n}) \right] | \Omega \rangle$$

$G(x_1 x_2 \ldots x_{2n})$ is translation invariant, i.e. if we shift origin of all fields, we get same answer:

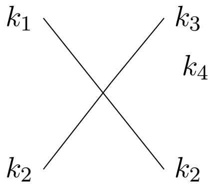
$$G(x_1 + a_1, \, x_2 + a_2 \ldots x_{2n} + a_{2n}) = G(x_1 \, x_2 \ldots x_{2n})$$

$\downarrow$
momentum conservation come from Translational invariance.

We can convert it using fourier transform —

$$\delta^4\left(\sum k_i\right) \times \widetilde{G}\left(k_1\, k_2 \cdots k_{2n}\right) = \int \prod_{i=1}^{2n} \left(d^4 x_i\right)\; e^{ik_i\cdot x_i}\, G\left(x_1\, x_2 \cdots x_{2n}\right)$$

Now it become extremely easy, — (We don't use $x_1\, x_2 \cdots x_{2n}$ anymore. We use momentum lines $k_1\, k_2 \cdots k_{2n}$.)



$$\widetilde{G}(k_1\, k_2\, k_3\, k_4) = \left(\tfrac{-i\lambda}{4!}\right) \delta^4\left(k_1+k_2+k_3+k_4\right)\; \tfrac{i}{k_1^2-m^2}\; \tfrac{i}{k_2^2-m^2}\; \tfrac{i}{k_3^2-m^2}\; \tfrac{i}{k_4^2-m^2} \times 4!$$

There are 4! such ways.

$$\widetilde{G}(k_1 \cdots k_4) = \left(\frac{-i\lambda}{4!}\right) \delta^4\left(k_1+k_2-k_3-k_4\right)\; \frac{i}{k_1^2-m^2}\; \frac{i}{k_2^2-m^2}\; \frac{i}{k_3^2-m^2}\; \frac{i}{k_4^2-m^2}\; \times 4!$$

The above is contribution to first order in λ. Note that, we don't have any integral, it is the final answer to first order contribution.

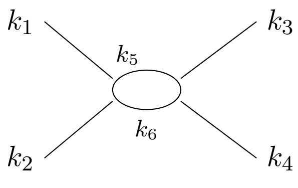
Note that in position space we had integral over 'y' but that integral gave us $\delta^4\left(\sum k_i\right)$ term.

$$\langle\Omega|T[\varphi(x_1)\varphi(x_2)\varphi(x_3)\varphi(x_4)]|\Omega\rangle = \left(\frac{-i\lambda}{4!}\right) 4! \int d^4y\, D_F(x_1-y)\, D_F(x_2-y)\, D_F(x_3-y)\, D_F(x_4-y)$$

(RHS is first order expansion. Not $\mathcal{M}$)

$$\begin{aligned}
&= (-i\lambda) \int d^4y \prod_{i=1}^{4} \frac{d^4k_i}{(2\pi)^4}\; \frac{i}{k_1^2-m^2}\; \frac{i}{k_2^2-m^2}\; \frac{i}{k_3^2-m^2}\; \frac{i}{k_4^2-m^2} \\
&\qquad \times\; e^{ik_1\cdot(x_1-y)}\; e^{ik_2\cdot(x_2-y)}\; e^{ik_3\cdot(x_3-y)}\; e^{ik_4\cdot(x_4-y)} \\
&= (-i\lambda) \int d^4y\; e^{-iy\cdot(k_1+k_2+k_3+k_4)}\; e^{i(k_1x_1+k_2\cdot x_2+k_3\cdot x_3+k_4\cdot x_4)} \\
&\qquad \times\; \frac{i}{k_1^2-m^2}\; \frac{i}{k_2^2-m^2}\; \frac{i}{k_3^2-m^2}\; \frac{i}{k_4^2-m^2}
\end{aligned}$$

Lets move to more difficult example —



At each vertex we have $\delta^4\left(\Sigma k\right)$, So, we have

$$\delta^4\left(k_1+k_2-k_5+k_6\right) \times \delta^4\left(k_5-k_6+k_3+k_4\right)$$

We have to integrate over undetermined momenta. Using 2^{nd} dirac delta we have

$$k_6 = k_5 + k_3 + k_4$$

So 2^{nd} order correction to $\langle\Omega|\, T\left[\varphi(x_1)\,\varphi(x_2)\,\varphi(x_3)\,\varphi(x_4)\right]|\Omega\rangle$ is given by

$$\int d^4k_5\, \frac{i}{k_5^2 - m^2}\, \frac{i}{(k_5 + k_3 + k_4)^2 - m^2}$$

Note that loop diagrams involve one momentum integral per loop. k_5 "do not" satisfy $k_5^2 = m^2$, that is why we have poles in above integral.

External particle satisfy $k_i^2 = m^2$ (on-shell)
Internal particles (loop) "do not" satisfy $k^2 = m^2$ (off-shell)

It means when we try to calculate physical processes in perturbation theory we can express them as if there are some free particles propagating in loops which are virtual, which does not have physical momenta (off shell $k^2 \neq m^2$), and whose only job is to communicate the interaction between incoming and outgoing particles. So they contribute to the process, it is interpretation of diagram. Its not that we are writing diagrams knowing there are virtual particles., we write diagram using Wick's theorem & we interprete the diagram in language of virtual particles.

Lecture 7 — Vector fields

Under lorentz transformation —

$$\begin{aligned} \text{coordinates} - \quad & x^\mu \longrightarrow x'^\mu = \Lambda^\mu_{\ \nu}\, x^\nu \\ \text{scalar field} - \quad & \varphi(x') = \varphi(x) \\ \text{vector field}: \quad & A^\mu(x) \longrightarrow A'^\mu(x') = \Lambda^\mu_{\ \nu}\, A^\nu(x) \end{aligned}$$

Exercise:

$$A'_\mu(x') = \left(\Lambda^{-1}\right)^{T\ \nu}_{\ \mu}\ A_\nu(x)$$

<u>Example:</u> $\partial_\mu \varphi$ (occur in K.G lagrangian) $\hookrightarrow$ transform like vector field.

$$\begin{aligned} \partial_\mu \varphi'(x') &= \left(\Lambda^{-1}\right)^{T\ \nu}_{\ \mu}\ \partial_\nu \varphi(x) \\ \partial^\mu \varphi'(x') &= \Lambda^\mu_{\ \nu}\, \partial^\nu \varphi(x) \qquad \text{with no other fields.} \end{aligned}$$

We want to study independent vector fields (A_μ). The idea is to find EOM of A_μ & write $\mathcal{L}$ reverse but we can guess it as well.

Guessing a lagrangian density —

$$\mathcal{L} = \frac{1}{2}\,\partial_\mu A_\nu\, \partial^\mu A^\nu \qquad \text{(For massless field)} \qquad \begin{cases} \text{just like K.G. lagrangian} \\ \varphi \longrightarrow A_\nu \end{cases}$$

$$\text{e.o.m}: \quad \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \varphi)} \right) = \frac{\partial \mathcal{L}}{\partial \varphi}$$

$$\partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu A_\nu)} \right) = \frac{\partial \mathcal{L}}{\partial A_\nu} \ \Rightarrow\ \partial_\mu \left(\partial^\mu A^\nu\right) = 0$$

It turns out e.o.m is $\partial^\mu \partial_\mu A_\nu = 0$ (one independent K.G. eq$^\text{n}$ for each component).

$$\boxed{\partial^2 A_\nu = 0}$$

This theory as soon as we quantize, have serious problem.

So to quantize we will first write hamiltonian and impose canonical commutation relations.

To write $\mathcal{H}$, we require $\mathcal{L} = \frac{1}{2}\dot{A}_\mu \dot{A}^\mu + \cdots$

$$\pi^\mu = \frac{\partial \mathcal{L}}{\partial \dot{A}_\mu} = \dot{A}^\mu \qquad \left| \qquad \begin{aligned} \dot{A}^\mu &= \eta^{\mu\nu}\dot{A}_\nu \\ \mathcal{L} &= \frac{1}{2}\dot{A}_\mu\, \eta^{\mu\nu}\, \dot{A}_\nu \\ \pi^\mu = \frac{\partial \mathcal{L}}{\partial \dot{A}_\mu} &= \frac{1}{2}\eta^{\mu\nu}\dot{A}_\nu + \frac{1}{2}\dot{A}_\nu\, \eta^{\nu\mu} \qquad \uparrow (\mu \leftrightarrow \nu) \\ &= \dot{A}^\mu \end{aligned} \right.$$

Commutator

$$\left[A_\mu(t,\vec{x}),\ \dot{A}_\nu(t,\vec{x}')\right] = i\,\eta_{\mu\nu}\,\delta^3(x - x')$$

We will write $A_\mu(t,\vec{x})$ in expansion of oscillators — we get —

$$\left[a_{\mu,\vec{k}}\,,\ a^\dagger_{\nu,\vec{k}'}\right] = (2\pi)^3\,\eta_{\mu\nu}\,\delta^3(\vec{k} - \vec{k}')$$

Note that $\eta_{\mu\nu} = \text{diag}(1,-1,-1,-1)$; for $\mu,\nu \neq 0$: $\eta_{\mu\nu} = 0$ or -1. We get

$$\begin{aligned}\left[a_{i,\vec{k}}\,,\ a^\dagger_{j,\vec{k}'}\right] &= (2\pi)^3\,(-1)\,\delta^3(\vec{k}-\vec{k}') \\ &= -\text{ve}\,!\end{aligned}$$

This is not good as it leads to

$$a^\dagger_{i,\vec{k}}\,|0\rangle \qquad (i = 1,2,3) \ —①$$

$\uparrow$ should create 3 possible types of particle depending upon value of i, with momentum $\vec{k}$.

Let us find norm of this state — —③

$$\text{Norm} \to \langle 0|\, a_{i,k}\, a^\dagger_{i,\vec{k}}\,|0\rangle \qquad \text{using} \begin{cases} & \left[a_{i,k},\, a^\dagger_{j,\vec{k}'}\right] = (2\pi)^3\,\delta^3(k-k')\,\eta_{ij} \\ \text{for } i=j: & \left[a_{i,\vec{k}},\, a^\dagger_{i,k'}\right] = (2\pi)^3(-1)\,\delta^3(k-k') \end{cases}$$

$$= \langle 0|\, a^\dagger_{i,\vec{k}}\, a_{i,\vec{k}}\,|0\rangle + \langle 0|\left[a_{i,k},\, a^\dagger_{i,\vec{k}}\right]|0\rangle$$

$$\text{Norm} = 0 + (-\text{ve})$$

$$\boxed{\text{Norm} < 0}$$

Physically this is not acceptable!

As it means that the theory have negative probabilities and it will be inconsistent.

So we write a slightly better lagrangian

$$\mathcal{L} = -\frac{1}{2}\,\partial_\mu A_\nu\,\partial^\mu A^\nu$$

$$\text{E.O.M.} \longrightarrow \partial^2 A_\nu = 0 \qquad \text{(same)}$$

$$\text{quantisation} \Rightarrow \pi^\mu = \frac{\partial \mathcal{L}}{\partial \dot{A}_\mu} = -\dot{A}^\mu$$

$$\Rightarrow \quad \left[A_\mu(t,\vec{x}),\, A_\nu(t,\vec{x}')\right] = -i\,\eta_{\mu\nu}\,\delta^3(\vec{x}-\vec{x}')$$

$$\Rightarrow \quad \left[a_{\mu,\vec{k}}\,,\ a^\dagger_{\nu,\vec{k}'}\right] = -(2\pi)^3\,\eta_{\mu\nu}\,\delta^3(\vec{k}-\vec{k}')$$

This solves $\left[a_{i,\vec{k}},\, a^\dagger_{j,k}\right] = +\text{ve}$. But now the problem is transferred to

$$\left[a_{0,\vec{k}}\,,\ a^\dagger_{0,k'}\right] = (2\pi)^3\,(-1)\,\delta(\vec{k}\,\vec{k}') = -\text{ve}\,!$$

$$\Rightarrow \quad \text{Norm of } a^\dagger_{0,\vec{k}}|0\rangle \ \text{ becomes } -\text{ve!}$$

We still have inconsistency!

We have a particle (photon) which is described by vector field $A_\mu(\vec{x},t)$. Maxwell's E.O.M for $A_\mu = (\phi, \vec{A})$ are

$$\partial^\mu F_{\mu\nu} = 0 \ \Rightarrow\ \partial^\mu\left(\partial_\mu A_\nu - \partial_\nu A_\mu\right) = 0 \qquad \begin{array}{l}\text{(set of 4 eq}^{\text{n}}\text{)} \\ \text{to determine } A_\mu \text{ (4 components)}\end{array}$$

These E.O.M. came from lagrangian

$$\begin{aligned}\mathcal{L} &= -\frac{1}{4}\,F_{\mu\nu}\,F^{\mu\nu} \\ &= -\frac{1}{4}\left(\partial_\mu A_\nu - \partial_\nu A_\mu\right)\left(\partial^\mu A^\nu - \partial^\nu A^\mu\right)\end{aligned}$$

$$= -\frac{1}{4}\Big(\underline{\partial_\mu A_\nu\, \partial^\mu A^\nu} - \overline{\partial_\mu A_\nu\, \partial^\nu A^\mu} - \overline{\partial_\nu A_\mu\, \partial^\mu A^\nu} + \underline{\partial_\nu A_\mu\, \partial^\nu A^\mu}\Big)$$
$$= -\frac{1}{4}\left(2\,\partial_\mu A_\nu\, \partial^\mu A^\nu - 2\,\partial_\mu A_\nu\, \partial^\nu A^\mu\right)$$
$$= -\frac{1}{2}\ \partial_\mu A_\nu\, \partial^\mu A^\nu + \frac{1}{2}\boxed{\partial_\mu A_\nu\, \partial^\nu A^\mu} \qquad \{ \text{ These terms are different } \checkmark$$

This closely resembles to our guessed lagrangian but it has some extra term:

$$\text{our new guess}\Big] \qquad \mathcal{L} = -\frac{1}{2}\,\partial_\mu A_\nu\, \partial^\mu A^\nu + \boxed{\frac{1}{2}\left(\partial^\mu A_\mu\right)^2} \qquad \underline{\text{How?}}$$

Let us try quantisation with this $\mathcal{L}$:

$$\pi^\mu = \frac{\partial \mathcal{L}}{\partial \dot{A}_\mu} = -F^{0\mu} = -\left(\dot{A}^\mu - \partial^\mu A^0\right)$$
$$\Rightarrow \quad \pi^0 = -\left(\dot{A}^0 - \dot{A}^0\right) = 0 \quad \text{— ①}$$
$$\Rightarrow \quad A_0(t,x) \ \text{does not have conjugate momentum.}$$

also, $A_0(t,\vec{x})$ is <u>not</u> a *dynamical* field.

Gauge Invariance

$$\begin{aligned}
A'_\mu(x) &= A_\mu(x) + \partial_\mu \lambda(x)\\
F'_{\mu\nu} &= \partial_\mu A'_\nu - \partial_\nu A'_\mu\\
&= \partial_\mu\left(A_\nu + \partial_\nu\lambda\right) - \partial_\nu\left(A_\mu + \partial_\mu\lambda\right)\\
&= \partial_\mu A_\nu + \cancel{\partial_\mu\partial_\nu\lambda} - \partial_\nu A_\mu - \cancel{\partial_\nu\partial_\mu\lambda}\\
&= \partial_\mu A_\nu - \partial_\nu A_\mu\\
F'_{\mu\nu} &= F_{\mu\nu} \qquad \Rightarrow \ \mathcal{L} \text{ is invariant under gauge transformation.}
\end{aligned}$$

Configuration A_μ and $A_\mu + \partial_\mu\lambda$ has same lagrangian, therefore A_μ inherit same dynamics.

where $\lambda =$ any arbitrary fun. of x.

It is better to call it gauge invariance rather than gauge symmetry.

This is redundancy of description: $A_\mu(x)$ and $A_\mu + \partial_\mu\lambda$ are same physical configuration.

We could have choosen $\mathcal{L} \propto F^2$ where $F_{\mu\nu}$ is basic field rather than A_μ, then

$$\text{EOM} \ \Rightarrow \ \partial_\mu\left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu F)}\right) = \frac{\partial\mathcal{L}}{\partial F} \ \Rightarrow \ 0 = F_{\mu\nu}$$
$$\Rightarrow \ F_{\mu\nu} = 0 \ \Rightarrow \ \vec{E} = 0 = \vec{B}$$

Which is not consistent with maxwells eq$^{\text{n}}$. So we have to write/guess right lagrangian in order to get correct E.O.M.

<u>Note</u>: Gauge invariance is only way to have vector fields and get rid of negative norm states.

In eq$^{\text{n}}$ ①: $\pi^0 = 0$ arise from gauge invariance.

— We will see that gauge invariance implies absence of negative norm states.

<u>Note 1</u>:

— Even considering tensor fields rather than vector fields results into occurence of negative norm states. The problem of $-$ve norm comes because time & space have different signatures that itself comes from relativity.

$\Rightarrow$ Any theory involving vector or tensor fields must have gauge invariance in order to be consistent. (No consistent quantum theory without gauge invariance, for such fields.)

Note 2: putting mass term $-\frac{1}{2}m^2 A_\mu A^\mu$ — but this term is not gauge invariant.

"It suggest that vector fields should be massless." (Any particle which is described by a vector field has to be massless)

$\left\{\text{massless ness of photon is due to gauge invariance.}\right\}$ except gluons

Unfortunately we know 11 more vector particles, $W^\pm, Z$ and 8 gluons and none of those is massless. That is due to 2 mechanism which are higgs mechanism and confinement.

$$\begin{cases} \text{Higgs mechanism — } W^\pm, Z \quad \text{weak interaction} \\ \quad\text{Confinement — gluons} \quad \text{strong "} \end{cases}$$

This will be discussed in end of course.

Note 3: We can perform multiple gauge transformations in different order —

$$\begin{aligned} A_\mu &\longrightarrow A_\mu + \partial_\mu \lambda_1(x) \longrightarrow (A_\mu + \partial_\mu \lambda_2(x)) + \partial_\mu \lambda_1(x) \\ \text{(or)} \quad A_\mu &\longrightarrow A_\mu + \partial_\mu \lambda_2(x) \longrightarrow (A_\mu + \partial_\mu \lambda_1(x)) + \partial_\mu \lambda_2(x) \end{aligned}$$

We are getting same answer in RHS.

$\boxed{\text{This is known as abelian gauge invariance.}}$

So maxwell lagrangian has abelian gauge invariance.

So we will study/consider

$$\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} \qquad \{ \text{ where } A_\mu(x) \text{ is field variable rather than } F^{\mu\nu}$$

We will quantize it, but we know $\pi^0 = 0$: we can't quantize.

In order to quantize we must fix the gauge. Here fixing the gauge means, given any configuration of $A_\mu(x)$ we apply a gauge transformation & bring it to some standard form, after that we are not able to make more gauge transformation.

It will enable us to work with genuine physical degrees of freedom. It will also exhibit for me which d.o.f. can be taken away by gauge transformations, & which remains. ✓

There are many ways to fix the gauge, also we have to make a choice wheather we want to respect manifest lorentz invariance or not, both of these have advantages. Now we will see two ways of fixing the gauge:

1) violates manifest lorentz Invariance (Coulomb gauge)

2) Preserves manifest lorentz Invariance (Lorentz gauge)

Finally, all physical results will be lorentz invariant, (Coulomb gauge violates during intermediate steps).

(1) Coulomb gauge:
Lets look at gauge transformation seperately for A_0 and A_i:

$$\begin{aligned} A_0 &\longrightarrow A_0 + \partial_0 \lambda \qquad \text{— (a)} \\ A_i &\longrightarrow A_i + \partial_i \lambda \qquad \text{— (b)} \end{aligned}$$

We can make $A_0 = 0$ (to get rid of $\pi^0 = 0$), i.e. choose $\lambda(x)$ such that $\partial_0 \lambda = -A_0(t, \vec{x})$

$$\lambda(t, \vec{x}) = -\int^t dt' \, A_0(t', \vec{x}) \qquad \uparrow \text{ put back in eq (b) to transform } A_i$$

Using this $\lambda(t, \vec{x})$, we "gauge away" A_0.

But still we can make more gauge transformations which will bring back non zero A_0 (time dependent $\lambda(x,t)$). But we only allow for time independent $\lambda(\vec{x})$ for further work. If we consider time independent $\lambda(\vec{x})$ we still have more gauge transformations:

$$A_i \longrightarrow A_i + \partial_i \lambda(\vec{x})$$

$$\text{E.O.M} \Rightarrow \partial^\mu F_{\mu\nu} = 0 \qquad \text{where } (A_0 = 0)$$

We have two sets of eq$^{\text{n}}$: 1) $\nu = 0$; 2) $\nu = j$.

$$1) \; \nu = 0 \Rightarrow \partial^i F_{i0} = 0 \qquad \left(F_{i0} = \partial_i \cancelto{0}{A_0} - \partial_0 A_i \right)$$

$$\begin{aligned} -\partial^i \left(\partial_0 A_i \right) &= 0 \\ \boxed{\partial_0 \left(\partial^i A_i \right) = 0} & \text{ — (2)} \\ \Rightarrow \quad \partial^i A_i & \text{ is time independent} \end{aligned}$$

(We first got rid of A^0, by doing gauge transformation; then we further do time independent gauge transformation to such that $\partial^i A_i$ is time independent.)

$$2) \; \nu = j : \qquad \text{E.O.M} \Rightarrow \partial^\mu F_{\mu\nu} = 0$$

$$\partial^0 F_{0j} + \partial^i F_{ij} = 0$$

$$\Rightarrow \quad \partial^0 \left(\partial_0 A_j - \partial_j \cancelto{0}{A_0} \right) + \partial^i \left(\partial_i A_j - \partial_j A_i \right) = 0$$

$$\Rightarrow \quad \boxed{\partial^0 \partial_0 \, A_j + \partial^i \partial_i \, A_j - \partial_j \left(\partial^i A_i \right) = 0} \text{ — (3)}$$

If $\partial^i A_i$ is set to zero, we get

$$\partial^0 \partial_0 \, A_j + \partial^i \partial_i \, A_j = 0$$

$$\left(\partial^0 \partial_0 + \partial^i \partial_i \right) A_j = \partial^\mu \partial_\mu \, A_j = 0 \text{ — 2}$$

which is K.G. equation for 3 components of $\vec{A}$.

$$\text{i.e.} \quad \begin{cases} \partial^2 A_1 = 0 \\ \partial^2 A_2 = 0 \\ \partial^2 A_3 = 0 \end{cases} \quad \text{— 3}$$

This suggest we should set $\partial^i A_i = 0$ by a gauge transformation.

Under gauge transformation:

$$\begin{gathered}
A_i \longrightarrow A_i + \partial_i \lambda \\
\partial^i A_i \longrightarrow \partial^i A_i + \partial^i \partial_i \lambda \\
\text{For} \quad \partial^i A_i = 0 = \partial^i A_i + \partial^i \partial_i \lambda \\
\Rightarrow \quad \nabla \cdot A + \nabla^2 \lambda = 0 \\
\nabla^2 \lambda = -\nabla \cdot A \qquad (\text{Poisson's eq}^{\text{n}})
\end{gathered}$$

$\Big($Solving for λ from above eq$^{\text{n}}$ we set $\partial^i A_i = 0\Big)$, and $\partial^i A_i$ remains zero as $\partial_0 \left(\partial^i A_i\right) = 0$.

Final equations of motion becomes — (eq$^{\text{n}}$ (3) becomes) —

$$\Box A_j = 0 \quad (\text{or}) \quad \partial^2 A_j = 0 \quad (\text{or}) \quad \partial^i \partial_i A_j = 0, \qquad \text{subject to cond}^{\text{n}} \quad \partial^j A_j = 0.$$

We have already set $A_0 = 0$ (to get rid of $\pi^0 = 0$), now one more constraint $\partial \cdot A = 0$ ($\partial^j A_j = 0$) implies there are only two independent oscillators.

That means vector particle has two independent polarisations. (A_0 was non physical we gauged it away very first, then we are left with three with one constraint. So we are left with 2 independent d.o.f of A_μ.)

— Note that there is an appearent problem b/w lorentz invariance & two polarisations. Four vector has to have 4 components. Gauge invariance removed two degrees of freedom.

Now we will see how these two d.o.f. are removed in a different gauge. Before that we would like to calculate Angular momentum of operators J_i ($J_1\, J_2\, J_3$) by using Noether's theorem and rotational invariance of system.

When we quantize the system we get vacuum state, 1 particle, 2 particle ... n particle states.

Let's act with J^2 on one particle states. One particle states can not have orbital angular momentum. (There is nothing to orbit around, so it is purely the spin total angular momentum.) If we find J^2:

$$J^2 \left|1\right\rangle = j(j+1)\, \hbar^2 \left|1\right\rangle$$

Spin of state $|1\rangle$ is 1: Vector field describe spin 1 particle.

Lorentz gauge —
In this gauge we don't distinguish $A_0\, A_1\, A_2\, A_3$; we make A_0 dynamical and carried along and get rid of at the end.

Starting from

$$\mathcal{L} = -\frac{1}{2}\, \partial_\mu A_\nu\, \partial^\mu A_\nu + \frac{1}{2} \left(\partial^\mu A_\mu\right)^2 \qquad \hookrightarrow \text{ this term created gauge invariance!}$$

As a choice of gauge we choose (lorentz gauge)

$$\partial^\mu A_\mu = 0 \qquad \big(\text{setting 4 divergences to 0 rather than 3 divergences } \partial^i A_i = 0 \text{ (Coulomb gauge)}\big)$$

Now;

$$\mathcal{L} = -\frac{1}{2}\, \partial_\mu A_\nu\, \partial^\mu A_\nu$$

$$\pi^0 = -\dot{A}^0 \qquad \Big(\text{But we saw in start of lecture that this leads to negative Norm of } a^\dagger_{0,k}|0\rangle\Big)$$

Given;

$$A_\mu \longrightarrow A_\mu + \partial_\mu \lambda$$
$$\partial^\mu A_\mu \longrightarrow \partial^\mu A_\mu + \partial^2 \lambda$$
$$\text{if } \partial^2\lambda = 0 \ \text{ then } \partial^\mu A_\mu \text{ is fixed.}$$

If $\partial^\mu A_\mu = 0$, then All such gauge transformations $A_\mu \to A_\mu + \partial_\mu\lambda(x)$ fixes $\partial^\mu A_\mu = 0$ iff $\partial^2\lambda = 0$.

If $\lambda(x)$ satisfies massless K.G eq$^{\text{n}}$ then it preserves $\partial^\mu A_\mu = 0$.

Analysis in momentum space —
Instead of $A_\mu(x)$ we have $\widetilde{A}_\mu(k)$ or $A_\mu(k)$. Gauge field: $\widetilde{A}_\mu(k)$. Lorentz Gauge condition in k-space:

$$k^\mu \widetilde{A}_\mu(k) = 0 \ \text{—} (1)$$

Residual gauge freedom (Remaining gauge transf.):

$$\widetilde{A}_\mu(k) \longrightarrow \widetilde{A}_\mu(k) + k_\mu \lambda(k)$$

$\partial^2\lambda = 0$ in k space becomes

$$k^\mu k_\mu = k^2 = 0 \ \text{—} (2) \qquad \text{i.e. only } \lambda(k) \text{ are allowed where } k^2 = 0.$$

These two conditions remove two out of 4 polarisations. (Removing A^0, A^3 & left with A^1 & A^2 (i.e. 2 transverse polarisations)).

1 particle state $\quad a^\dagger_{\mu,k}|0\rangle$:

$$(1) \text{ set } \ k^\mu a^\dagger_{\mu,k}|0\rangle \sim 0 \qquad \left(k^\mu a^\dagger_{\mu,k}|0\rangle \text{ should be identified with zero.}\right)$$

$\downarrow$ To remove component along k^μ.

$$\left\{\begin{array}{l}\text{Analogy is to have component of } \vec{V} = x\hat{i} + y\hat{j} + z\hat{k} \\ \text{on } xy \text{ plane; we simply do } \vec{V}\cdot\hat{k} = 0 \ \text{ or } \ \text{set } z = 0 \\ \text{to have } \vec{v}\,' = x\hat{i} + y\hat{j}\end{array}\right.$$

ex: We need $k^2 = 0$ (to maintain $k^\mu \cdot \widetilde{A}_\mu(k) = 0$).
So, choose k such that $k^2 = 0$ (light like k^μ)

$$\Rightarrow \quad \begin{aligned} k_\mu &= k(1,0,0,1) \\ k^\mu &= k(1,0,0,-1) \end{aligned}$$

$$k^\mu a_{\mu,k}|0\rangle = k\, a^\dagger_{0,k}|0\rangle - k\, a^\dagger_{3,k}|0\rangle = 0$$
$$\Rightarrow \quad \underline{a^\dagger_{0,k}|0\rangle \sim a^\dagger_{3,k}|0\rangle}$$

Physical states —
Taking arbitrary linear combination

$$\xi^\mu a^\dagger_{\mu,\vec{k}}|0\rangle$$

Lecture 8 — Scalar QED

Scalar QED: Complex scalar field coupled to a vector field.

$$\mathcal{L} = \partial_\mu \varphi^* \, \partial^\mu \varphi - m^2 \varphi^* \varphi$$
$$\varphi = \varphi_1 + i\varphi_2 \,, \qquad \varphi^* = \varphi_1 - i\varphi_2$$

$$\begin{aligned}
\mathcal{L} &= \partial_\mu \left(\varphi_1 - i\varphi_2\right) \partial^\mu \left(\varphi_1 + i\varphi_2\right) - m^2 \left(\varphi_1^2 + \varphi_2^2\right) \\
&= \partial_\mu \varphi_1 \, \partial^\mu \varphi_1 - \cancel{i \, \partial_\mu \varphi_2 \, \partial^\mu \varphi_1} + \cancel{i \, \partial_\mu \varphi_1 \, \partial^\mu \varphi_2} + \partial_\mu \varphi_2 \, \partial^\mu \varphi_2 - m^2 \left(\varphi_1^2 + \varphi_2^2\right) \\
\mathcal{L} &= \partial_\mu \varphi_1 \, \partial^\mu \varphi_1 + \partial_\mu \varphi_2 \, \partial^\mu \varphi_2 - m^2 \left(\varphi_1^2 + \varphi_2^2\right)
\end{aligned}$$

which is $\mathcal{L}$ for two real scalar fields.

Interaction term is $\lambda \dfrac{(\varphi^* \varphi)^2}{6}$ $\hookrightarrow$ will be clear later. So that;

$$\mathcal{L} = \partial_\mu \varphi^* \, \partial^\mu \varphi - m^2 \, \varphi^* \varphi - \lambda \frac{(\varphi^* \varphi)^2}{6}$$

EOM:

$$\partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \varphi^*)} \right) = \frac{\partial \mathcal{L}}{\partial \varphi^*} \;\Rightarrow\; \partial_\mu \partial^\mu \varphi = -m^2 - \frac{\lambda}{3} \left(\varphi^* \varphi\right) \varphi$$

i.e.

$$\text{complex conjugate} \, (\curvearrowright) \quad \left.\begin{aligned}
\partial^2 \varphi + m^2 + \frac{\lambda \, \varphi^2 \varphi^*}{3} &= 0 \\
\partial^2 \varphi^* + m^2 + \frac{\lambda \, \varphi^{*2} \varphi}{3} &= 0
\end{aligned}\right\} \begin{array}{l} \text{2 set of EOM.} \\ \text{For } \underline{\varphi \,\&\, \varphi^*}. \end{array}$$

Our $\mathcal{L}$ is invariant under Global symmetry. Initially even this global symmetry (which presently is known as phase symmetry) was called as gauge transformation. i.e. we rescale field φ:

$$\varphi(x) \mapsto e^{\alpha} \, \varphi(x)$$
$$\varphi^*(x) \mapsto e^{\alpha} \, \varphi^*(x)$$

but action/$\mathcal{L}$ is not invariant so we are forced to have phase transformations

$$\varphi(x) \longrightarrow e^{i\alpha} \, \varphi(x)$$
$$\varphi^*(x) \longrightarrow e^{-i\alpha} \, \varphi(x)$$

to have invariant $\mathcal{L}$ under this phase symmetry. (or) Global gauge transformation ($\alpha = \text{const.}$).

Under infinitesimal transformation ($\alpha \ll 1$):

$$\begin{aligned}
\varphi(x) &\longrightarrow (1 + i\alpha) \, \varphi(x) &\quad&\Rightarrow \quad \delta\varphi(x) = i\alpha \, \varphi(x) \\
\varphi^*(x) &\longrightarrow (1 - i\alpha) \, \varphi^*(x) &\quad&\Rightarrow \quad \delta\varphi^*(x) = -i\alpha \, \varphi(x)
\end{aligned}$$

Conserved current of this symmetry is —

$$J_\mu = i \left(\varphi^* \, \partial_\mu \varphi - \varphi \, \partial_\mu \varphi^*\right)$$

↓ Called as current as it is a 4 vector made from fields.

To see if J_μ is conserved, we find:

$$J_\mu \left(\varphi, \varphi^*\right) = J_\mu \left(\varphi', \varphi'^*\right)$$

<u>RHS</u>:

$$
\begin{aligned}
J_\mu\left(\varphi', \varphi'^*\right) &= i\left(\varphi'^*\,\partial_\mu\varphi' - \varphi'\,\partial_\mu\varphi'^*\right) \\
&= i\left(e^{-i\alpha}\varphi^*\,\partial_\mu\left(e^{i\alpha}\varphi\right) - \left(e^{i\alpha}\varphi\right)\partial_\mu\left(e^{-i\alpha}\varphi^*\right)\right) \\
&= i\left(\varphi^*\,\partial_\mu\varphi - \varphi\,\partial_\mu\varphi^*\right) \\
&= J_\mu\left(\varphi, \varphi^*\right).
\end{aligned}
$$

(or)

$$
\partial^\mu J_\mu = 0 \qquad \text{(using EOM)}.
$$

$$
\begin{aligned}
\partial^\mu J_\mu &= \partial^\mu(i)\left(\varphi^*\partial_\mu\varphi - \varphi\,\partial_\mu\varphi^*\right) \\
&= i\left(\cancel{\partial^\mu\varphi^*\,\partial_\mu\varphi} + \varphi^*\,\partial^2\varphi - \cancel{\partial^\mu\varphi\,\partial_\mu\varphi^*} - \varphi\,\partial^2\varphi^*\right) \\
&= i\left(\varphi^*\partial^2\varphi - \varphi\,\partial^2\varphi^*\right) \\
&= i\left(-\varphi^*\left(m^2 + \frac{\lambda}{3}\varphi^*\varphi^2\right) + \varphi\left(m^2 + \frac{\lambda}{3}\varphi\left(\varphi^*\right)^2\right)\right) \\
&= i\left(-m^2\varphi^* - \cancel{\frac{\lambda}{3}\varphi^2\left(\varphi^*\right)^2} + m^2\varphi + \cancel{\frac{\lambda}{3}\varphi\varphi^{*2}}\right) \\
&= i\,m^2\left(\varphi - \varphi^*\right) \\
&= 0\,! \qquad \text{(Complete it)}.
\end{aligned}
$$

$$
Q = \int d^3x\; J_\mu = i\int d^3x\left(\varphi^*\,\partial_\mu\varphi - \varphi\,\partial_\mu\varphi^*\right)
$$

Symmetry is generated by $e^{-i\alpha Q}$:

$$
\begin{aligned}
\varphi(x) &\longrightarrow e^{-i\alpha Q}\,\varphi(x)\,e^{i\alpha Q} \\
\varphi^*(x) &\longrightarrow e^{-i\alpha Q}\,\varphi^*(x)\,e^{i\alpha Q} \\
\delta\varphi &= +i\alpha\left[Q, \varphi\right] \\
\delta\varphi^* &= -i\alpha\left[Q, \varphi^*\right]
\end{aligned}
$$

$$
\begin{aligned}
\left[\varphi(t,\vec{x})\,,\ \dot{\varphi}^*(t,\vec{y})\right] &= i\,\delta^3(\vec{x}-\vec{y}) \\
\left[\varphi^*(t,\vec{x})\,,\ \dot{\varphi}(t,\vec{y})\right] &= i\,\delta^3(\vec{x}-\vec{y})
\end{aligned}
$$

$$
\text{Symmetry} \Rightarrow \text{current} \Rightarrow \text{Charge} \curvearrowleft
$$

If we apply translational invariance; this leads to current $T_{\mu\nu}$ (energy–momentum tensor) that leads to conserved charge (4-momentum P_μ); & four momentum generates back the translational invariance.

Suppose $\alpha = \alpha(x)$ (local gauge invariance):

$$
\begin{aligned}
\partial_\mu\varphi \longrightarrow\; & \partial_\mu\left(e^{i\alpha(x)}\,\varphi\right) \\
&= e^{i\alpha(x)}\,\partial_\mu\varphi + i\,\partial_\mu\alpha(x)\,e^{i\alpha(x)}\,\varphi \\
&= e^{i\alpha(x)}\left(\partial_\mu\varphi + i\,\partial_\mu\alpha(x)\;\varphi(x)\right)
\end{aligned}
$$

$$
\alpha = \alpha(t,\vec{x}) = \alpha(x)\,, \qquad \varphi = \varphi(t,\vec{x}) = \varphi(x)
$$

So under,

$$
\varphi \longrightarrow e^{i\alpha(x)}\,\varphi(x)
$$

$$\varphi^* \longrightarrow e^{-i\alpha(x)}\,\varphi^*(x)$$

the (kinetic term) $\mathcal{L}_T = \partial_\mu\varphi\,\partial^\mu\varphi^*$ is not invariant!
But; $\mathcal{L}_m = m\,\varphi^*\varphi$ is invariant!

We then have to make $\mathcal{L}$ which is invariant under local gauge transformation. We look for term

$$\mathcal{L}_{\text{kinetic}} = D_\mu\varphi\; D^{*\mu}\varphi^*$$

So that

$$D_\mu\varphi \longrightarrow e^{i\alpha(x)}\,D_\mu\varphi$$
$$D_\mu\varphi^* \longrightarrow e^{-i\alpha(x)}\,D_\mu\varphi^*$$

We look for D_μ of type —

$$D_\mu = \partial_\mu - ie\,A_\mu$$

$$\begin{aligned}
D_\mu\varphi \longrightarrow D'_\mu\varphi' &= e^{i\alpha(x)}\,D_\mu\varphi \\
D'_\mu\,e^{i\alpha(x)}\varphi &= e^{i\alpha(x)}\,D_\mu\varphi \\
D'_\mu &= e^{i\alpha(x)}\,D_\mu\,e^{-i\alpha(x)} \\
\partial_\mu - ie\,A'_\mu &= e^{i\alpha(x)}\,(\partial_\mu - ieA_\mu)\,e^{-i\alpha(x)} \\
\cancel{\partial_\mu} - ie\,A'_\mu &= \cancel{\partial_\mu} - i\,\partial_\mu\alpha(x) - ie\,A_\mu
\end{aligned}$$

$$A'_\mu = \frac{1}{e}\,\partial_\mu\alpha(x) + A_\mu$$

$$\delta A_\mu = \frac{1}{e}\,\partial_\mu\alpha(x)$$

i.e. If we replace $\partial_\mu \to D_\mu$ & introduce a vector field A_μ which transform like

$$A'_\mu = A_\mu + \frac{1}{e}\,\partial_\mu\alpha(x)$$

then $\mathcal{L}_{\text{kinetic}} = D_\mu\varphi\; D^{*\mu}\varphi^*$ is invariant under local gauge transformations.

So, Invariant $\mathcal{L}$ becomes; ✓ to make real $\mathcal{L}$.

$$\mathcal{L} = D_\mu\varphi\; D^{*\mu}\varphi^* - m\,\varphi\varphi^* - \frac{\lambda}{6}\,(\varphi\varphi^*)^2$$

We also know from (Lecture-7) that $-\frac{1}{4}F_{\mu\nu}F^{\mu\nu}$ is invariant under local gauge transformations. This term act as propagation term for A_μ. So (scalar QED $\mathcal{L}$)

$$\boxed{\mathcal{L} = D_\mu\varphi\; D^{*\mu}\varphi^* - m^2\,\varphi^*\varphi - \frac{\lambda}{6}\,(\varphi^*\varphi)^2 - \frac{1}{4}\,F_{\mu\nu}\,F^{\mu\nu}}$$

The term $-\frac{1}{4}F_{\mu\nu}F^{\mu\nu}$ contains derivative of A which allows A to propagate to have canonical momenta. Without the term $-\frac{1}{4}F_{\mu\nu}F^{\mu\nu}$:

$$\pi_A = \frac{\partial\mathcal{L}}{\partial\dot{A}(x)} = 0$$

So we needed lorentz invariant; gauge invariant term which has non zero π_A; which was $-\frac{1}{4}F_{\mu\nu}F^{\mu\nu}$.

Feynman Rules for scalar QED

1) First step is to look for propagators; seperate $\mathcal{L}$ into

$$\mathcal{L} = \mathcal{L}_{\text{free}} + \mathcal{L}_{\text{int}}$$

To identify $\mathcal{L}_{\text{free}}$; without λ term:

a) EOM are linear $\Rightarrow \mathcal{L}_{\text{free}}$ shall be quadratic. So, all 2 field terms are free terms & everything else is interacting term.

$$\mathcal{L} = + \underset{\hookrightarrow \text{free term.}}{\varphi_1\varphi_2} + \cdots$$

So,

$$\mathcal{L}_{\text{free}}\Big)_{\text{scalar QED}} = \underbrace{\partial_\mu\varphi\,\partial^\mu\varphi^* - m^2\varphi^*\varphi}_{\text{Free K.G. theory}} - \underbrace{\frac{1}{4}F_{\mu\nu}F^{\mu\nu}}_{\text{free gauge theory}}$$

Everything which is left over is $\mathcal{L}_{\text{int}}$:

$$\mathcal{L}_{\text{int}} = ie\, A^\mu\left(\varphi^*\partial_\mu\varphi - \varphi\,\partial_\mu\varphi^*\right) + e^2 A_\mu A^\mu\,\varphi^*\varphi - \frac{\lambda}{6}(\varphi^*\varphi)^2 \qquad \{\ (A^\mu) \text{ vector field couples to conserved current (}$$

$$= ie\, A^\mu\, J_\mu + e^2\, A_\mu A^\mu\,\varphi^*\varphi - \frac{\lambda}{6}(\varphi^*\varphi)^2$$

where; 1) $A^\mu J_\mu$ is qubic term; 2) $e^2 A_\mu A^\mu \varphi^*\varphi$ involves 4 fields so quartic term; 3) $(\varphi^*\varphi)^2$ is also a quartic term.

So,

$$\mathcal{H} = \sum_a \pi_a\,\dot{\phi}_a - \mathcal{L}$$

$$\mathcal{H} = \pi_A\,\dot{A} + \pi_\varphi\,\dot{\varphi} + \pi_{\varphi^*}\,\dot{\varphi}^* - \mathcal{L}$$

$\mathcal{H}_{\text{int}}$ involves qubic quartic & higher order terms, & $\sum p_a\pi_a$ involves only quadratic terms. (Since $\pi_A = \frac{\partial\mathcal{L}}{\partial\dot{A}}$ & $\dot{A}$ comes from $\mathcal{L}_{\text{free}}$.) So

$$\mathcal{H}_{\text{int}} = -\mathcal{L}_{\text{int}}$$

$$= -ie\,A^\mu J_\mu \qquad \underset{\substack{\downarrow \\ -\text{ve sign is also} \\ \text{correct; as } \varphi^*\varphi>0 \\ A_\mu A^\mu = A_0^2 - \vec{A}^2 \\ = -\vec{A}^2 \text{ we can gauge away } A_0. \\ \text{i.e. } -e^2 A_\mu A^\mu\varphi^*\varphi \text{ is +ve quartic term}}}{- e^2\,A_\mu A^\mu\,\varphi^*\varphi} \qquad \underset{\substack{\downarrow \\ + \text{ Sign} \Rightarrow \text{+ve quartic potential} \\ (\varphi^*\varphi)^2>0}}{+\frac{\lambda}{6}(\varphi^*\varphi)^2}$$

+ve quartic potential is bounded from below.

a) The propagator —

$$\langle 0|\,T\left(\varphi_0^\dagger(x)\,\varphi_0(y)\right)|0\rangle = D_F(x-y) \qquad \bullet\!\longrightarrow\!\bullet$$

$$\langle 0|\,T\ A_\mu(x)\ A_\nu(y)\,|0\rangle = -\eta_{\mu\nu}\ D_F(x-y)\Big|_{m=0} \qquad x \sim\!\sim\!\sim\!\sim\ y \qquad \text{(Lorentz gauge)}$$

A_0 & A_3 are removed by fixing the gauge, so for A_1 & A_2 we have

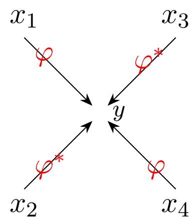
$$\langle 0|\,T\ A_1(x)\ A_1(y)\,|0\rangle = D_F(x-y).$$
$$\langle 0|\,T\ A_2(x)\ A_2(y)\,|0\rangle = D_F(x-y)$$

(A_μ here is free field.)

(2) Second step is to draw interactions —
For $\mathcal{L}_{\text{int}}$ term $\frac{\lambda}{6}\,(\varphi^*\varphi)^2$: $\quad e^{+i\int \frac{\lambda}{6}(\varphi^*\varphi)^2\, d^4y}$
(For first order) this term gives

$$+\frac{i\lambda}{6}\,(\varphi^*(y)\,\varphi(y))^2$$

which is:



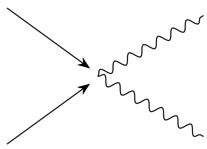
Sample calculation:

$$\frac{\lambda}{6}\int \langle 0|\, T\left(\varphi(x_1)\,\varphi^*(x_2)\,\varphi(x_3)\,\varphi^*(x_4)\,(\varphi^*(y)\,\varphi(y))^2\right)|0\rangle\; d^4y$$

Rule is — φ can only contract with φ^*.

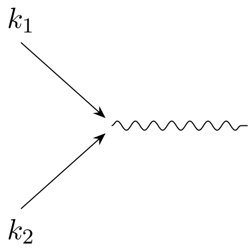
$$= \frac{\lambda}{6}\cdot(2)\cdot(2)\int D_F(x_1-y)\;D_F(x_3-y)\;D_F(x_2-y)\;D_F(x_4-y)\;d^4y\ldots$$

Other quartic term $-e^2 A_\mu A^\mu\,(\varphi^*\varphi)$ provides:



$(-e^2)$

Cubic term $= ie\,A^\mu J_\mu$ (gives) $= ie\,A^\mu\,(\varphi^*\partial_\mu\varphi - \varphi\,\partial_\mu\varphi^*)$:

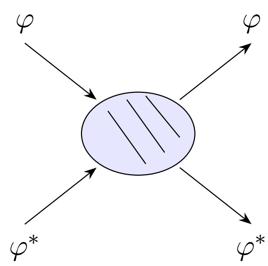


$e\,(k_1-k_2)_\mu$

Now we can calculate interacting propagators

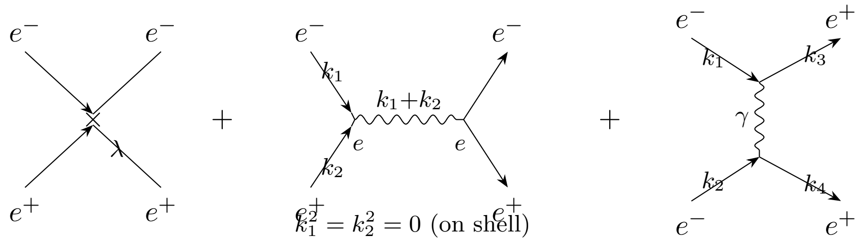
$$\langle\Omega|\,T\,(\varphi^*(x)\,\varphi(y))\,|\Omega\rangle \quad \& \quad \langle\Omega|\,T\,(A_\mu(x)\,A_\nu(y))\,|\Omega\rangle \quad \& \quad n\text{-point functions, in scalar QED.}$$

<u>Sample processes</u>:



$\to$ represents all possible such processes.

say, $\varphi \to$ particle, $\varphi^* \to$ antiparticle.



$k_1^2 = k_2^2 = 0$ (on shell)

$(k_1 + k_2)^2 \neq 0$ (off shell)

(No exchange of γ) $\downarrow$ e^- & e^+ annihilated and later photon creates e^- & e^+.

(exchange of γ) $\downarrow$ e^- somewhere emits virtual photon which is absorbed by positron during their propagation.

<u>Fermions</u> (QED) $\longrightarrow$ Algebra 1 $\longrightarrow$ Algebra 2

Lecture 9 — Algebra

Lorentz Algebra — (Mnemonic)

$$\begin{pmatrix} 0 & M_{01} & M_{02} & M_{03} \\ & 0 & M_{12} & M_{13} \\ & & 0 & M_{23} \\ & & & 0 \end{pmatrix} \qquad M_{\mu\nu} \longrightarrow \begin{array}{ll} M_{01} & \text{boost in } x \\ M_{02} & \text{'' '' } y \\ M_{03} & \text{'' '' } z \end{array} \qquad \begin{array}{l} 0 \to \text{time component} \\ \hookrightarrow \text{has to be boost} \end{array}$$

$$\left.\begin{array}{l} M_{\mu\nu} \text{ is antisymmetric} \\ \text{so has only 6-independent} \\ \text{parameters} \end{array}\right\} \quad \left.\begin{array}{ll} M_{12} & \text{Rotation in } x\text{-}y \text{ plane} \\ M_{23} & \text{'' '' } y\text{-}z \text{ plane} \\ M_{13} & \text{'' '' } x\text{-}z \text{ plane} \end{array}\right\} \begin{array}{l} \mu,\nu = 1,2,3 \\ \hookrightarrow \text{space} \\ \text{(Rotation)} \end{array}$$

When we talk about algebra we distinguish two concepts — one is abstract algebra and another one is actual representation.

We learned that a particle of spin J has $2J+1$ states. That means the angular momentum generators $\vec{J}\cdot\hat{n}$ act on a wave-fun. of particle of spin J, as $(2J+1)\times(2J+1)$ matrix. So for spin 1/2 particles we have generators of 2×2 matrix.

So, for particle of spin 1 we need 3×3 representation of angular momentum $\vec{J}\cdot\hat{n}$.

So, for particle of spin J we need $(2J+1)\times(2J+1)$ representation of angular momentum $(\vec{J}\cdot\hat{n})$.

Note that each such representations of (J_x, J_y, J_z) will obey

$$[J_x, J_y] = i\,J_z \qquad \text{or} \qquad [J_i, J_j] = i\,J_k\,\epsilon_{ijk} \qquad (\hbar = 1)$$

For higher spin we require higher dimensional matrices. So, there are many kinds/dimensions of matrices satisfying same algebra.

There are many sets of matrices $S = \{J_x, J_y, J_z\}_{2\times 2},\ S_{3\times 3}\dots$ (or $S = \{M_1, M_2 \dots M_n\}_{n\times n}$) satisfying a given algebra, each set is called a representation.

So Algebra is an abstract thing, $M_{\mu\nu}$ need not be matrices. They only define the algebra. Its like a rule telling to look for matrices which satisfy the rule.

So the analogue of angular momentum algebra $[J_i, J_j] = i\,\epsilon_{ijk}J_k$ is

$$\boxed{[M_{\mu\nu},\, M_{\lambda\rho}] = \underbrace{M_{\mu\rho}\,\eta_{\nu\lambda} + M_{\nu\lambda}\,\eta_{\mu\rho}}_{1)\ \mu\nu\curvearrowright\lambda\rho} - \underbrace{(M_{\mu\lambda}\,\eta_{\nu\rho} + M_{\nu\rho}\,\eta_{\mu\lambda})}_{2)\ \mu\nu\curvearrowright\lambda\rho}} \longleftarrow \text{Lorentz Algebra.}$$

We can also derive angular momentum algebra from lorentz algebra:

$$\begin{aligned} [M_{23},\, M_{21}] &= (M_{21}\,\eta_{33} + M_{33}\,\eta_{21}) - (M_{23}\,\eta_{21} + M_{31}\,\eta_{23}) \\ &= M_{21}(-1) + M_{33}(0) - (M_{23}(0) + M_{31}(0)) \\ &= -M_{21} \end{aligned}$$

So far we have just written abstract algebra (of Lorentz Algebra). What about its representation — (16 generators?)

A representation would be set of matrices $(M_{\mu\nu})^A{}_B$ which satisfy the lorentz algebra (above rule).

— Any such choice of matrices is a representation, and the range of values of A and B tells us dimension or size of matrices.

Our favourite example: For $M_{\mu\nu} = J_x\ J_y\ J_z$,

$$\underline{\text{If}} \quad A, B = 1, 2 \quad \text{then matrices are} \quad M_{\mu\nu} = \sigma_1,\ \sigma_2,\ \sigma_3$$

$$\text{If} \quad A, B = 1,2,3 \qquad M_{\mu\nu} = S_1, S_2, S_3 \text{ —}$$

Relation b/w $\Lambda^\mu_{\ \nu}$ & $M_{\mu\nu}$ —

We know, $(x')^\mu = \Lambda^\mu_{\ \nu}\, x^\nu$ (how coordinates & vector transform under lorentz transformation).

— $\Lambda^\mu_{\ \nu}$ is 4×4 matrix, so it must be a 4×4 representation of lorentz algebra.

— First of all $\Lambda^\mu_{\ \nu}$ generate finite lorentz transformation while $M_{\mu\nu}$ generates infinitesimal lorentz transformatio
(Because algebra is always among infinitesimal generators).

So we must make a finite lorentz transformation.

Remember, $M_{\mu\nu}$ is anti-symmetric so has only 6-independent parameters $\omega_{\mu\nu}$ (3 boost + 3 rotation).

To generate finite lorentz transformation —

$$\underbrace{e^{\left(\frac{1}{2}\right)\omega_{\mu\nu}M^{\mu\nu}}}_{\downarrow \text{ representation}} \qquad \left\{ \begin{array}{l} \text{factor of } \left(\frac{1}{2}\right) \text{ comes} \\ \text{from fact that} \\ \omega_{01}M^{01} = \omega_{10}M^{10} \\ \text{since } M^{\mu\nu} \text{ \& } \omega^{\mu\nu} \\ \text{are anti-symmetric.} \end{array} \right.$$

No '$i = \sqrt{-1}$' in exponential, since these are real things.

For any given representation we can have field which transform under that representation, that is why it is useful for field theory.

Ex:

(1) Representation of lorentz algebra which acts upon scalar fields (or what is infinitesimal transformation on scalar fields). Actually there is no such transformation. ($M_{\mu\nu} = 0$) ($M_{\mu\nu}$ is rep^s of $M_{\mu\nu}$),

$$\phi'(x') = \Lambda\, \phi(x)$$

$$\phi'(x') = 1 \cdot \phi(x) \qquad \text{(spin 0 rep}^{\text{n}} \text{ of } \Lambda)$$

$$\text{or,} \qquad e^{\frac{1}{2}\omega_{\mu\nu}M^{\mu\nu}} = e^{\frac{1}{2}\omega_{\mu\nu}(0)} = e^0 = 1$$

(2) Find rep^s of lorentz algebra for transformation of vector fields.

$$A'_\mu(x') = \Lambda^\mu_{\ \nu}\, A_\nu(x)$$

reps: $(M^{\mu\nu})^\alpha_{\ \beta}$ (or) $(V^{\mu\nu})^\alpha_{\ \beta}$ (V: vector)

$$(V^{\mu\nu})^\alpha_{\ \beta} = \eta^{\mu\alpha}\,\delta^\nu_\beta - \eta^{\nu\alpha}\,\delta^\mu_\beta$$

$\downarrow$ set of 6 matrices a) $(V^{01})^\alpha_{\ \beta}$, $(V^{02})^\alpha_{\ \beta}$, $(V^{03})^\alpha_{\ \beta}$ b) $(V^{12})^\alpha_{\ \beta}$, $(V^{13})^\alpha_{\ \beta}$, $(V^{23})^\alpha_{\ \beta}$

where each of 6 matrices generates their corresponding transformation. (a) generates Boost (b) generates rotations for vector fields.

Since R.H.S include $\eta = \begin{pmatrix} 1 & & & \\ & -1 & & \\ & & -1 & \\ & & & -1 \end{pmatrix}_{4\times4}$ a product with 4×4 matrices, $V^{\mu\nu}$ has to be 4×4 matrices. For such case where dimension of $(\mu,\nu) = \dim(\alpha,\beta)$ we call it fundamental rep^s.

The finite lorentz transformation thus becomes

$$= e^{\frac{1}{2}\,\omega_{\mu\nu} V^{\mu\nu}}$$
$$= e^{\frac{1}{2}\,\omega_{\mu\nu}\left(\eta^{\mu\alpha}\delta^{\nu}{}_{\beta}-\eta^{\nu\alpha}\delta^{\mu}{}_{\beta}\right)}$$
$$= e^{(\eta\,\mu)^{\alpha}{}_{\beta}} \;=\; \eta^{\alpha\gamma}\,\omega_{\gamma\beta} \qquad (\gamma \text{ gets contracted}).$$

So $\quad \Lambda^{\alpha}_{\beta} = e^{(\eta\,\omega)^{\alpha}{}_{\beta}}$ $\qquad$ (spin 1 rep$^{\text{n}}$)

Check if; $\quad e^{\eta\omega}\,\eta\,(e^{\eta\omega})^{T} = \eta \qquad$ (as Λ satisfies $\Lambda\eta\Lambda^{T} = \eta$)

We will see that, we will find a representation which will describe spin $\frac{1}{2}$ particles.

We look for spin $\frac{1}{2}$ rep$^{\text{n}}$ of $M^{\mu\nu}$ particles

We will look for new rep$^{\text{n}}$ of lorentz algebra $[M^{\mu\nu}, M^{\lambda\rho}] = \cdots$ we will see that this rep$^{\text{n}}$ describe or act upon new class of fields which have half integer spin.

One motivation to look for new rep$^{\text{n}}$ is that so far we have found spin 0 & spin 1

$$\phi'(x') = 1\,\phi(x) \qquad\qquad V'^{\mu}(x') = \Lambda^{\mu}_{\nu}\,V^{\nu}(x)$$

we can add indices to fields $\longrightarrow B_{\mu\nu}(x)$ which will transform like

$$B'^{\mu\nu}(x') = \Lambda^{\mu}_{\rho}\,\Lambda^{\nu}_{\sigma}\,B^{\rho\sigma}(x) \qquad \text{or} \quad B^{\rho\sigma}(x).$$

But this rep$^{\text{n}}$ $\left(\Lambda^{\mu}_{\rho}\,\Lambda^{\nu}_{\sigma} = \exp\left(\frac{1}{2}\,\omega_{\mu\nu}\,M^{\mu\nu}\right)\right)$ will describe integer spin particles.

Terminology: $\;$ rep$^{\text{n}}$ $= M^{\mu\nu}$ (or) $\Lambda^{\mu}_{\rho}\,\Lambda^{\nu}_{\sigma}$ (or) $B^{\sigma\nu}(x)$

In Nature we are only interested in spin 0 & spin 1 (as higher spin requires more energy to discover/probe).

But spin $\frac{1}{2}$ is of crucial interest as $e^{-}, \mu, \tau, \ldots$ quarks (u, d, s, c, t, b) are spin $\frac{1}{2}$ particles.

Clifford Algebra —

The idea here is to change from lorentz algebra to new easier algebra called clifford algebra (it is simpler than lorentz algebra.)

Any rep$^{\text{s}}$ of clifford algebra automatically satisfies lorentz algebra (i.e. is also a rep$^{\text{s}}$ of lorentz algebra). So it is an intermediate tool to make reps of lorentz algebra.

In any algebra we have to have abstract generators

$$\{\Gamma^{\mu},\, \Gamma^{\nu}\} = 2\,\eta^{\mu\nu} \qquad (\Gamma^{\mu},\ \Gamma^{\nu} \text{ are generators of clifford algebra})$$

$\downarrow$

Notice that it is way simpler than lorentz algebra.

Any representation would be a "set" of matrices, $\left(\gamma^{\mu}_{ab}\right)$ satisfying

$$\gamma^{\mu}_{ab}\,\gamma^{\nu}_{bc} + \gamma^{\nu}_{ab}\,\gamma^{\mu}_{bc} = 2\,\eta^{\mu\nu}\,\delta_{ac} \qquad (\text{to make } a = c)$$
$$\text{or} \quad \{\gamma^{\mu},\, \gamma^{\nu}\} = 2\,\eta^{\mu\nu}\cdot I$$

$\downarrow$ Since LHS are matrices, RHS should be multiplied with an identity $\;(\eta^{\mu\nu} = \text{number})$

(Q) Why we are interested in clifford algebra?

Ans: There is a theorem why we are interested,

Theorem — Clifford algebra reps provide lorentz algebra reps via

$$\boxed{\frac{1}{4}\,[\gamma^{\mu},\, \gamma^{\nu}]_{ab} = (S^{\mu\nu})_{ab}}$$

This is called
$\hookrightarrow$ spinor reps.

Find LHS, then, the matrices $(S^{\mu\nu})_{ab}$ satisfy —

$$\text{This is lorentz algebra} \longleftarrow \qquad \left[S^{\mu\nu},\, S^{\lambda\rho}\right] = \left[S^{\mu\rho}\,\eta^{\nu\lambda} + S^{\nu\lambda}\,\eta^{\mu\rho} - \left(S^{\mu\lambda}\,\eta^{\nu\rho} + S^{\nu\rho}\,\eta^{\mu\lambda}\right)\right]$$

$\mu\nu\,,\ \lambda\rho\ -\ (\mu\nu\,,\ \lambda\rho)$ (hence $S^{\mu\nu}$ satisfies lorentz algebra)

Above is easy to remember, but can be derived from using $\left\{\begin{array}{r} S^{\mu\nu} = \frac{1}{4}\left[\gamma^{\mu},\gamma^{\nu}\right] \\ \&\ S^{\lambda\rho} = \frac{1}{4}\left[\gamma^{\lambda},\gamma^{\rho}\right] \end{array}\right\}$ in above expression.

To put in context of group theory in general, only orthogonal groups have spinor reps. & lorentz group is an orthogonal group.

We were missing lorentz algebra reps of spin $\frac{1}{2}$ particles & lorentz group is $SO(3,1)$, $\Rightarrow$ if we study clifford algebra we will find a new class of reps. That's how we found lorentz reps. of spin $\frac{1}{2}$ particles.

This allows us to define fields $\Psi_a(x)$, which transform under lorentz transformation as

$$\Psi_a \longrightarrow \Psi'_a = \left(e^{\frac{1}{2}\,\omega_{\mu\nu}\,M^{\mu\nu}}\right)_{ab}\Psi_b$$

where $\quad M^{\mu\nu} = S^{\mu\nu}$ here

$$\Psi_a \ \rightarrow\ \Psi'_a = \left(e^{\frac{1}{2}\,\omega_{\mu\nu}\,S^{\mu\nu}}\right)_{ab}\Psi_b$$

& such fields $\Psi_a(x)$ are called spinors.

In next lecture, we will see how to make an eq$^{\text{n}}$ of motion and an action for such spinors. (which are invariant under lorentz transformation, i.e. the transformation will cancel out in final e.o.m & action.)

Once we have reps of clifford algebra $(\gamma^{\mu})_{ab}$ we can construct $S^{\mu\nu} = \frac{1}{4}\left[\gamma^{\mu}\,\gamma^{\nu}\right]$ which are also reps of lorentz algebra (or satisfy lorentz algebra)

$$\text{i.e.}\quad \left[S^{\mu\nu},\, S^{\lambda\rho}\right] = S^{\mu\lambda}\,\eta^{\nu\rho} + S^{\nu\rho}\,\eta^{\mu\lambda} - \left(S^{\mu\rho}\,\eta^{\nu\lambda} + S^{\nu\lambda}\,\eta^{\mu\rho}\right)$$

$\boxed{\text{The reps we get } (S^{\mu\nu})_{ab} \text{ are called spinor reps.}}$

It happens due to property of $SO(\ )$ type algebra. Each $SO(\ )$ type algebra/group has spinor reps. (i.e. we know that lorentz algebra/lorentz group are of orthogonal type $SO(3,1)$, where $SO(3)$ is group of rotations, and $SO(3,1)$ is group of rotation & boost.)

So any rotation group $\rightarrow SO(3)$ or orthogonal group $O(n)$ has spinor reps.

Spinor reps of rotation group shall satisfy $\{\gamma^{\mu}\,\gamma^{\nu}\} = 2\eta^{\mu\nu} I$

$$\mu = i = 1,2,3$$

$$\gamma^{\mu} \longrightarrow \sigma^{i}\,, \qquad \eta^{\mu\nu} \longrightarrow \delta^{ij}\,, \qquad \left\{\sigma^{i},\,\sigma^{j}\right\} = 2\,\delta^{ij}\,\underline{I}$$

$\hookrightarrow$ Identity matrix.

So $\{\sigma_1, \sigma_2, \sigma_3\}$ satisfy clifford algebra, it is spinor reps. of rotation group. $(SO(3))$

Whereas $\frac{1}{4}\left[\gamma^{\mu},\,\gamma^{\nu}\right] = S^{\mu\nu}$ shall be spinor reps of lorentz group. $(SO(3,1))$

$$\underline{\text{Remarks}} \text{ — } \quad \text{(i)} \quad (\gamma^{\mu})^2 = \pm I \qquad \left\{\begin{array}{r} \{\gamma^{\mu},\gamma^{\nu}\} = 2\eta^{\mu\nu} I \\ \gamma^{\mu}\gamma^{\mu} + \gamma^{\mu}\gamma^{\mu} = 2\eta^{\mu\mu}\, I \\ 2\,(\gamma^{\mu})^2 = 2\,\eta^{\mu\mu}\, I \\ \text{for } \mu = 0 \quad (\gamma^{\mu})^2 = +I \\ \text{for } \mu = 1,2,3 \ \ (\gamma^{\mu})^2 = -I \end{array}\right.$$

(ii) All distinct γ^μ anticommute.

$$\{\gamma^\mu,\, \gamma^\nu\} = 2\eta^{\mu\nu} I$$

$$\text{For } \mu \neq \nu \qquad \eta^{\mu\nu} = 0 \quad \Rightarrow \quad \boxed{\{\gamma^\mu,\, \gamma^\nu\} = 0}$$

Similarly we can write for $S^{\mu\nu} = \frac{1}{4}\,[\gamma^\mu,\, \gamma^\nu]$

(1) $S^{\mu\mu} = 0$

(2) $\mu \neq \nu \qquad S^{\mu\nu} = \frac{1}{4}\,(\gamma^\mu\gamma^\nu - \gamma^\nu\gamma^\mu)$

But for $\mu \neq \nu$ $\{\gamma^\mu, \gamma^\nu\} = 0 \Rightarrow$

$$S^{\mu\nu} = \frac{1}{4}\,(\gamma^\mu\gamma^\nu + \gamma^\mu\gamma^\nu)$$

$$\boxed{S^{\mu\nu} = \frac{1}{2}\,\gamma^\mu\gamma^\nu}$$

Now let's introduce this idea in field theory —

We introduced a field $\Psi_a(x)$, which will have the property that it will transform under lorentz transformations using this matrix $\left(e^{\frac{1}{2}\,\omega_{\mu\nu}\,S^{\mu\nu}}\right)$

$$\Psi_a(x) \;\longrightarrow\; \Psi'_a(x') = \left(e^{\frac{1}{2}\,\omega_{\mu\nu}\,S^{\mu\nu}}\right)_{\underset{\text{matrix indices}}{ab}} \underbrace{\Psi_b(x)}_{\hookrightarrow\text{ column matrix reps of } \Psi(x)}$$

Just to remind — For scalars $m^{\mu\nu} = 0$; For vectors $m^{\mu\nu} = V^{\mu\nu}$; For spinors $m^{\mu\nu} = S^{\mu\nu}$

Construction of $(\gamma^\mu)_{ab}$ matrices using Pauli matrices —

We start with pauli matrices satisfying a kind of clifford algebra.

$$\left\{\sigma^i,\, \sigma^j\right\} = 2\,\delta^{ij}$$

and then we take tensor product : $\sigma^i \otimes \sigma^j$

ex
$$\sigma^1 \otimes \sigma^2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \otimes \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$$
$$= \begin{pmatrix} 0 \cdot \sigma^2 & 1(\sigma^2) \\ (1)\,\sigma^2 & 0\,(\sigma^2) \end{pmatrix} = \begin{pmatrix} 0 & \sigma^2 \\ \sigma^2 & 0 \end{pmatrix}$$
$$= \begin{pmatrix} 0 & 0 & 0 & -i \\ 0 & 0 & i & 0 \\ 0 & -i & 0 & 0 \\ i & 0 & 0 & 0 \end{pmatrix} = \left(\begin{array}{c|c} 0 & \begin{matrix} 0 & -i \\ i & 0 \end{matrix} \\ \hline \begin{matrix} 0 & -i \\ i & 0 \end{matrix} & 0 \end{array}\right)$$

$$— \quad \underbrace{(A \otimes B)\,(C \otimes D)}_{\text{matrix product}} = \underbrace{(AC \otimes BD)}_{\text{direct product}}$$

To make 4×4 reps of γ^μ matrices, we need $\rightarrow$ (1) $(\gamma^\mu)\,(\gamma^\mu) = \pm I$ (2) & $(\mu \neq \nu)$ $\{\gamma^\mu, \gamma^\nu\} = 0$

Start with,

$$1) \qquad \sigma_1 \otimes \sigma_1 = \begin{pmatrix} 0 & \sigma_1 \\ \sigma_1 & 0 \end{pmatrix} \quad \text{as} \quad (\sigma_1 \otimes \sigma_1)\,(\sigma_1 \otimes \sigma_1) = \sigma_1^2 \otimes \sigma_1^2 = I_{2\times 2} \otimes I_{2\times 2} = I_{4\times 4}$$

So, $\boxed{\sigma_1 \otimes \sigma_1 = \gamma^0}$

$$2) \qquad \sigma_1 \otimes \sigma_2 \qquad (\sigma_1 \otimes \sigma_2)\,(\sigma_1 \otimes \sigma_2) = \left(\sigma_1^2 \otimes \sigma_2^2\right)_{2\times 2} = I \underset{2\times 2}{\otimes} I_{2\times 2} = I_{4\times 4}$$

Since we require $-I_{4\times4}$ for $\mu = 1,2,3$, we have to multiply with 'i'. So,

$$\boxed{i\,\sigma_1 \otimes \sigma_2 = \gamma^1}$$

Since $[\sigma_1\,\sigma_1] = 0$ but $\{\sigma_1\,\sigma_2\} = 0 \;\Rightarrow\; \{\sigma_1 \otimes \sigma_1\,,\; \sigma_1 \otimes \sigma_2\} = 0$

(3) We require γ^2 & γ^3 such that $\left(\gamma^2\right)^2 = \left(\gamma^3\right)^2 = -I$ and they should anticommute with each other i.e. $\{\gamma^0\,\gamma^2\} = \{\gamma^0\,\gamma^3\} = \{\gamma^1\,\gamma^2\} = \{\gamma^1\,\gamma^3\} = 0$

guess — $\sigma_1 \otimes \sigma_3$ shall anticommute with all above γ^0 & γ^1

Since $\{\sigma_3,\,\sigma_2\} = \{\sigma_3,\,\sigma_1\} = 0$

also

$$(\sigma_1 \otimes \sigma_3)^2 = (\sigma_1 \otimes \sigma_3)\,(\sigma_1 \otimes \sigma_3) = \left(\sigma_1^2 \otimes \sigma_3^2\right) = I_{4\times4}$$

again we have to multiply by 'i' to get $(-I_{4\times4})$. So

$$\boxed{i\,(\sigma_1 \otimes \sigma_3) = \gamma^2}$$

(4) We have ran out of σ matrices, we write

$$\boxed{\gamma^3 = i\,(\sigma_2 \otimes I)} \quad \text{since its square} = -I_{4\times4}$$

and it anticommute with all $\gamma^0, \gamma^1, \gamma^2$. {we could have chosen $i\,(\sigma_3 \otimes I)$ as well!}

$$A \otimes B \;\neq\; B \otimes A$$

We also have $\gamma_5 = i\,\gamma^0\gamma^1\gamma^2\gamma^3$ with $(\gamma^\mu)^2 = +I$ and which anticommute with all $\gamma^0, \gamma^1, \gamma^2, \gamma^3$. Also it is hermitian. $\Big((\gamma_5)^\dagger = \gamma_5\Big)$.

(i) $(\gamma_5)^2 = +I$

$$(\gamma_5)^2 = i^2\,\left(\gamma^0\gamma^1\,\gamma^2\,\gamma^3\right)\left(\gamma^0\gamma^1\gamma^2\gamma^3\right)$$

$\{\gamma^0\,\gamma^2\} = 0 \;\Rightarrow\; \gamma^0\gamma^3 = -\gamma^3\gamma^0$

So when we take γ^ν to left to $(\gamma^\mu \neq \gamma^\nu)$

take one more minus sign

$$
\begin{aligned}
&= (-1)(-1)^3\left(\gamma^1\gamma^2\gamma^3\,\gamma^1\gamma^2\gamma^3\right)\\
&= (-1)^2\left(\gamma^1\right)^2\left(\gamma^2\gamma^3\,\gamma^2\gamma^3\right)\\
&= (-1)(-1)\cancel{\left(\gamma^2\right)^2}\left(\gamma^3\right)^2\\
(\gamma_5)^2 &= (-1)(-1)(-1)(-1) \;=\; +I
\end{aligned}
$$

$\left(\gamma^1\right)^2 = -I$

In 5-d we require 5-gamma matrices in that case $\gamma_5 = \gamma_0\gamma_1\gamma_2\gamma_3$ which anticommutes with $\gamma_0\gamma_1\gamma_2\gamma_3$ & $\gamma_5^2 = -1$ $\left(\gamma_5\,\gamma_5 = \gamma_5^2\,\eta^{55},\;\; \eta = (+1,-1,-1,-1,-1)\right)$

(ii) $(\gamma_5)^\dagger = \left(i\,\gamma^0\gamma^1\gamma^2\gamma^3\right)^\dagger = -i\left(\gamma^3\right)^\dagger\left(\gamma^2\right)^\dagger\left(\gamma^1\right)^\dagger\left(\gamma^0\right)^\dagger = \gamma$

<u>Gamma matrices algebra</u> —

$$\Big\{\underset{1}{\mathbb{I}}\,,\; \underset{+4}{\gamma^\mu}\,,\; \underset{+4}{\gamma_5\,\gamma^\mu}\,,\; \underset{+6}{\gamma^\mu\gamma^\nu\;(\text{or } S^{\mu\nu})}\,,\; \underset{+1}{\gamma_5}\Big\} = 16 \text{ such matrices.}$$

<u>Exercise</u> —

$$\gamma_5\,\underbrace{\gamma^0\gamma^1}_{S^{01}} = +\left(i\,\gamma^0\gamma^1\gamma^2\gamma^3\right)\gamma^0\gamma^1$$

$$
\begin{aligned}
&= -i\,\gamma^1\gamma^2\gamma^3\gamma^1 \\
&= i\,\gamma^2\gamma^3 \\
&= 2i\,S^{23} \qquad\qquad \left\{ \frac{1}{2}\,\gamma^{\mu\nu} = S^{\mu\nu} \right. \\
\gamma_5\,S^{01} &= 2i\,S^{23}
\end{aligned}
$$

Lets get back —

We have field $\Psi_a(x)$ which transform like

$$\Psi'_a(x') = \left(e^{\frac{1}{2}\,\omega_{\mu\nu}\,S^{\mu\nu}}\right)_{ab}\Psi_b$$

This shall be a 4 component field because gamma matrices are 4×4 matrices.

These are 4 component field but do not transform the same way as 4 component vector field.

These are called spinor field because they transform under spinor representation.

<u>EOM</u> — We could think that $\left(\Box^2 + m^2\right)\Psi_a(x) = 0$ could be e.o.m but K.G. eq$^{\text{n}}$ leads to $-$ve probability current. So Dirac took an approach to write e.o.m. which are first order in space & time.

The possibility of 1$^{\text{st}}$ order diff. eq$^{\text{n}}$ arose for the first time, because when we had scalar $\varphi(x)$ & vector fields $A_\mu(x)$, there was simply no possibility ($\partial\varphi$ or $\partial A_\mu(x)$,) we don't get anything interesting & "lorentz invariant".

Here we have new operator: $\underline{(\gamma^\mu)_{ab}\,\partial_\mu}$ ↙ first order in derivatives.

$(\gamma^\mu)_{ab}\,\partial_\mu\,\Psi_b(x)$ shall also transform like spinors under lorentz transformation (i.e. to act $(\gamma^\mu)_{ab}$ on Ψ_b, the result shall behave like spinor & thus transform like spinor under L.T.)

Given Ψ_a, we can construct a new object —

$$\chi_a = (\gamma^\mu)_{ab}\,\partial_\mu\,\Psi_b \qquad \text{—(1)}$$

Now, if $\quad \Psi \longrightarrow \left(e^{\frac{1}{2}\,\omega_{\mu\nu}\,S^{\mu\nu}}\right)\Psi$

does $\quad \chi \longrightarrow \left(e^{\frac{1}{2}\,\omega_{\mu\nu}\,S^{\mu\nu}}\right)\chi\,?$

If it is true that χ transform like spinor then we can set $\chi = 0$ and say that

$$\boxed{(\gamma^\mu)_{ab}\,\partial_\mu\,\Psi_b = 0} \quad \text{is equation of motion.}$$

(Because EOM have to be lorentz Invariant!)

Under lorentz transformation —

$$\gamma^\mu\,\partial_\mu\Psi \;\longrightarrow\; \gamma^\mu\,\underbrace{\left(e^{-\frac{1}{2}\,\omega_{\lambda\rho}\,V^{\lambda\rho}}\right)^{\nu}_{\ \mu}}\,\partial_\nu\left(e^{\frac{1}{2}\,\omega\,S}\right)\Psi = \left(e^{\frac{1}{2}\,\omega_{\lambda\rho}\,S^{\lambda\rho}}\right)\gamma^\mu\,\partial_\mu\Psi$$

Since $\quad A'_\mu(x') = \left(\Lambda^{-1}\right)^T A_\mu(x) \quad$ → Transpose

$$\Lambda^{-1} = \left(e^{-\frac{1}{2}\,\omega_{\lambda\rho}\,V^{\lambda\rho}}\right)^{\nu}_{\ \mu} \qquad \left(\text{So,}\quad \partial_\mu \to \left(e^{-\frac{1}{2}\,\omega\,V^T}\right)\partial_\mu\right)$$

if

$$\left(e^{-\frac{1}{2}\,\omega_{\lambda_1\rho_1}\,V^{\lambda_1\rho_1}}\right)^{\mu}_{\ \nu}\left(e^{-\frac{1}{2}\,\omega_{\lambda_2\rho_2}\,S^{\lambda_2\rho_2}}\right)_{ac}\left(e^{\frac{1}{2}\,\omega_{\lambda_3\rho_3}\,S^{\lambda_3\rho_3}}\right)\gamma^{\nu}_{\ cd\,db} = (\gamma^\mu)_{ab} \qquad \uparrow \text{ Matrix indices.}$$

$$\begin{cases} \lambda_i, \rho_i = 0,1,2,3 \\ \quad i = 1,2,3 \end{cases}$$

H.W. (Q) Show that the above identity implies :-

$$\boxed{[\gamma^\mu, S^{\lambda\rho}] = (V^{\lambda\rho})^\mu{}_\nu \, \gamma^\nu}$$

So, since $\chi = (\gamma^\mu)_{ab} \, \partial_\mu \, \Psi_b$ transform like spinor we can set $\chi = 0$.

So, we found a lorentz invariant e.o.m —

$$\gamma^\mu \, \partial_\mu \Psi + m\Psi = 0 \qquad \hookrightarrow \text{ Because } m\Psi \text{ also transform like } \Psi \text{ under L.T.}$$

or

$$i \, \gamma^\mu \, \partial_\mu \Psi + m\Psi = 0 \qquad (\text{Free Dirac eq}^{\text{n}}) \quad \downarrow \quad (\text{linear in } \Psi) \text{ or (quadratic in } \mathcal{L})$$

$\downarrow$ will explain later. $\quad\downarrow$ (lorentz invariant eq$^{\text{n}}$)

We know γ^μ are in general complex (σ_2 = complex). So we must look for complex solution.

i.e. Ψ should be complex.

Only in reps where γ^μ are real, we can look for Ψ to be real only.

Since all γ^μ are complex $\Rightarrow S^{\mu\nu} = \frac{1}{4}\,[\gamma^\mu, \gamma^\nu]$ shall also be complex.

$$\Rightarrow \quad \Psi \text{ must be complex}$$

as if we take real Ψ, then $\left(e^{\frac{1}{2}\,\omega_{\mu\nu}\,S^{\mu\nu}}\right)_{ab} \Psi_b$ will be complex after lorentz transformation.

Lagrangian —

Once we have lagrangian (free (quadratic) lagrangian), then we can think of all higher order interactions to act with. & we will then find an interacting theory.

We need to guess lagrangian —

Normally, e.o.m obey

$$\partial_\mu \left(\frac{\partial \mathcal{L}}{\partial\,(\partial_\mu \varphi)} \right) = \frac{\partial \mathcal{L}}{\partial \varphi}$$

If we have e.o.m as $(i\,\gamma^\mu\,\partial_\mu \Psi + m\Psi) = 0$ probably lagrangian density should have Ψ^2 term to have above e.o.m.

We want that term to be lorentz invariant. As generally e.o.m remain same if $\mathcal{L}' = \mathcal{L} + \partial_\mu \chi \;\hookrightarrow$ total derivative.

but here we just want $\mathcal{L}$ to be invariant,

Puzzle — Is $\Psi_a^T \Psi_a$ lorentz invariant?

$$\Psi_a \longrightarrow \left(e^{\frac{1}{2}\,\omega_{\mu\nu}\,S^{\mu\nu}}\right)_{ab} \Psi_b$$

So

$$\begin{aligned} \Psi_a^T \, \Psi_a \;\longrightarrow\; & \left(e^{\frac{1}{2}\,\omega S}\right)^T_{ab} \Psi_b^T \, \left(e^{\frac{1}{2}\,\omega S}\right)_{ac} \Psi_c \\ = & \left(e^{\frac{1}{2}\,\omega S}\right)^T_{ba} \left(e^{\frac{1}{2}\,\omega S}\right)_{ac} \; \Psi_b^T \, \Psi_c \\ = & \left(\left(e^{\frac{1}{2}\,\omega S}\right)^T \left(e^{\frac{1}{2}\,\omega S}\right) \right)_{bc} \; \Psi_b^T \, \Psi_c \end{aligned}$$

For lorentz invariance we require

$$\left(e^{\frac{1}{2}\,\omega S}\right)^T e^{\frac{1}{2}\,\omega S} = I$$
$$\Rightarrow \quad e^{\frac{1}{2}\,\omega S^T} e^{\frac{1}{2}\,\omega S} = I$$

Only possible if $\left(S^T = -S\right)$

But S is not antisymmetric, so $S^T \neq -S$ so $\Psi_a^T \Psi_a$ is not lorentz invariant.

(Q) How do we know 'S' is not antisymmetric?

So the guess should be to check if $\Psi_a^* \Psi_a$ is lorentz invariant. (the logic is if Ψ_a is complex $\Psi_a^* \Psi_a$ is real to add in lagrangian)

Digression :- For vector representation of lorentz transformation, we can write it in two ways.

$$e^{\frac{1}{2}\,\omega V} \qquad\qquad e^{\eta\,\omega}$$

let's write with indices.

$$\left(e^{\frac{1}{2}\,\omega_{\mu\nu}\,V^{\mu\nu}}\right)^{\alpha}{}_{\beta} \qquad\qquad \left(e^{\eta\,\omega}\right)^{\alpha}{}_{\beta}$$

expand to 1st order —

$$\left[1 + \frac{1}{2}\,\omega_{\mu\nu}\,(V^{\mu\nu})^{\alpha}{}_{\beta} + \cdots\right] \qquad\qquad \left[1 + \eta^{\alpha\mu}\,\omega_{\mu\beta} + \cdots\right]$$

$$\left(1 + \frac{1}{2}\,\omega_{\mu\nu}\left(\eta^{\mu\alpha}\,\delta^{\nu}_{\ \beta} - \eta^{\nu\alpha}\,\delta^{\mu}_{\ \beta}\right) + \cdots\right)$$
$$\left(1 + \left(\frac{1}{2}\,\omega_{\mu\nu}\,\delta^{\nu}_{\ \beta}\,\eta^{\mu\alpha} - \frac{1}{2}\,\omega_{\mu\nu}\,\eta^{\nu\alpha}\,\delta^{\mu}_{\ \beta}\right) + \cdots\right)$$
$$\left(1 + \left(\frac{1}{2}\,\omega_{\mu\beta}\,\eta^{\mu\alpha} - \frac{1}{2}\,\omega_{\mu}{}^{\alpha}\,\delta^{\mu}_{\ \beta}\right) + \cdots\right) \quad \longleftarrow \text{ will be equal to}$$

For transpose: $\;1 + \frac{1}{2}\,\omega_{\mu\nu}\,(V^{\mu\nu})^{T\,\alpha}{}_{\beta}\,; \qquad 1 + \omega^T \eta^T$

Is $\Psi_a^*\,\Psi_a$ lorentz invariant —

$$\begin{aligned}
\Psi_a^*\,\Psi_a \;\longrightarrow\; & \left(e^{\frac{1}{2}\,\omega S}\right)^*_{ab} \Psi_b^* \left(e^{\frac{1}{2}\,\omega S}\right)_{ac} \Psi_c \\
&= \left(e^{\frac{1}{2}\,\omega S}\right)^{*T}_{ba} \left(e^{\frac{1}{2}\,\omega S}\right)_{ac} \Psi_b^*\,\Psi_c \\
&= \left(\left(e^{\frac{1}{2}\,\omega S}\right)^{\dagger} \left(e^{\frac{1}{2}\,\omega S}\right)\right)_{bc} \Psi_b^*\,\Psi_c \\
&= \left(\left(e^{\frac{1}{2}\,\omega S^{\dagger}}\right) \left(e^{\frac{1}{2}\,\omega S}\right)\right)_{bc} \Psi_b^*\,\Psi_c
\end{aligned}$$

We know, we require $S^{\dagger} = -S$ for lorentz invariance.

$$\begin{aligned}
S = \frac{1}{4}\,[\gamma^{\mu},\,\gamma^{\nu}] \;\Rightarrow\; S^{\dagger} &= \frac{1}{4}\,(\gamma^{\mu}\gamma^{\nu} - \gamma^{\nu}\gamma^{\mu})^{\dagger} \\
&= \frac{1}{4}\left((\gamma^{\nu})^{\dagger}\,(\gamma^{\mu})^{\dagger} - (\gamma^{\mu})^{\dagger}\,(\gamma^{\nu})^{\dagger}\right) \\
&= \frac{1}{4}\left[(\gamma^{\nu})^{\dagger}\,,\,(\gamma^{\mu})^{\dagger}\right]
\end{aligned}$$

Is $(\gamma^\mu)^\dagger = (\gamma^\mu)$?

$$\begin{cases} & (\gamma^0)^\dagger = (\sigma_1 \otimes \sigma_1)^\dagger = \gamma^0 \\ & (\gamma^1)^\dagger = (i\,\sigma_1 \otimes \sigma_2)^\dagger = -\gamma^1 \\ \text{Similarly} & (\gamma^2)^\dagger = -\gamma^2 \\ & (\gamma^3)^\dagger = -\gamma^3 \end{cases}$$

γ^0 is hermitian, γ^i is anti-hermitian

So for $\mu = 0,\ \nu = 0$: $\quad S^\dagger = S = 0$

for $\mu = 0,\ \nu \neq 0$: $\quad (\gamma^\nu)^\dagger = -\gamma^\nu$

$$\begin{aligned} S^\dagger &= \frac{1}{4}\left[-\gamma^\nu,\, \gamma^0\right] \\ &= \frac{1}{4}\left[\gamma^0,\, \gamma^\nu\right] = S^{0\nu} \\ \left(S^{0\nu}\right)^\dagger &= S^{0\nu} \end{aligned}$$

For $\mu \neq \nu,\ \ \mu, \nu = 1, 2, 3$

$$\begin{aligned} \left(S^{ij}\right)^\dagger &= \frac{1}{4}\left[-\gamma^j,\, -\gamma^i\right] \\ &= -\frac{1}{4}\left[\gamma^i,\, \gamma^j\right] \\ &\boxed{\left(S^{ij}\right)^\dagger = -S^{ij}} \end{aligned}$$

$$(\gamma^\mu)^\dagger = \begin{cases} \gamma^\mu & \mu = 0 \\ -\gamma^\mu & \mu \neq 0 \end{cases} \qquad \text{So} \quad S^\dagger = -S \quad \forall\ \mu, \nu.$$

We need an operation which treats all γ^μ similar. Some analogue of γ^μ which is always hermitian or anti-hermitian.

So, $\quad S^\dagger = -S$ only for $\mu, \nu \neq 0$.

So $\quad \Psi_a^*\, \Psi_a$ is also not lorentz invariant.

Our recent experience with real fields & complex fields does not help us make a lagrangian.

We need something which anticommute with γ^i but does not anticommute with γ^0, which is γ^0 itself.

$$\left\{\gamma^0,\, \gamma^i\right\} = 0$$

We check for $\quad \overline{\gamma}^\mu = \gamma^0\,(\gamma^\mu)^\dagger\,\gamma^0 = \begin{cases} \gamma^0\,(-\gamma^\mu)\,\gamma^0 = \left(\gamma^0\right)^2\gamma^\mu = \gamma^\mu \ (\mu \neq 0) \\ \gamma^0\gamma^0\,\gamma^0 = \gamma^0 \qquad \mu = 0 \end{cases}$

$$\overline{\gamma}^\mu = \gamma^\mu$$

$$\begin{aligned} \Rightarrow \quad \overline{S}^{\mu\nu} = \gamma^0\,(S^{\mu\nu})^\dagger\,\gamma^0 &= \gamma^0\,\frac{1}{4}\left[(\gamma^\nu)^\dagger,\, (\gamma^\mu)^\dagger\right]\gamma^0 \\ &= (-)\,\gamma^0\,\frac{1}{4}\left[(\gamma^\mu)^\dagger,\, (\gamma^\nu)^\dagger\right]\gamma^0 \\ &= (-1)\,\frac{1}{4}\left[\gamma^0\,(\gamma^\mu)^\dagger\,\gamma^0,\ \gamma^0\,(\gamma^\nu)^\dagger\,\gamma^0\right] \\ &= -\frac{1}{4}\left[\gamma^\mu,\, \gamma^\nu\right] \ = \ -S^{\mu\nu} \end{aligned}$$

$$\boxed{\overline{S}^{\mu\nu} = -S^{\mu\nu}}$$

Finally we found an operation for which $\overline{S}^{\mu\nu} = -S^{\mu\nu}$

If we define, $\overline{\Psi}_b = \Psi_a^\dagger \left(\gamma^0\right)_{ab}$ then,

$$\overline{\Psi}_a\, \Psi_a \text{ is lorentz invariant.}$$

$$\begin{aligned}
\overline{\Psi}\,\Psi \;=\; \Psi^\dagger \left(\gamma^0\right)\Psi \;\longrightarrow\; & \underbrace{\left(e^{\frac{1}{2}\,\omega S}\right)^\dagger \gamma^0 \left(e^{\frac{1}{2}\,\omega S}\right)} \;\Psi^\dagger\Psi \\
&= \gamma^0\, e^{-\frac{1}{2}\,\omega S}\; e^{\frac{1}{2}\,\omega S}\;\Psi^\dagger\Psi \\
&= \gamma^0\,\Psi^\dagger\Psi \\
&= \overline{\Psi}\,\Psi
\end{aligned}$$

There are probably no other combination which are Lorentz invariant.

So

$$\begin{aligned}
\mathcal{L} &= \overline{\Psi}\left(i\,\gamma^\mu\partial_\mu - m\right)\Psi \\
&= \Psi_a^\dagger\,\gamma^0_{ab}\left(i\,\gamma^\mu_{bc}\,\partial_\mu - m\,\delta_{bc}\right)\Psi_c \\
\mathcal{L} &= \overline{\Psi}\left(i\,\partial\!\!\!/ - m\right)\Psi
\end{aligned}$$

What are equation of motion of this $\mathcal{L}$? We can vary Ψ & $\overline{\Psi}$ independently.

$$\frac{\delta\mathcal{L}}{\partial\overline{\Psi}} = \left(i\,\partial\!\!\!/ - m\right)\Psi \;=\; \partial_\mu\left(\frac{\partial\mathcal{L}}{\partial\left(\partial_\mu\overline{\Psi}\right)}\right) = \partial_\mu(0) = 0$$

taking $(\overline{})$ $\Rightarrow$ $\dfrac{\partial\mathcal{L}}{\partial\Psi} = \overline{\Psi}\left(-i\,\gamma^\mu\overleftarrow{\partial}_\mu - m\right) = 0$

Canonical momentum to Ψ $\quad\left(\dot{\Psi}\text{ occurs in }\gamma^\mu\partial_\mu\text{ at }\mu = 0\right)$

$$\Pi_a = \frac{\delta\mathcal{L}}{\delta\dot{\Psi}_a} = \left(\Psi_a^\dagger\,\gamma^0\right) i\,\gamma^0 = i\,\Psi^\dagger\left(\gamma^0\right)^2 = i\,\Psi_a^\dagger$$

<u>Weyl representation</u> —

$$\gamma^0 = \sigma^1 \otimes I = \begin{pmatrix} 0 & I \\ I & 0 \end{pmatrix}$$

$$\gamma^i = \begin{bmatrix} 0 & \sigma^i \\ -\sigma^i & 0 \end{bmatrix}
\begin{cases}
\gamma^1 = i\,\sigma^2 \otimes \sigma^1 = i\begin{pmatrix} 0 & -i\sigma^1 \\ i\sigma^1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & \sigma^1 \\ -\sigma^1 & 0 \end{pmatrix} \\
\gamma^2 = i\,\sigma^2 \otimes \sigma^2 = i\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \otimes \sigma^2 = \begin{pmatrix} 0 & \sigma^2 \\ -\sigma^2 & 0 \end{pmatrix} \\
\gamma^3 = i\,\sigma^2 \otimes \sigma^3 = i\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \otimes \sigma^3 = \begin{pmatrix} 0 & \sigma^3 \\ -\sigma^3 & 0 \end{pmatrix}
\end{cases}$$

Check that $\left(\gamma^\mu\right)^2 = \begin{cases} 1 & \mu = 0 \\ -1 & \mu \neq 0 \end{cases}$ & $\{\gamma^\mu, \gamma^\nu\} = 0 \;\; (\mu \neq \nu)$

$$\begin{aligned}
\gamma_5 &= i\,\gamma^0\gamma^1\gamma^2\gamma^3 \\
&= i\begin{pmatrix} 0 & I \\ I & 0 \end{pmatrix}\begin{pmatrix} 0 & \sigma^1 \\ -\sigma^1 & 0 \end{pmatrix}\begin{pmatrix} 0 & \sigma^2 \\ -\sigma^2 & 0 \end{pmatrix}\begin{pmatrix} 0 & \sigma^3 \\ -\sigma^3 & 0 \end{pmatrix}
\end{aligned}$$

$$= i \begin{pmatrix} -\sigma^1 & 0 \\ 0 & \sigma^1 \end{pmatrix} \begin{pmatrix} -\sigma^2\sigma^3 & 0 \\ 0 & -\sigma^2\sigma^3 \end{pmatrix}$$

$$\left| \begin{aligned} \sigma_1\sigma_2\sigma_3 &= i\,\sigma_3\sigma_3 \\ &= i\,\sigma_3^2 \\ &= i\,(I) \end{aligned} \right.$$

$$= i \begin{pmatrix} \underline{\sigma^1\sigma^2\sigma^3} & 0 \\ 0 & -\underline{\sigma^1\sigma^2\sigma^3} \end{pmatrix}$$

$$= i \begin{pmatrix} i(I) & 0 \\ 0 & -i(I) \end{pmatrix}$$

$$\gamma_5 = \begin{pmatrix} -I & 0 \\ 0 & I \end{pmatrix}$$

γ_5 is diagonal. Is called as Weyl reps. [any reps. in which γ_5 is diagonal

We can find $S^{\mu\nu} = \frac{1}{4}[\gamma^\mu, \gamma^\nu]$ (for weyl rep.)

$$\begin{aligned} &= \frac{1}{4}(\gamma^\mu\gamma^\nu - \gamma^\nu\gamma^\mu) \\ &= \frac{1}{4}(2\,\gamma^\mu\gamma^\nu) \qquad (\mu \neq \nu) \end{aligned}$$

$$\begin{cases} \text{for } (\mu \neq \nu), & S^{\mu\nu} = \dfrac{1}{2}\gamma^\mu\gamma^\nu \\ \text{For } (\mu = \nu) & S^{\mu\nu} = 0 \end{cases}$$

$$S^{0i} = \frac{1}{2}\gamma^0\gamma^i = \frac{1}{2}\begin{pmatrix} 0 & I \\ I & 0 \end{pmatrix}\begin{pmatrix} 0 & \sigma^i \\ -\sigma^i & 0 \end{pmatrix}$$

$$S^{0i} = \frac{1}{2}\begin{pmatrix} -\sigma^i & 0 \\ 0 & \sigma^i \end{pmatrix} \qquad \longrightarrow \text{represents boost}$$

$(i, j = 1, 2, 3)$

$$\begin{aligned} S^{ij} = \frac{1}{2}\gamma^i\gamma^j &= \frac{1}{2}\begin{pmatrix} 0 & \sigma^i \\ -\sigma^i & 0 \end{pmatrix}\begin{pmatrix} 0 & \sigma^j \\ -\sigma^j & 0 \end{pmatrix} \\ &= \frac{1}{2}\begin{pmatrix} -\sigma^i\sigma^j & 0 \\ 0 & -\sigma^i\sigma^j \end{pmatrix} \\ &= \frac{1}{2}\begin{pmatrix} -i\,\epsilon_{ijk}\,\sigma^k & 0 \\ 0 & -i\,\epsilon_{ijk}\,\sigma^k \end{pmatrix} \end{aligned}$$

$$S^{ij} = \left(-\frac{1}{2}\right) i\,\epsilon_{ijk}\,\sigma_k \begin{pmatrix} +I & 0 \\ 0 & I \end{pmatrix} \qquad \downarrow \text{Represents rotations}$$

Note that S^{0i}, S^{ij} (Boost & rotation reps) are block diagonal form $\left(\begin{array}{c|c} 3 & 0 \\ \hline 0 & 3 \end{array}\right)$ in weyl basis. (i.e. all continuous lorentz transformations are block diagonal). i.e. these are reducible reps of lorentz transformations.

i.e. Top two components only transform among themselves and bottom two components of Ψ will only transform among themselves.

This shows that 4 component spinors are reducible. (if they are reducible in one basis then they are reducible in all basis, but weyl representation makes it manifest.)

Note — This is the 4 component representation of lorentz algebra

$$\Psi = \begin{bmatrix} \Psi_1 \\ \Psi_2 \\ \Psi_3 \\ \Psi_4 \end{bmatrix};$$ it will be reducible if any subset of it transforms within itself under any transformation

and it is reducible under lorentz transformations so under weyl reps.

$$\begin{pmatrix}\Psi_1\\ \Psi_2\\ 0\\ 0\end{pmatrix} \longrightarrow \begin{pmatrix}\Psi_1'\\ \Psi_2'\\ 0\\ 0\end{pmatrix}$$

$$e^{\frac{1}{2}\omega_{\mu\nu}S^{\mu\nu}} \begin{pmatrix}\Psi_1\\ \Psi_2\\ 0\\ 0\end{pmatrix} = \begin{pmatrix}\Psi_1'\\ \Psi_2'\\ 0\\ 0\end{pmatrix}$$

where we will use weyl reps. of $S^{\mu\nu}$.

$$\begin{aligned}(i\,\partial\!\!\!/ - m) = (i\,\gamma^\mu\partial_\mu - m) &= i\,\gamma^0\partial_0 + i\,\gamma^i\partial_i - m\,I\\ &= i\begin{pmatrix}0 & I\\ I & 0\end{pmatrix}\partial_0 + i\begin{pmatrix}0 & \sigma^i\\ -\sigma^i & 0\end{pmatrix}\partial_i - m\,I\\ &= \begin{pmatrix}-m & i\left(\partial_0 + \sigma^i\partial_i\right)\\ i\left(\partial_0 - \sigma^i\partial_i\right) & -m\end{pmatrix}\end{aligned}$$

The dirac eq$^{\text{n}}$ $\left(i\,\partial\!\!\!/ - m\right)\Psi = 0 \qquad \Psi = \begin{pmatrix}\Psi_L\\ \Psi_R\end{pmatrix}$

$$\Rightarrow \quad \begin{pmatrix}-m & i\left(\partial_0 + \sigma^i\partial_i\right)\\ i\left(\partial_0 - \sigma^i\partial_i\right) & -m\end{pmatrix}\begin{pmatrix}\Psi_L\\ \Psi_R\end{pmatrix}$$

where L & R label eigen value under $\gamma_5 = \begin{pmatrix}-I & 0\\ 0 & I\end{pmatrix}$

$$\gamma_5\,\Psi = \begin{pmatrix}-\Psi_L\\ \Psi_R\end{pmatrix}$$

Ψ_L & Ψ_R transform independently & don't mix under lorentz transformation.

However dirac eq$^{\text{n}}$ mixes them —

$$\left(i\,\partial\!\!\!/ - m\right)\Psi = 0$$

$$\begin{pmatrix}-m & i\left(\partial_0 + \sigma\,\partial_i\right)\\ i\left(\partial_0 - \sigma^i\partial_i\right) & -m\end{pmatrix}\begin{pmatrix}\Psi_L\\ \Psi_R\end{pmatrix} = \begin{pmatrix}0\\ 0\end{pmatrix}$$

$$\left.\begin{aligned} i\left(\partial_0 + \sigma^i\partial_i\right)\Psi_R \;-\; m\,\Psi_L = 0\\ i\left(\partial_0 - \sigma^i\partial_i\right)\Psi_L \;-\; m\,\Psi_R = 0\end{aligned}\right\} \quad \begin{array}{l}\text{even dirac eq}^{\text{n}}\\ \text{can't mix them}\\ \text{if } m = 0.\end{array}$$

For $m = 0$: $\qquad \left(\partial_0 + \sigma^i\partial_i\right)\Psi_R = 0\,; \qquad \left(\partial_0 - \sigma^i\partial_i\right)\Psi_L = 0$

Suppose we have a particle with right circular polarisation its travelling with some velocity (it is massive), so we can slow it & bring it to rest. If we move it in other way we will see left circular polarisation, so right & left spin can be transformed into each other by just boost.

So it must be that dynamics mixes Ψ_L & Ψ_R for massive particle.

For $m = 0$, we can't bring particle to rest by going in any frame or by any means, its motion can not be reversed. Therefore for $m = 0$, Ψ_L & Ψ_R shall be decoupled in e.o.m.

Sometimes people like to take a P.O.V. that mass is interaction term which flips $\Psi_L \leftrightarrow \Psi_R$. (in EOM).

Majorana representation —

Lecture 10

While it may not be visible/manifest for other representation than weyl basis, but the 4 component spinor is always reducible. It is not just visible to the eye that matrices has probability of reducibility in other basis.

$$\text{For weyl basis;} \qquad \gamma_5\,\Psi = \Psi \qquad \Psi = \underbrace{\begin{pmatrix} 0 \\ \Psi_R \end{pmatrix}}_{\text{right handed}}$$

$$\gamma_5\,\Psi = -\Psi \qquad \Psi = \underbrace{\begin{pmatrix} \Psi_L \\ 0 \end{pmatrix}}_{\text{left handed}}$$

In any basis dirac equation does not couple modes which satisfy $\gamma_5\Psi = \Psi$ & $\gamma_5\Psi = -\Psi$, (when it is massless) if it is massive Ψ, then it couples through mass term. So it is representation independent physics, physics has to be representation independent.

All representations of clifford algebra are related by unitary transformation.

Let's go to another representation where a different property is visible/manifest. Again the property hold in this reps. will also hold in all reps. but won't be manifest in those reps.

Majorana representation —

$$\gamma^0 = \sigma^1 \otimes \sigma^2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \otimes \sigma_2 = \begin{pmatrix} 0 & \sigma_2 \\ \sigma_2 & 0 \end{pmatrix}$$

$$\gamma^1 = i\left(\sigma^3 \otimes \sigma^1\right) = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} \otimes \sigma^1 = \begin{pmatrix} i\sigma^1 & 0 \\ 0 & -i\sigma^1 \end{pmatrix}$$

$$\gamma^2 = -i\,\sigma^2 \otimes \sigma^2 = -i\begin{pmatrix} 0 & -i\sigma^2 \\ i\sigma^2 & 0 \end{pmatrix} = \begin{pmatrix} 0 & -\sigma^2 \\ \sigma^2 & 0 \end{pmatrix}$$

$$\gamma^3 = -i\,I \otimes \sigma^1 = -i\begin{pmatrix} \sigma^1 & 0 \\ 0 & \sigma^1 \end{pmatrix} = \begin{pmatrix} -i\sigma^1 & 0 \\ 0 & -i\sigma^1 \end{pmatrix}$$

We can check that $\quad (\gamma^\mu)^2 = \begin{cases} -1 & \mu \neq 0 \\ +1 & \mu = 0 \end{cases}$

and all γ^μ anticommute with each other.

$$\text{i.e.} \quad \{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu} = 0 \qquad (\mu \neq \nu)$$
$$\text{as} \quad \{\sigma_j\,,\ \sigma_k\} = 2\,\delta_{jk}\,I$$

We can write another rep. in majorana basis.

$$\left.\begin{aligned} \gamma^0 &= \sigma^2 \otimes \sigma^1 \\ \gamma^1 &= i\,\sigma^3 \otimes \sigma^1 \\ \gamma^2 &= i\,\sigma^1 \otimes \sigma^1 \\ \gamma^3 &= i\,I \otimes \sigma^3 \end{aligned}\right\} \quad \begin{aligned} &\text{All } \gamma^\mu \text{ are purely imaginary} \\ &\text{satisfying} \quad (\gamma^\mu)^2 = +1 \quad \mu = 0 \\ &\qquad\qquad\qquad\quad = -1 \quad \mu \neq 0 \\ &\&\ \{\gamma^\mu, \gamma^\nu\} = 0 \quad \mu \neq \nu. \end{aligned}$$

$S^{\mu\nu} = \frac{1}{4}\,[\gamma^\mu,\, \gamma^\nu]$ are real.

Dirac eq$^{\text{n}}$: $\qquad i\left(\underline{\gamma^\mu}\,\partial_\mu - m\right)\Psi = 0$

Since γ^μ are purely imaginary, $i\gamma^\mu$ has to be purely real.

So operator $\underline{i\,(\gamma^\mu\partial_\mu - m)\text{ is real.}}$

So, solutions of dirac equation 'Ψ' can be chosen to be real. $\Big\{$Since $S^{\mu\nu} = \frac{1}{4}[\gamma^\mu, \gamma^\nu]$ has to be real ($\frac{1}{4}(\gamma^\mu\gamma^\nu - \gamma^\nu\gamma^\mu)$ is product of two purely imaginary numbers).$\Big\}$ The lorentz transformation of Ψ

$$\Psi \longrightarrow \Psi' = \left(e^{\frac{1}{2}\omega S}\right)_{ab} \Psi_b \quad \text{is always real. (If we choose } \Psi_b \text{ to be real).}$$

then, 'Ψ' remains real in any frame of reference.

So, we can choose real solutions, and those solutions which are real in majorana basis are called majorana spinors.

That means there must be some property, that I can choose in any basis, such that the property is preserved, and it reduces to being real in this basis, that property has a name called charge conjugation.

In general basises of gamma matrices, majorana spinors are not real, but they satisfy a condition:

$$\Psi_m^C = \Psi_m \swarrow \text{ majorana} \qquad\qquad C: \text{ charge conjugate}$$

(i.e. they are same as their antiparticle in any basis)

For real Ψ, C is replaced by just '$*$' (i.e. $\Psi^* = \Psi \Rightarrow \Psi = \text{real}$)

Q) Majorana spinors are real in majorana basis 'Ψ_m', suppose we want to find 'Ψ' in some other basis, (weyl or dirac or any).. also we know they satisfy $\Psi_m^C = \Psi_m$ in any basis.

(or) Suppose we want to see the condition of being majorana spinors when we have 'Ψ' of weyl basis reps.

Ans) Since all representations of gamma matrices are equivalent this means, $\exists$ a similarity transformation from one gamma matrices rep to another.

$$\text{then} \quad \boxed{S\,\Psi_m = (\Psi)_{\text{in other basis}}} \qquad S: \text{ Similarity transformation.}$$

With that we can derive cond$^\text{n}$ of charge conjugation in another choice.

$$\Psi_m^T = \Psi_m$$
$$\left(S^{-1}\,\Psi_{\text{other basis}}\right)^T = S^{-1}\,\Psi_{\text{other basis}}$$
$$\Psi^T\left(S^{-1}\right)^T = S^{-1}\,\Psi$$

Lecture (10)

Note: When we talk about 'γ' matrices, we say sometimes weyl representation and majorana reps, & dirac rep. They all are equivalent, which means, they are similarity transforms of each other. So it is better to call them basis rather than representation, because representation of clifford algebra (Γ^μ) {which satisfies $[\Gamma^\mu, \Gamma^\nu] = 2\eta^{\mu\nu}$} are dimension dependent. We can have 2×2, 8×8 reps. of γ^μ as well, so same reps. in different basis is what we studied (dirac basis, weyl basis & majorana basis).

— The condition that γ^0 is hermitian & γ^i is anti-hermitian is also basis dependent. (Only thing which is basis independent is $\left(\gamma^0\right)^2 = +1$, $\left(\gamma^i\right)^2 = -1$ & $\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}$.) It turns out that γ^0 & γ^i remains hermitian & antihermitian if similarity transformations are unitary. So we always look for unitary similarity transforms of γ matrices so that γ^0 is always hermitian & γ^i is always antihermitian.

— Comment on Majorana reps (Remember that majorana basis is basis where γ^μ are imaginary therefore spinors can be real, does not mean they have to be real, but if we choose them real then they are called majorana spinors.) $(\Psi_a^* = \Psi_a)$

Majorana spinors have a property that they are "real" in majorana basis, we will have analogue of that property in other basis.

(Majorana spinors physically mean Particle = antiparticle) & in majoran basis they have property that $\Psi^* = \Psi$ (i.e. Ψ is real) and in other basis they have property like $\Psi^* = (\ )\,\Psi \hookrightarrow 4$ matrix.

$\Rightarrow \quad \Psi, \Psi^*$ are "not" independent fields. It means physically particle = antiparticle. Since particle & antiparticle have opposite charge this implies majorana spinor has no-charge (neutral).

There is another way to see it, charge is related to a symmetry under change of phase of Ψ.

$$\Psi \longrightarrow \Psi' = e^{i\alpha}\,\Psi$$
$$\Psi^* \longrightarrow \left(\Psi'\right)^* = e^{-i\alpha}\,\Psi^*$$
$$\text{So,} \quad (\Psi^*)' = \Psi' \,; \qquad e^{-i\alpha}\,\underline{\Psi^*} = (\ )\,e^{i\alpha}\,\Psi$$

But this eq$^{\text{n}}$ can not be correct. as $\underline{\Psi^* = (\ )\Psi}$,

$$\Rightarrow \qquad e^{-i\alpha}\,(\ )\,\Psi \neq (\ )\,e^{i\alpha}\,\Psi$$

$\Rightarrow$ symmetry under such phase don't exist $\Rightarrow$ charge does not exist.

So, only neutrinos have possibility of being majorana spinors.

<u>Mass term</u> — mass term for a majorana spinor = ?

For any spinor, mass term in dirac lagrangian is $m\overline{\Psi}\Psi$.

$$\begin{aligned}
\mathcal{L}_{\text{mass}} &= m\,\overline{\Psi}\,\Psi = m\,\Psi_a^\dagger\,\gamma^0_{ab}\,\Psi_b \\
\mathcal{L}_{\text{mass}}\big)_{\text{majorana}} &= m\,\Psi_a^\dagger\,\gamma^0_{ab}\,\Psi_b \qquad \left(\Psi_a^\dagger = \Psi_a \ \ \text{majorana basis}\right) \\
&= m\,\Psi_a\left(\gamma^0\right)_{ab}\Psi_b \qquad \left(\begin{array}{l} \gamma^0 = \sigma_2 \otimes \sigma_1 \\ \quad = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \otimes \sigma^1 = \begin{pmatrix} 0 & -i\sigma^1 \\ i\sigma^1 & 0 \end{pmatrix} \\ \gamma^0 = \text{anti symmetric} \end{array}\right) \\
&= m\,(\Psi_1\ \Psi_2\ \Psi_3\ \Psi_4)\left(\begin{array}{cc|cc} \multicolumn{2}{c|}{\multirow{2}{*}{0}} & 0 & -i \\ & & -i & 0 \\ \hline 0 & i & \multicolumn{2}{c}{\multirow{2}{*}{0}} \\ i & 0 & & \end{array}\right)\begin{pmatrix}\Psi_1\\ \Psi_2\\ \Psi_3\\ \Psi_4\end{pmatrix} \\
&= m\,(\Psi_1\ \Psi_2\ \Psi_3\ \Psi_4)\begin{pmatrix}-i\,\Psi_4\\ -i\,\Psi_3\\ i\,\Psi_2\\ i\,\Psi_1\end{pmatrix} = -i\,m\,(\Psi_1\Psi_4 + \Psi_2\Psi_3 - \Psi_3\Psi_2 - \Psi_4\Psi_1)
\end{aligned}$$

$$\mathcal{L}_{\text{mass}} = \underline{\ 0\ } \qquad (?) \quad (\times).$$

<u>Iff</u> $\Psi_a\Psi_b = \Psi_b\Psi_a$ ie. $(\Psi_1\Psi_2 = \Psi_2\Psi_1)$ (later we will see that in order to have $\mathcal{L}$ Ψ_a's should be grassmann numbers.)

$\mathcal{L}_{\text{mass}}\big)_{\text{majorana}} = 0$ is not correct conclusion.

We can say that probably such particles are massless.

<u>Kinetic term</u> —

$$\begin{aligned}
\mathcal{L}_{\text{kinetic}} &= i\,\overline{\Psi}\,\gamma^\mu\,\partial_\mu\Psi_b \\
&= i\,\Psi^\dagger\left(\gamma^0\gamma^\mu\right)\partial\,\Psi
\end{aligned}$$

$$\mathcal{L}_{\text{kinetic}}\Big)_{\text{majorana}} = i\,\Psi_a^a\,(\gamma^0\gamma^\mu)_{ac}\,\partial^\mu{}_2\Psi_c^2 \qquad \begin{pmatrix}\Psi_a^\dagger = \Psi_a \\ \text{for majorana}\end{pmatrix}$$
$$= \frac{i}{2}\,\partial_\mu\left(\Psi_a\,(\gamma^0\gamma^\mu)_{ab}\,\Psi_b\right)$$

$\Rightarrow$ kinetic term is total derivative. (True for every basis)

$$\Rightarrow \quad \text{Action} = \left(\frac{i}{2}\,\Psi_a\,(\gamma^0\gamma^\mu)_{ab}\,\Psi_b\right)_{x_1}^{x_2}$$

at boundary $\Psi_a(x_1) = \Psi_b(x_2) = 0$

$\Rightarrow$ No lagrangian. This is true for majorana spinors in any basis.

This is worrying as $\mathcal{L} = 0$ in all basis.

Solution — Assume classical fields $\Psi(x)$ satisfies

$$\Psi_a(x)\,\Psi_b(x) = -\Psi_b(x)\,\Psi_a(x) \qquad \text{(for all spinors)}$$

Grassmann numbers — Fermion fields should be valued in grassmann numbers. (classically)

When we quantize we should use anti-commutators. i.e.

$$\{\Pi_a\,(t,\vec{x})\;,\;\Psi_a\,(t,\vec{y})\} = i\hbar\,\delta^3\,(\vec{x}-\vec{y}) \qquad \Pi_a = i\,\Psi_a^\dagger$$
$$\Pi_a = \Psi_a^\dagger\,(t,\vec{x}) \qquad \text{For majorana basis } \Psi_a^\dagger = \Psi_a\Big\}\; i\,\{\Psi(x),\Psi(y)\} = i\hbar\,\delta^3(x-y)$$
$$\Rightarrow \quad \Psi_a^\dagger\,\Psi_a + \Psi_a\,\Psi_a^\dagger = 0 \qquad (?$$

In Quantum mechanics $[x,p] = i\hbar$; in classical limit we use $\hbar = 0$ thus $xp = px$ (in classical limit). But in classical limit ($\hbar = 0$) we require that the fields anticommute.

We require $\Psi_a\,\Psi_b = -\Psi_b\,\Psi_a$

So we use anticommutators for fermionic fields quantisation.

$$\{\Psi_a(x)\,,\;\Pi_b(y)\} = i\hbar\,\delta^4(x-y)\,\delta_{ab}$$
$$\Rightarrow \quad \left\{\Psi_a(x)\,,\;\Psi_b^\dagger(y)\right\} = \hbar\,\delta_{a,b}\,\delta^3(x-y)$$

in classical limit we set $\hbar = 0$ at $x = y \Rightarrow \underline{\Psi_a(x)\,\Psi_b^\dagger(x) = -\Psi_b^\dagger(x)\,\Psi_a(x)}$ $\downarrow$ Classically.

Solutions of Dirac eq$^{\text{n}}$ —

Remember that for scalar fields we found solution as superposition of all sol$^{\text{n}}$ for fixed momenta $\vec{k}$.

$$\phi(t,x) = \sum_k a\,\underbrace{e^{-ik\cdot x}}\, + \left(a^\dagger\cdots\right) \text{ free particle sol}^{\text{n}}. \qquad \hookrightarrow \text{Also called mode expansion of field.}$$

it satisfies K.G. eq$^{\text{n}}$ if $k^2 = m^2$.

For fermions we need to find free particle solution $\Psi_a(x)$, multiply them with oscillators & quantize the oscillators by anticommutation relation.

Now diff. is that since fermion has index, $\Psi_a(x)$, the free particle sol$^{\text{n}}$ (for fixed momentum $\vec{k}$) are not just $e^{-ik\cdot x}$.

$$\Psi_a \longrightarrow \sum_k \underbrace{u_a(k)}_{\downarrow}\,e^{-ik\cdot x} \qquad \text{assuming } k^2 = m^2$$

as field satisfying dirac eq$^{\text{n}}$ also satisfies K.G. eq$^{\text{n}}$. $\hookleftarrow$ prove it!

mode expansion. Spinor coefficient (to carry label; & this coefficient u_a shall depend on k) (as in end we will do summation over k)

$$\Rightarrow \quad (i\,\gamma^\mu \partial_\mu - m)\,\Psi_a = 0$$
$$(i\,\gamma^\mu \partial_\mu - m)\,u_a(k)\,e^{-ik\cdot x} = 0$$
$$(i(-i)\,\gamma^\mu\,k_\mu - m)_{ab}\,u_b(k) = 0$$

$$\left|\begin{aligned} k\cdot x &= k_\mu\,x^\mu \\ \partial_\mu\,e^{-ik\cdot x} &= \partial_\mu\,e^{-ik_\mu x^\mu} \\ &= \partial_\mu\,e^{-i\left(k_0x^0+k_1x^1+\cdots\right)} \\ &= -i\,k_\mu\,e^{-ik\cdot x} \end{aligned}\right.$$

We can solve it by going into the rest frame

$$k_\mu = (m,0,0,0)$$

above eq$^{\text{n}}$ becomes —

$$\left(\gamma^0 k_0 + 0 + 0 + 0 - m\right)_{ab} u_b(k) = 0$$
$$m\left(\gamma^0 - I\right) u_b(k) = 0$$
$$\left(I - \gamma^0\right) u_b(k) = 0$$

So we see that free particle sol$^{\text{n}}$ of Dirac eq$^{\text{n}}$ depend upon basis of gamma matrices.

Weyl basis —

$$\gamma^0 = \begin{pmatrix} 0 & I \\ I & 0 \end{pmatrix}$$

$$I - \gamma^0 = \begin{pmatrix} I & -I \\ -I & I \end{pmatrix} \quad \text{i.e.} \quad \underset{4\times4}{\begin{pmatrix} I & 0 \\ 0 & I \end{pmatrix}} \qquad \begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & -1 \\ -1 & 0 & 1 & 0 \\ 0 & -1 & 0 & 1 \end{bmatrix}$$

$$\left(I - \gamma^0\right) u_a(k) = 0 \;=\; 0\left(I - \gamma^0\right) \qquad \downarrow \text{ seems like eigen value eq}^{\text{n}}.$$

→ Implies that $u_a(k)$ is zero eigen vector of $\left(I - \gamma^0\right)$.

$\left(I - \gamma^0\right)$ has two eigenvector for eigenvalue zero, which are

$$u = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \end{bmatrix} \cdots \qquad \text{(true for rest frame of particle)}$$

For in general/any frame —

'u' is linear combination of two zero eigen kets.

$$u = a\begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix} + b\begin{pmatrix} 0 \\ 1 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} a \\ b \\ a \\ b \end{pmatrix} = \begin{pmatrix} \chi \\ \chi \end{pmatrix}$$

So, $\quad u =$ two — (two component spinor) =

where $\chi = \begin{pmatrix} a \\ b \end{pmatrix}$ a two component spinor.

Any general 4 component spinor can be written as $\begin{pmatrix}\chi \\ \chi'\end{pmatrix}$ i.e. two different 2-component spinors.

But for free particle we see that those two component spinors are same.

$$u = \begin{pmatrix}\chi \\ \chi\end{pmatrix}$$

We normalize $\quad u(k) = \sqrt{m}\begin{pmatrix}\chi \\ \chi\end{pmatrix} \qquad$ where $\chi^\dagger \chi = 1 \quad \downarrow \; |a|^2 + |b|^2 = 1$

How (?) $\qquad$ Normalization of χ

We can go to general lorentz frame.

$k^\mu = (E,\, 0,\, 0,\, k_3) \qquad$ Any arbitrary $\vec{k}$ can be rotated to just have momentum in z-direction.

Then $\Psi_a(x) = u_a(k)\, e^{-ik\cdot x}$ will satisfy K.G. eq$^\text{n}$ as well $\Rightarrow k^2 = m^2,\; E^2 - k_3^2 = m^2$

From same procedure $\left(i\partial\!\!\!/ - m\right)\Psi = 0$ results in eigen value eq$^\text{n}$, which gives us

$$u^{(1)}(k) = \begin{pmatrix}\sqrt{\sigma^\mu k_\mu}\,\chi^{(1)} \\ \sqrt{\bar{\sigma}^\mu k_\mu}\,\chi^{(1)}\end{pmatrix} \qquad u^{(2)}(k) = \begin{pmatrix}\sqrt{\sigma^\mu k_\mu}\,\chi^{(2)} \\ \sqrt{\bar{\sigma}^\mu k_\mu}\,\chi^{(2)}\end{pmatrix} \qquad \left|\begin{array}{l}\chi^{(1)} = \begin{pmatrix}1 \\ 0\end{pmatrix} \\ \chi^{(2)} = \begin{pmatrix}0 \\ 1\end{pmatrix}\end{array}\right.$$

where σ^μ & $\bar{\sigma}^\mu$ can help us expressing dirac eq$^\text{n}$.

$$\begin{aligned} \sigma^\mu &= (1,\, \vec{\sigma}) && \sigma^0 = 1,\; \sigma^i = \sigma^i \\ \bar{\sigma}^\mu &= (1,\, -\vec{\sigma}) && \bar{\sigma}^0 = 1,\; \bar{\sigma}^i = -\sigma^i \\ \{\sigma^\mu, \sigma^\nu\} &\neq 2\eta^{\mu\nu} && \text{does not satisfy clifford algebra.} \\ \text{But} \quad \{\sigma^\mu,\, \bar{\sigma}^\nu\} &= 2\eta^{\mu\nu} && \text{satisfies clifford algebra.} \end{aligned}$$

where,

$$\left.\begin{aligned} \sqrt{\sigma^\mu k_\mu} &= \sqrt{E + k_3}\left(\frac{1-\sigma^3}{2}\right) + \sqrt{E - k_3}\left(\frac{1+\sigma^3}{2}\right) \\ \& \quad \sqrt{\bar{\sigma}^\mu k_\mu} &= \sqrt{E + k_3}\left(\frac{1+\sigma^3}{2}\right) + \sqrt{E - k_3}\left(\frac{1-\sigma^3}{2}\right) \end{aligned}\right\} \text{ left as Exercise.}$$

where

$$\sigma^3 \chi^{(1)} = \begin{pmatrix}1 & 0 \\ 0 & -1\end{pmatrix}\begin{pmatrix}1 \\ 0\end{pmatrix} = \begin{pmatrix}1 \\ 0\end{pmatrix} \;\Rightarrow\; \left(1 - \sigma^3\right)\chi^{(1)} = 0 \;\Rightarrow\; \frac{\left(1+\sigma^3\right)}{2}\chi^{(1)} = \chi^{(1)}$$

$$\sigma^3 \chi^{(2)} = \begin{pmatrix}1 & 0 \\ 0 & -1\end{pmatrix}\begin{pmatrix}0 \\ 1\end{pmatrix} = -\begin{pmatrix}0 \\ 1\end{pmatrix} = -\chi^{(2)}$$

$$\Rightarrow \frac{\left(1-\sigma^3\right)}{2}\chi^{(2)} = \chi^{(2)} \quad \& \quad \left(\frac{1+\sigma^3}{2}\right)\chi^{(2)} = 0$$

So,

$$u^{(1)}(k) = \begin{bmatrix}\sqrt{E - k_3} \\ 0 \\ \sqrt{E + k_3} \\ 0\end{bmatrix} \qquad \& \qquad u^{(2)}(k) = \begin{bmatrix}0 \\ \sqrt{E + k_3} \\ 0 \\ \sqrt{E - k_3}\end{bmatrix}$$

So, there are 2 solutions to dirac eq$^\text{n}$ $\left(i\partial\!\!\!/ - m\right)\Psi = 0$

When we make a mode expansion of Ψ to quantize it, we should add $u^{(1)}(k)\, e^{-ik\cdot x}\, a^\dagger_{(1)}$ & $u^{(2)}\, e^{-ik\cdot x}\, a^\dagger_{(2)}$, where $a^\dagger_{(1)}$ & $a^\dagger_{(2)}$ shall create spin up & spin down states, of same particle.

So, we should think label (1)&(2) in $u^{(1)}$ & $u^{(2)}$ as some spin index $u^{(s)}(k)$.

$$u^{(s)}(k) = \text{Free particle sol}^{\text{n}} \text{ for momentum } k \text{ \& spin } s$$

$$s = +1/2 \;\text{ or }\; -1/2 \qquad\qquad \text{here } s \text{ represents spin state } m_s.$$

When we learned scalar fields we noticed that there was no spin degeneracy in mode expansion, that's why we said that they must be scalar fields.

Here due to spin degeneracy, we can be sure that these are not scalar particles.

We define weyl Basis ↓

$$\begin{aligned}
\overline{u}^{(1)}(k) = \left(u^{(1)}\right)^{\dagger}(k)\cdot\gamma^0 &= \left(u^{(1)}\right)^{T}(k)\,\gamma^0 \qquad \left(\gamma^0 = \sigma_1 \otimes I\right)\\
&= \begin{bmatrix}\sqrt{E-k_3} & 0 & \sqrt{E+k_3} & 0\end{bmatrix}\begin{bmatrix}0&0&1&0\\0&0&0&1\\1&0&0&0\\0&1&0&0\end{bmatrix}\\
&= \begin{pmatrix}\sqrt{E+k_3} & 0 & \sqrt{E-k_3} & 0\end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
\overline{u}^{(2)}(k) = u^{\dagger(2)}(k)\,\gamma^0 = u^{T(2)}(k)\,\gamma^0 &= \begin{pmatrix}0 & \sqrt{E+k_3} & 0 & \sqrt{E-k_3}\end{pmatrix}\begin{bmatrix}0&0&1&0\\0&0&0&1\\1&0&0&0\\0&1&0&0\end{bmatrix}\\
&= \begin{pmatrix}0 & \sqrt{E-k_3} & 0 & \sqrt{E+k_3}\end{pmatrix}
\end{aligned}$$

Note that,

1) Just swap the nonzero numbers to get the $\overline{u}$ from u

2)

$$\overline{u}^{(r)}_{\vec{k}}\, u^{(s)}_{\vec{k}} = 2m\,\delta^{rs} \;\Rightarrow\; 2\sqrt{(E^2-k^2)}\,\delta^{rs} \qquad \Rightarrow\; \overline{u}^{(1)}\,u^{(2)} = 2m\,\delta^{12} = 0$$

$$(r,s = 1,2)$$

Antiparticle sol$^{\text{n}}$ —

We can get antiparticle solution from complex conjugate of dirac eq$^{\text{n}}$.

$$\begin{gathered}
\left(i\partial\!\!\!/ + m\right)\Psi = 0 \qquad\qquad \Psi = V(k)\,e^{-ik\cdot x}\\
(i\,\gamma^{\mu}\partial_{\mu} + m)\,V(k)\,e^{-ik\cdot x} = 0\\
\Rightarrow \qquad (\gamma^{\mu}k_{\mu} + m)\,V(k) = 0\\
\hookrightarrow \text{ eigen value eq}^{\text{n}} \text{ gives again two eigen vectors}
\end{gathered}$$

$$V^{(1)}(k)\;\&\;V^{(2)}(k) \;\longrightarrow\; \text{free antiparticle sol}^{\text{n}}.$$

We can find $V^{(r)}(k)$ & $\overline{V}^{(r)}(k)$, they will satisfy

$$\overline{V}^{r}(k)\,V^{(s)}(k) = -2m\,\delta^{rs} \quad\Rightarrow\quad \overline{V}^{(1)}\,V^{(2)} = -2m\,\delta^{12} = 0$$

To summarize we found 4 free particle solution, (particle spin up, particle spin down, antiparticle spin up, antiparticle spin down)

Identity —

$$\sum_{s=1}^{2} u_a^s \, \overline{u}_b^s = (\not{k} + m)_{ab}$$
$$\sum_{s=1}^{2} V_a^s \, \overline{V}_b^s = (\not{k} - m)_{ab}$$

Mode Expansion —

$$\Psi_a(x) = \int \frac{d^3k}{(2\pi)^3} \frac{1}{\sqrt{2\omega_k}} \sum_{s=1}^{2} \left[a_{\vec{k}}^{(s)} \, u_{(k)}^{(s)} \, e^{-ik\cdot x} + b_{\vec{k}}^{\dagger} \, V_{(k)}^{(s)} \, e^{ik\cdot x} \right]$$

$\hookrightarrow$ creates antiparticle. $\nearrow$ Annihilates particle

Both charges charge by same amount;

$$\overline{\Psi}_a(x) = \int \frac{d^3k'}{(2\pi)^3} \frac{1}{\sqrt{2\omega_{k'}}} \sum_{s=1}^{2} \left(a_{\vec{k}'}^{\dagger} \, \overline{u}_{k'}^{(s)} \, e^{ik\cdot x} + b_{\vec{k}'} \, \overline{V}^{(s)}(k) \, e^{-ik\cdot x} \right)$$

u — spin of particle; v — " " Antiparticle

To quantise, we impose anti-commutation relation.

$$\{\Psi_a\,(t,\vec{x})\,,\ \Pi_b\,(t,\vec{y})\} = i\,\delta^3\,(\vec{x}-\vec{y}) \qquad ; \quad \Pi_a = \frac{\partial \mathcal{L}}{\partial \dot{\Psi}_a} = \overline{\Psi}\,(i\gamma^0) = i\,\Psi^{\dagger}\,(\gamma^0)^2 = i\,\Psi^{\dagger}$$

and

$$\left\{\Psi_a\,(t,\vec{x})\,,\ \Psi_b^{\dagger}\,(t,\vec{y})\right\} = \delta^3\,(\vec{x}-\vec{y})\ \delta_{ab}$$

if we impose this, we find.

$$\left\{a_{\vec{k}}^{(s)},\ a_{\vec{k}'}^{\dagger(s')}\right\} = \delta^{ss'}\,(2\pi)^3\ \delta^3\left(\vec{k}-\vec{k}'\right) = \left\{b_{\vec{k}}^{s},\ b_{\vec{k}'}^{s'}\right\}$$

$$\Rightarrow \qquad \{a,a\} = \{b,b\} = \left\{a^{\dagger},a^{\dagger}\right\} = \left\{b^{\dagger},b^{\dagger}\right\}$$
$$= \{a,b\} = \left\{a^{\dagger},b\right\} = \left\{a,b^{\dagger}\right\} = 0$$

Hamiltonian —

$$H = \int d^3x \left(\Pi_a\,\dot{\Psi}_a - \mathcal{L}\right) \qquad\qquad \mathcal{L} = \overline{\Psi}\,(i\not{\partial} - m)\,\Psi$$
$$= \int d^3x \left(i\,\Psi_a^{\dagger}\,\dot{\Psi}_a - i\,\overline{\Psi}\,\gamma^{\mu}\,\partial_{\mu}\Psi + m\,\overline{\Psi}\Psi\right)$$

$\boxed{\text{Is } \gamma^0 = I\ ?}$

$$= \int d^3x \left(i\,\Psi_a^{\dagger}\dot{\Psi}_a - \cancel{i\,\overline{\Psi}_a\gamma^0\,(\partial_0\Psi)} - i\,\overline{\Psi}_a\,\gamma^i\partial_i\Psi + m\,\overline{\Psi}\Psi\right)$$

$$H = \int d^3x\ \left(-i\,\overline{\Psi}_a\,\gamma^i\partial_i\Psi + m\,\overline{\Psi}\Psi\right) \qquad \begin{cases} \text{using } \partial_0\Psi = \dot{\Psi} \\ \&\ \ (\gamma^0)^2 = +I \end{cases}$$

Can't as 'I' commutes with all operators/matrices. But we need γ^0 such that it satisfies clifford algebra $\left\{\gamma^0,\gamma^i\right\} = 2\eta^{0i} = 0$ must anti-commute

$$= \int \frac{d^3k}{(2\pi)^3}\ \omega_k \sum_{s=1}^{2} \left(a_k^{\dagger(s)}\,a_k^{(s)}\ +\ b_k^{\dagger(s)}\,b_k^{(s)}\right)\ -\ \underbrace{\sum_{k=0}^{\infty} \text{const.}}_{\nearrow\infty}$$

$\downarrow$ fermionic creation operators

Normalisation:

$$\left|\vec{k}, s\right\rangle = \sqrt{2\omega_k}\ \ a_k^{\dagger(s)}\,|0\rangle$$

$\uparrow$ one particle state with spin s & momentum $\vec{k}$ (here, s means m_s). $\quad \langle k', r| = \sqrt{2\omega_{k'}}\ \ \langle 0|\, a_{k'}^{(r)}$

$$\begin{aligned}
\left\langle \vec{k}', r \,\middle|\, \vec{k}, s\right\rangle &= \sqrt{2\omega_{\vec{k}}}\,\sqrt{2\omega_{\vec{k}'}}\ \left\langle 0\middle|\, a_{k'}^{(r)}\, a_k^{\dagger(s)}\,\middle|0\right\rangle \\
&= 2\sqrt{\omega_k\omega_{k'}}\ \left\langle 0\middle|\left\{a_{k'}^{(r)},\, a_k^{\dagger(s)}\right\} - a_k^{\dagger(s)}\, a_{k'}^{(r)}\,\middle|0\right\rangle \\
&= 2\sqrt{\omega_k\omega_{k'}}\ \left(2\pi\right)^3 \delta^{rs}\,\delta^3\left(k' - k\right)\ -\ 0 \\
&= 2\sqrt{\omega_k\omega_{k'}}\ \left(2\pi\right)^3\ \delta^3\left(\vec{k}' - \vec{k}\right)\ \delta^{rs}
\end{aligned}$$

Dirac propagator —

$$\begin{aligned}
T\left(\Psi(x)\,\overline{\Psi}(y)\right) &= \Psi(x)\,\overline{\Psi}(y) \qquad \text{if}\quad x^0 > y^0 \\
&= -\,\overline{\Psi}(y)\,\Psi(x) \qquad \text{if}\quad y^0 > x
\end{aligned}$$

as $\Psi(x)$ & $\overline{\Psi}(y)$ are valued in Grassmann numbers.

We would like to find

$$\langle 0|\, T\left(\Psi_a(x)\,\overline{\Psi}_b(y)\right)|0\rangle\ = ?$$

Just like scalar field :

$$\langle 0|\, T\left(\phi(x)\,\phi(y)\right)|0\rangle = D_F(x-y)$$

$D_F(x-y)$ is Greens fun. for operator $\left(-\left(\partial^2 + m^2\right)\right)$.

$$-\left(\partial^2 + m^2\right) D_F(x-y) = i\,\delta^4(x-y)$$

Similarly, $\;\Rightarrow$

$$\left(i\partial\!\!\!/ - m\right)\hat{S}_F(x-y) = i\,\delta^4(x-y)$$

$$\left(i\partial\!\!\!/ + m\right)\left(i\partial\!\!\!/ - m\right) S_F(x-y) = i\left(i\partial\!\!\!/ + m\right) i\,\delta^4(x-y)$$

$$\underbrace{\left(-\gamma^\mu\partial_\mu\,\gamma^\nu\partial_\nu - m^2\right)}_{\Downarrow\, -(\partial^2+m^2)\ \text{How?}} S_F(x-y) = \underline{i\left(i\partial\!\!\!/ + m\right)\delta^4(x-y)} \qquad \begin{aligned} &= -\left(i\partial\!\!\!/ + m\right) \\ &\left(\partial^2 + m^2\right) D_F(x-y)\end{aligned}$$

$$\Rightarrow \qquad -\left(\partial^2 + m^2\right) S_F(x-y) = -\left(i\partial\!\!\!/ + m\right)\left(\partial^2 + m^2\right) D_F(x-y)$$

$$\boxed{S_F(x-y)\ =\ \left(i\partial\!\!\!/ + m\right) D_F(x-y)}$$

We have solved fermion propagator in terms of scalar propagator.

$$\begin{aligned}
S_F(x-y) &= \left(i\partial\!\!\!/ + m\right) D_F(x-y) \\
&= \lim_{\epsilon\to 0}\ \left(i\partial\!\!\!/ + m\right)\int \frac{d^4k}{(2\pi)^4}\,\frac{i\,e^{-ik\cdot(x-y)}}{k^2 - m^2 + i\epsilon} \\
&= \lim_{\epsilon\to 0}\ \int \frac{d^4k}{(2\pi)^4}\,\frac{i\left(i(-i)\,\gamma^\mu k_\mu + m\right)e^{-ik\cdot(x-y)}}{\left(k^2 - m^2 + i\epsilon\right)}
\end{aligned}$$

$$\boxed{S_F(x-y)\ =\ \lim_{\epsilon\to 0}\int\frac{d^4k}{(2\pi)^4}\boxed{\frac{i\left(k\!\!\!/ + m\right)}{\left(k^2 - m^2 + i\epsilon\right)}}\,e^{-ik\cdot(x-y)}} \quad \longleftarrow \quad \begin{aligned}&\text{position space}\\ &\text{rep of } S_F(x-y)\end{aligned}$$

$\downarrow$ momentum space rep$^{\text{n}}$ of $S_F(x-y)$

$$S_F(k) = \frac{i\left(k\!\!\!/ + m\right)}{k^2 - m^2} = \frac{i\left(k\!\!\!/ + m\right)}{\left(k\!\!\!/ - m\right)\left(k\!\!\!/ + m\right)} = \frac{i}{k\!\!\!/ - m}$$

We have done every thing in detail so far. From here we have to change gears & things will become sketchy.

What does $\langle\Omega|\, T\left(\varphi(x_1)\, \varphi(x_2) \cdots \varphi(x_n)\right) |\Omega\rangle$ has to do with measurable quantities?

<u>S- matrix</u> - It is not a matrix but function of incoming and outgoing momenta.

$$\text{i.e.} \qquad S_{(\vec{k}_1 \vec{k}_2 \cdots \vec{k}_n \,|\, \vec{p}_1 \vec{p}_2)} = \underset{\text{out}}{\left\langle \vec{k}_1\, \vec{k}_2 \cdots \vec{k}_n \right|} |\vec{p}_1\, \vec{p}_2\rangle_{\text{in}} \qquad \begin{array}{l} \to \text{state at } t = -\infty \\ \text{Interacting states } \swarrow \end{array}$$

$$= \langle f | i \rangle \qquad \hookrightarrow \text{ state at } t = +\infty.$$

$\downarrow$ Probability amplitude for $\langle f|i\rangle$

Assume we have state $|\vec{p}_1\, \vec{p}_2\rangle_{\text{in}}$ of interacting theory at $t = -\infty$.

$$\left(\text{For free theory} \quad |\vec{p}_1\, \vec{p}_2\rangle_0 = a^\dagger_{p_1}\, a^\dagger_{p_2}\, |0\rangle \right).$$

We can produce $|\vec{p}_1\, \vec{p}_2\rangle_{\text{in}}$ from $(\vec{p}_1\, \vec{p}_2)_0$ if we assume that particles are so far apart that they don't interact in far past. (Assumption not valid in CFT, so CFT does not have S-matrix)

$$|\vec{p}_1\, \vec{p}_2\rangle_{\text{in}} = \lim_{T\to\infty(1-i\epsilon)} e^{-iHT}\, |\vec{p}_1\, \vec{p}_2\rangle_0$$

$$\underset{\text{out}}{\left\langle \vec{k}_1\, \vec{k}_2 \cdots \vec{k}_n \right|} = \lim_{T\to\infty(1-i\epsilon)} \left\langle \vec{k}_1\, \vec{k}_2 \cdots \vec{k}_n \right| e^{-iHT}$$

$$\boxed{S_{(\vec{k}_1 \vec{k}_2 \cdots \vec{k}_n | \vec{p}_1 \vec{p}_2)} = \underset{0}{\left\langle \vec{k}_1\, \vec{k}_2 \cdots \vec{k}_n \right|} T\left(e^{-i\int_{-T}^{T} H_I(t')\, dt'} \right) |\vec{p}_1\, \vec{p}_2\rangle_0}$$

We have not devided by $\langle k_1 k_2 \cdots k_n | \vec{p}_1 \vec{p}_2\rangle_0$; instead we will drop disconnected diagrams.

We can apply same feynman diagrams but here we have two particle state $|\vec{p}_1, \vec{p}_2\rangle_0$, in place of $|0\rangle$ (i.e. free vacuum)

$$\text{But} \quad |\vec{p}_1, \vec{p}_2\rangle_0 = N\, a^\dagger_{p_1}\, a^\dagger_{p_2}\, |0\rangle \qquad \downarrow \text{ Normalization}$$

$$= \sqrt{2\omega_{p_1}}\, \sqrt{2\omega_{p_2}}\; a^\dagger_{p_1}\, a^\dagger_{p_2}\, |0\rangle$$

<u>For lowest order</u> : (For 2 particles in final state).

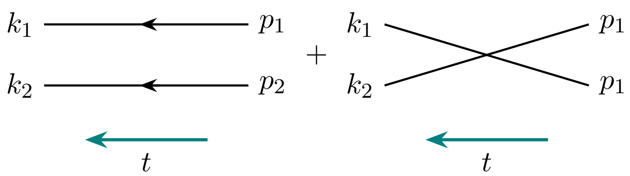
$$S_{(\vec{k}_1 \vec{k}_2, (\vec{p}_1 \vec{p}_2))} = \underset{0}{\left\langle \vec{k}_1\, \vec{k}_2 \right|} \vec{p}_1\, \vec{p}_2 \Big\rangle$$

$$= \sqrt{2\omega_{p_1}\, 2\omega_{p_2}\, 2\omega_{k_1}\, 2\omega_{k_2}}\; \left\langle 0 \right| a_{k_1}\, a_{k_2}\, a^\dagger_{p_1}\, a^\dagger_{p_2} \left| 0 \right\rangle$$

$$= \sqrt{2\omega_{p_1}\, 2\omega_{p_2}\, 2\omega_{k_1}\, 2\omega_{k_2}}\; \left\langle 0 \right| \underline{a_{k_1}\, a^\dagger_{p_1}}\; \underline{a_{k_2}\, a^\dagger_{p_2}} \left| 0 \right\rangle$$

$$= (2\pi)^3\, (2\pi)^3\, \sqrt{2\omega_{p_1}\, 2\omega_{p_2}\, 2\omega_{k_1}\, 2\omega_{k_2}}$$

$$\left(\delta^3\left(\vec{p}_1 - \vec{k}_1\right) \delta^3\left(\vec{p}_2 - \vec{k}_2\right) + \delta^3\left(\vec{p}_1 - \vec{k}_2\right) \delta^3\left(\vec{p}_2 - \vec{k}_1\right) \right)$$



$\to I \equiv$ No scattering

$$S = I + i\,T \qquad \hookrightarrow \text{ scattering (everything else)}$$

<u>(For First order)</u>

$$S_{(\vec{k}_1\vec{k}_2|\vec{p}_1\vec{p}_2)} = \underset{0}{\left\langle \vec{k}_1\,\vec{k}_2 \right|} - i\int_{-T}^{T} H_I(t')\,dt' \;\; |\vec{p}_1\,\vec{p}_2\rangle$$

$$\text{For} \qquad H_I(t') = \lambda\,\frac{\varphi^4(t')}{4!}$$

$$= \underset{0}{\left\langle \vec{k}_1\,\vec{k}_2 \right|} T\left(-\frac{i\lambda}{4!} \int_{-T}^{T} d^4x'\;\varphi^4(x') \right) |\vec{p}_1\,\vec{p}_2\rangle_0$$

using

$$T\left(\varphi^4(y)\right) = :\varphi^4(y): \; + 4_{c_2} :\varphi^2(y): \; D_F(y-y) \; + \; 3\,D_F^2(y-y)$$

$$\left\langle 0 \middle| :\varphi^4(y): \middle| 0 \right\rangle = 0 \qquad \text{but here we have}$$

$$\underset{0}{\left\langle \vec{k}_1\,\vec{k}_2 \right|} :\varphi^4(y): |\vec{p}_1\,\vec{p}_2\rangle \;\neq\; 0. \qquad \text{(we can't apply wicks theorem)}$$

In fact we can check if $\underset{0}{\left\langle \vec{k}\,\vec{k}_2 \right|} \varphi^4 \left| \vec{p}_1\vec{p}_2 \right\rangle$ survives.

it seems that only term which contributes is the term which destroys 2 particles & creates 2 particles. So in, $\varphi^4 = (\varphi^+ + \varphi^-)^4$

we have

$$\underset{0}{\left\langle \vec{k}_1\,\vec{k}_2 \right|} (\varphi^- + \varphi^+)^4\, |\vec{p}_1\,\vec{p}_2\rangle = 4_{c_2} \underset{0}{\left\langle \vec{k}_1\,\vec{k}_2 \right|} \varphi_-^2\,\varphi_+^2\, |\vec{p}_1\,\vec{p}_2\rangle_0 + 0 + 0\;\cdots$$

$$\varphi_+ \text{ destroys a particle} = a_k\,e^{-ik\cdot x}$$

$$\varphi_- \text{ creates a particle} = a_k^\dagger\,e^{ik\cdot x}$$

$$= 6\cdot \underset{0}{\left\langle \vec{k}_1\vec{k}_2 \right|} \varphi_-^2\;\varphi_+^2\,|\vec{p}_1\,\vec{p}_2\rangle_0$$

$$= 6\int \frac{d^3k}{(2\pi)^3}\,\frac{1}{2\omega_k} \int \frac{d^3k'}{(2\pi)^3}\,\frac{1}{2\omega_{k'}} \;\; \left\langle 0 \middle| a_{k_1}^\dagger\,a_{k_2}^\dagger\,(a_k)^2 \left(a_{k'}^\dagger\right)^2 a_{p_1}^\dagger a_{p_2}^\dagger \middle| 0 \right\rangle$$

Lecture 11

We saw that S matrix represents amplitude for set of particles in far past to evolve as set of some particles in far future via interactions.

There will always be a part in S which means particle travel freely from past to future. (I)

$$S = I + i\,T_{(\vec{k}_1\vec{k}_2\cdots\vec{k}_n|\vec{p}_1\vec{p}_2)}$$

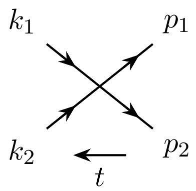
where,

$$T = (2\pi)^4\,\delta^4(\Sigma P)\,(i\mathcal{M})$$

We are majorly interested in $(i\mathcal{M})$.

For φ^4 scalar theory:

$i\mathcal{M}$ =



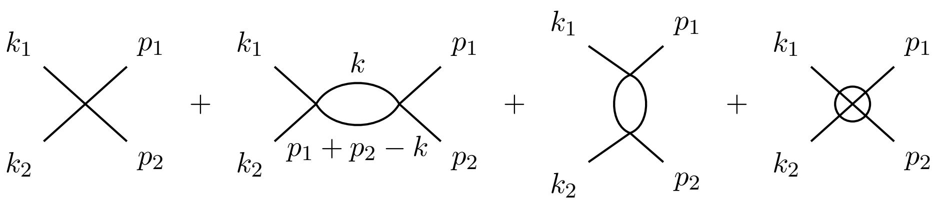
$= -i\lambda$

No propagator for external legs in (P-space) $(P_i^2 = m^2)$

Feynman diagram calculation is different for correlation functions and for scattering matrix element (For momentum space).

Correlation functions	S-matrix elements
1) off-shell	on-shell
2) external propagators	No ext. propagators
3) ("Building blocks")	(physical observables).
4) Gauge dependent	Gauge independent.
	(Gauge invariant).

$i\mathcal{M}$ =



$$= -i\lambda + \left(\frac{-i\lambda}{4!}\right)^2 \int \frac{d^4k}{(2\pi)^4}\,\frac{i\,e^{ik\cdot(x-y)}}{k^2 - m^2} + \cdots$$

Quantum Electrodynamics :- Fermions + photons.

$$\mathcal{L} = \overline{\Psi}(x)\,\big(i\not{D} - m\big)\,\Psi(x) \; - \; \frac{1}{4}\,F_{\mu\nu}(x)\,F^{\mu\nu}(x)$$

$$F_{\mu\nu}(x) = \partial_\mu A_\nu(x) - \partial_\nu A_\mu(x)$$

$$\not{D} = \gamma^\mu\,D_\mu = \gamma^\mu\,(\partial_\mu - ie\,A_\mu(x))$$

Digression:
Dirac $\mathcal{L} = \overline{\Psi}(x)\,\big(i\not{\partial} - m\big)\,\Psi(x)$ is invariant under global phase transformation.

$$\Psi(x) \to \Psi'(x) = e^{i\alpha}\,\Psi(x) \qquad \overline{\Psi}(x) \to \overline{\Psi}'(x) = \overline{\Psi}(x)\,e^{-i\alpha}$$

$$\mathcal{L}' = \overline{\Psi}'(x)\,\big(i\not{\partial} - m\big)\,\Psi'(x) = \overline{\Psi}(x)\,e^{-i\alpha}\,\big(i\not{\partial} - m\big)\,e^{i\alpha}\,\Psi(x) = \overline{\Psi}(x)\cancel{e^{0}}\,\big(i\not{\partial} - m\big)\,\Psi(x) = \mathcal{L}$$

If we impose the local gauge invariance

$$\Psi(x) \to \Psi'(x) = e^{i\alpha(x)}\,\Psi(x) \qquad \overline{\Psi}(x) \to \overline{\Psi}'(x) = \overline{\Psi}(x)\,e^{-i\alpha(x)}$$

$$\begin{aligned}
\mathcal{L}' &= \overline{\Psi}'(x)\,(i\,\gamma^\mu\partial_\mu - m)\,\Psi(x)' \\
&= \overline{\Psi}(x)\,e^{-i\alpha(x)}\,(i\,\gamma^\mu\partial_\mu - m)\,e^{i\alpha(x)}\,\Psi(x) \\
&= \overline{\Psi}(x)\,e^{-i\alpha(x)}\,\Big(i\,\gamma^\mu\,\partial_\mu\,\Big(e^{i\alpha(x)}\Psi(x)\Big)\Big) - m\,\overline{\Psi}(x)\,\Psi(x) \\
&= \overline{\Psi}(x)\,e^{-i\alpha(x)}\,\Big(i\,\gamma^\mu\,e^{i\alpha(x)}\,(\Psi(x)\,(i\,\partial_\mu\alpha(x)) + \partial_\mu\Psi(x))\Big) - m\,\overline{\Psi}(x)\,\Psi(x) \\
&= \overline{\Psi}(x)\,(i\,(\gamma^\mu\partial_\mu - m))\,\Psi(x) \;-\; \overline{\Psi}(x)\,\gamma^\mu\,\Psi(x)\,\partial_\mu\alpha(x) \\
&= \mathcal{L} - \overline{\Psi}(x)\,\gamma^\mu\,(\partial_\mu\alpha(x))\,\Psi(x)
\end{aligned}$$

So, we look for $\mathcal{L}$ which is gauge invariant (local); we require some $\mathcal{L}_{\text{kinetic}} = \overline{\Psi}(x)\,i\,\gamma^\mu D_\mu\Psi(x)$ such that under gauge transformation.

$$D_\mu\Psi(x) \;\longrightarrow\; e^{i\alpha(x)}\,(D_\mu\Psi(x))$$

If gauge transformation is $U(x) \in U(N)$ then;

$$\begin{aligned}
D_\mu\Psi(x) \;\longrightarrow\; D'_\mu\Psi'(x) &= U(x)\,(D_\mu\Psi(x)) \\
D'_\mu\,U(x)\,\Psi(x) &= U(x)\,D_\mu\,\Psi(x) \\
D'_\mu\,U(x) &= U(x)\,D_\mu \\
D'_\mu &= U(x)\,D_\mu\,U^{-1}(x)
\end{aligned}$$

If we look for $D_\mu = \partial_\mu + i\,A_\mu(x)$

$$\begin{aligned}
\cancel{\partial_\mu} + i\,A'_\mu(x) &= U(x)\,(\partial_\mu + i\,A_\mu(x))\,U^{-1}(x) \\
&= U(x)\,\partial_\mu\,U^{-1}(x) + \cancel{\partial_\mu} + i\,U(x)\,A_\mu(x)\,U^{-1}(x)
\end{aligned}$$

$$i\,A'_\mu(x) = U(x)\,\partial_\mu U^{-1}(x) + i\,U(x)\,A_\mu\,U^{-1}(x)$$

$$\boxed{A'_\mu(x) = U(x)\,A_\mu\,U^{-1}(x) - i\,U(x)\,\partial_\mu U^{-1}(x)} \quad \text{—1}$$

If we are looking for D_μ of type $D_\mu = \partial_\mu - ie\,A_\mu(x)$: we replace $A_\mu \to -eA_\mu$ in eq$^{\text{n}}$ 1

$$A'_\mu(x) = (-e)\,U(x)\,A_\mu\,U^{-1}(x) - i\,U(x)\,\partial_\mu U^{-1}(x)$$

Feynman Rules :

(1) Fermion propagator (S_F) : $\xrightarrow{k}$ $\qquad \dfrac{i}{\not{k} - m} = \dfrac{i\,(\not{k} + m)}{k^2 - m^2 + i\varepsilon}$

(1) $\xrightarrow{k}$ $\equiv$ charge & momentum flow to right (2) $\underset{k \rightarrow}{\longleftarrow}$ $\equiv$ charge flow to left & momentum flow to right.

in fact k can take any value; it just helps in writing momentum conservation. (momentum flow has no physical meaning; on the other hand charge flow has definite physical meaning).

$$S_F = \left(i\partial\!\!\!/ + m\right) D_F$$

$$S_F(x-y) = \left(i\partial\!\!\!/ + m\right) \int d^4k \, \frac{i}{k^2 - m^2 + i\varepsilon} \, e^{-ik\cdot(x-y)}$$

$$= \int d^4k \, \frac{i\,(k\!\!\!/ + m)}{k^2 - m^2 + i\varepsilon} \, e^{-ik\cdot(x-y)}$$

$$\widetilde{S}_F(k) = \frac{i\,(k\!\!\!/ + m)}{k^2 - m^2 + i\varepsilon} = \frac{i}{k\!\!\!/ - m}$$

(2) Vector field propagator —

 $\equiv \dfrac{-i\,\eta_{\mu\nu}}{k^2 + i\varepsilon}$ $\quad\downarrow$ (No mass term for A_μ, as $mA^\mu A_\mu$ is not gauge invariant)

(For vector field we chose a gauge; here we choose Feynman gauge) $\quad \partial_\mu A^\mu = 0$

When we do S-matrix calculations; we will have external lines; external lines are particles which are prepared or detected with definite polarisation (fermions are particle with spin). So we need info of polarisation of $|\text{in}\rangle$ and $|\text{out}\rangle$ states.

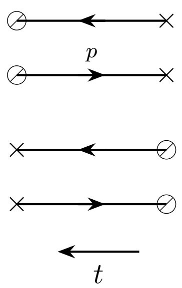
There are external polarisation factors.

In our convention the time flows from right to left and 'x' is boundary (external point).

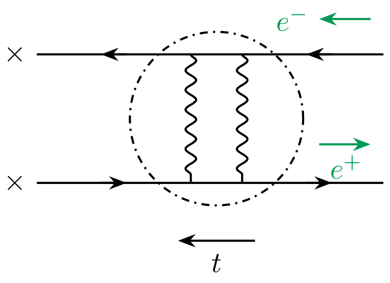
$$\Psi(x) = \sum_s \int \frac{d^3p}{(2\pi)^3} \, \frac{1}{\sqrt{2\omega_p}} \left(a_p^s\, u_p^s\, e^{-ip\cdot x} + b_p^{s\dagger}\, v_p^s\, e^{ip\cdot x}\right)$$

$$\overline{\Psi}(x) = \sum_s \int \frac{d^3p}{(2\pi)^3} \, \frac{1}{\sqrt{2\omega_p}} \left(b_p^s\, \overline{v}_p^s\, e^{-ip\cdot x} + a_p^{\dagger s}\, \overline{u}_p^s\, e^{ip\cdot x}\right)$$

u_p^s is associated with annihilation of e^-; $\overline{u}_p^s$ — creation of e^-; v_p^s is associated with — e^+; $\overline{v}_p^s$ — annihilation of e^- $\longrightarrow$ Annihilation of e^- at vertex.

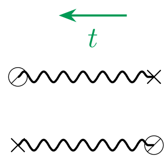


incoming electron	$u^s(p)$
incoming positron	$\overline{v}^s(p)$
outgoing e^-	$\overline{u}^s(p)$
outgoing e^+	$v^s(p)$



$e^-e^+ \to e^-e^+$ scattering

$$A_\mu(x) = \int \frac{d^3k}{(2\pi)^3} \, \frac{1}{\sqrt{2\omega_k}} \sum_{i=1}^{2} \left(\epsilon_\mu^i(k)\, a_{k,i}\, e^{-ikx} + \epsilon_\mu^{i\,*}(k)\, a_{k,i}^\dagger\, e^{ikx}\right)$$

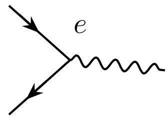


$\epsilon^s_\mu(p)$ (incoming)

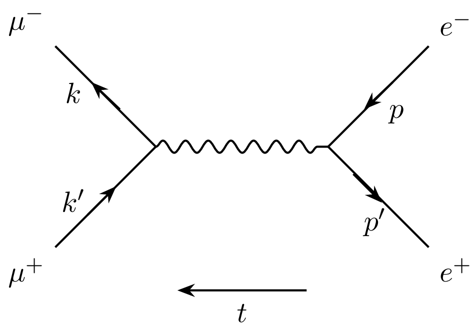
$\epsilon^{s\,*}_\mu(p)$ (outgoing)

$(s = 1, 2)$

<u>Interaction vertex</u> :



Let us consider a case :

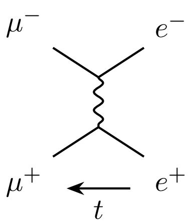


$e^+e^- \longrightarrow \mu^+\mu^-$

$$\mathcal{L} = i\,\overline{\Psi}_i\,\gamma^\mu\,(\partial_\mu - i\,e_i\,A_\mu)\,\Psi_i \;-\; m_i\,\overline{\Psi}_i\,\Psi_i \;-\; \frac{1}{4}\,F_{\mu\nu}\,F^{\mu\nu}$$

$$i = \underset{e^-}{1}\,,\;\; \underset{\mu^-}{2}\,,$$

We can't have t-channel as to have t-channel we need coupling of e^- & μ^- via photon.



$i\,\overline{\Psi}_1\,(\gamma^\mu\partial_\mu - i\,e_i\,A_\mu)\,\Psi_2$ which we don't have.

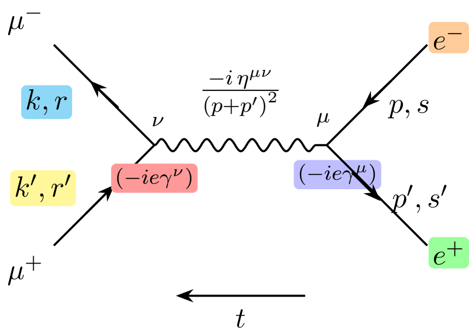
So, e^- number / lepton no. conservation holds here because of

$$\mathcal{L}_{\text{int}} = \overline{\Psi}_i\,e_i\,A_\mu\,\Psi_i \qquad \text{where } A_\mu \text{ couples to fermions of same species.}$$

For $e^-e^+ \longrightarrow e^-e^+$ $(s, t$ channel)

For $e^-e^- \longrightarrow e^-e^-$ $(t, u$ channel)

Similarly we need all diagrams for $e^-e^+ \to \mu^-\mu^+$ diagram. We only have s-channel for this process.



$$\mathcal{M} = \boxed{\overline{v}^{\,s'}(p')}\ \boxed{(-ie\gamma^\mu)}\ \boxed{u^s(p)} \cdot \frac{(-i\,\eta_{\mu\nu})}{(p+p')^2} \cdot \boxed{\overline{u}^r(k)}\ \boxed{(-ie\gamma^\nu)}\ \boxed{v^{r'}(k')}$$

$$= \frac{i\,e^2}{(p+p')^2}\ \overline{v}^{\,s'}(p')\,\gamma^\mu\,u^s(p)\ \cdot\ \overline{u}^r(k)\,\gamma_\mu\,v^{r'}(k')$$

Physical quantity it represents is scattering crossection.

$$\sigma\ \propto\ |\mathcal{M}|^2$$
$$\text{(or)}\qquad \sigma\ \propto\ \mathcal{M}\,\mathcal{M}^*$$

We know,

$$\frac{d\sigma}{d\Omega} = \frac{1}{2\omega_p}\ \frac{1}{2\omega_{p'}}\ \frac{1}{|\vec{v}-\vec{v}'|}\ \frac{\left|\vec{k}\right|}{16\pi^2\,E_{cm}}\ |\mathcal{M}|^2$$

where $\quad \vec{v} = \dfrac{\vec{p}}{\omega_p}\ ;\quad \vec{v}' = \dfrac{\vec{p}'}{\omega_{p'}}$

in COM frame $\quad (\vec{p}+\vec{p}' = 0)$

let us calculate $|\mathcal{M}|^2$

$$|\mathcal{M}|^2 = \frac{e^4}{(p+p')^4}\ \left(\overline{v}^{\,s'}(p')\,\gamma^\mu\,u^s(p)\right)\left(\overline{v}^{\,s'}(p')\,\gamma^\nu\,u^s(p)\right)^*$$
$$\cdot\ \left(\overline{u}^r(k)\,\gamma_\mu\,v^r(k')\right)\left(\overline{u}^r(k)\,\gamma_\nu\,v^{r'}(k')\right)^* \qquad \text{—}1$$

here

$$\left((p+p')^2\right)^2 = \left((p^0+p'^0)^2 - (\vec{p}+\vec{p}')^2\right)^2$$
$$\left(\overline{v}_a\,\gamma^\mu_{ab}\,u_b\right)^* = \overline{u}_c\,\gamma^\mu_{cd}\,v_d \qquad \text{(Prove it !)}$$

Suppose we do experiment with unpolarised beams —

For incoming unpolarised beams we take average

$$\boxed{\frac{1}{2}}\sum_{s=1}^{2} \times \boxed{\frac{1}{2}}\sum_{s'=1}^{2} \qquad \left(s, s' \text{ can take } \boxed{2} \text{ values}\right)$$

For outgoing beams; if we don't detect the polarisation — we include all possible states.

$$\text{(i.e.)}\quad \text{sum over final states.}\qquad \left(\sum_{r=1}^{2}\ \sum_{r'=1}^{2}\right)$$

Using

$$\sum_s u^s(p)\,\overline{u}^s(p) = (\not{p}+m)\ ;\qquad \sum_s v^s(p)\,\overline{v}^s(p) = \not{p}' - m$$

Expression ① becomes.

$$|\mathcal{M}|^2 = \frac{e^4}{(p+p')^4} \left(\overline{v}^{\,s'}(p')\,\gamma^\mu\, \underline{u^s(p)\,\overline{u}^s(p)}\,\gamma^\nu\,\overline{v}^{\,s'}(p')\right) \left(\overline{u}^r(k)\,\gamma_\mu\, \underline{v^r(k')\,\overline{v}^{\,r'}(k')}\,\gamma_\nu\, u^r(k)\right)$$

Lecture 12 — (Higgs Mechanism and Non Abelian gauge theories)

Let's talk about complex scalar fields (φ, φ^*)

$$\mathcal{L}\left(\varphi, \partial_\mu \varphi, \varphi^*, \partial_\mu \varphi^*\right) = \partial_\mu \varphi^* \, \partial^\mu \varphi \; - \; m^2 \, \varphi \, \varphi^* \; - \; \frac{\lambda}{6} \left(\varphi \, \varphi^*\right)^2$$

We define new variables (we can work in any form of variables)

New field variables: (Polar Form)

$$\begin{cases} \varphi(x) = \frac{1}{\sqrt{2}} \, R(x) \, e^{i\theta(x)} \\ \varphi^*(x) = \frac{1}{\sqrt{2}} \, R(x) \, e^{-i\theta(x)} \end{cases}$$

Any complex # can be written in polar form as $\left(z = R\, e^{i\theta}\right)$ $(\theta, R \in \mathbb{R})$

So,

$$\begin{aligned} \mathcal{L}\left(R, \theta\right) &= \frac{1}{2} \, \partial_\mu \left(R\, e^{-i\theta}\right) \partial^\mu \left(R\, e^{i\theta}\right) \; - \; \frac{1}{2} \, m^2 R^2 \; - \; \frac{\lambda}{4.6} \, R^4 \qquad \cdots (R, \theta = R(x), \theta(x)) \\ &= \frac{1}{2} \left(e^{-i\theta} \, \partial_\mu R + (-iR) \, e^{-i\theta} \, \partial_\mu \theta\right) \left(e^{i\theta} \, \partial^\mu R + iR\, e^{i\theta} \, \partial^\mu \theta\right) \; - \; \frac{m^2 R^2}{2} - \frac{\lambda}{4!} \, R^4 \\ &= \frac{1}{2} \left(\partial_\mu R - iR\, \partial_\mu \theta\right) \left(\partial^\mu R + iR\, \partial^\mu \theta\right) \; - \; \frac{m^2 R^2}{2} - \frac{\lambda}{4!} \, R^4 \end{aligned}$$

$$\begin{aligned} \mathcal{L}\left(R, \partial_\mu R, \theta, \partial_\mu \theta\right) &= \frac{1}{2} \left(\partial_\mu R \, \partial^\mu R + R^2 \, \partial_\mu \theta \, \partial^\mu \theta + \underline{iR\, \partial_\mu R \cancel{\partial^\mu \theta}} - \cancel{iR\, \partial_\mu \theta \, \partial^\mu R}\right) \\ &\qquad - \frac{m^2 R^2}{2} - \frac{\lambda}{4!} \, R^4 \\ &= \frac{1}{2}\Big(\underbrace{\partial_\mu R \, \partial^\mu R}_{\text{kinetic term for Real scalar field}} + R^2 \, \partial_\mu \theta \, \partial^\mu \theta \Big) \; - \; \underbrace{\frac{m^2 R^2}{2}}_{\text{free real}} - \frac{\lambda}{4!} \, R^4 \end{aligned}$$

Use of new variables gives us another way of thinking; & we might run into problems if new variables has some problems; here these plane polar coordinates have well known problem of being singular at origin.

These coordinates are not convenient as as $R \to 0$ the kinetic term for θ disappears. also $\varphi(x) \to 0$ as $R \to 0$.

— It is convenient because $\theta(x)$ field is not interacting with self. This is because of $\mathcal{L}_{\text{int}} \propto (\varphi^* \varphi)^2 \propto R^4$. So only $R(x)$ is interacting with self.

Consider global symmetry —

$$\begin{aligned} \varphi \; &\to \; e^{i\alpha} \, \varphi \\ \varphi^* \; &\to \; e^{-i\alpha} \, \varphi^* \end{aligned}$$

We know that this $U(1)$ symmetry implies conserved charge, and after coupling to photon that conserved charge became the electric charge.

Same global symmetry $\Rightarrow$ $R\, e^{i\theta} \longrightarrow e^{i\alpha} \, R\, e^{i\theta}$

$$\begin{aligned} \Rightarrow \qquad \theta \; &\to \; \theta + \alpha \\ \theta(x) \; &\to \; \theta(x) + \alpha \end{aligned}$$

$\alpha = [0, 2\pi)$ $\downarrow$ Const shift of angle

Such transformations form group $U(1)$.

If we can fix R (in term $\frac{R^2}{2} \, \partial_\mu \theta(x) \, \partial^\mu \theta(x)$ of $\mathcal{L}(R, \theta, \partial R, \partial\theta)$) to be some finite value then there will be a quadratic term for $\theta(x)$. ($R = \text{const.}$)

We will take value of R at which potential is minimum.

We take same $\mathcal{L}(R, \theta, \partial_\mu R, \partial_\mu \theta)$ but replace $m^2 \to -m^2$ Why ?

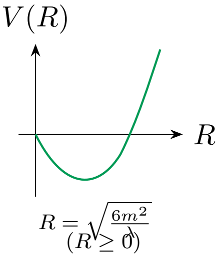
Initially

$$\begin{aligned}
\mathcal{L} &= T - V \\
&= \frac{1}{2}\,\partial_\mu R\,\partial^\mu R + \frac{R^2}{2}\,\partial_\mu \theta\,\partial^\mu \theta \;-\; \left(\frac{1}{2}\,m^2 R^2 + \lambda\,\frac{R^4}{4!}\right)
\end{aligned}$$

after replacing $m^2 \to -m^2$

$$\mathcal{L} = \frac{1}{2}\,\partial_\mu R\,\partial^\mu R + \frac{R^2}{2}\,\partial_\mu \theta\,\partial^\mu \theta \;-\; \left(\frac{-m^2 R^2}{2} + \lambda\,\frac{R^4}{4!}\right)$$

$$V(R) = \frac{-m^2 R^2}{2} + \frac{R^4}{4!}$$



We do perturbation theory by expanding around $R = 0$, but since we have replaced $m^2 \to -m^2$; $R = 0$ now has become unstable point, we must then find a stable point about which we do perturbation.

$$\begin{cases} R(x) = \text{Radion (particles which describe radial fluctuations in field)} \\ \theta(x) = \text{Axion (hypothetical particles which describe angular fluctuations in field space)} \end{cases}$$

$$\begin{aligned}
\left.\frac{\partial V}{\partial R}\right|_R = -\frac{2m^2 R}{2} + \lambda\,\frac{4\,R^3}{4\cdot 3!} \;&=\; 0 \\
-m^2 R + \lambda\,\frac{R^3}{3!} &= 0 \\
R\left(-m^2 + \lambda\,\frac{R^2}{3!}\right) &= 0 \\
\Rightarrow \qquad R = 0 \qquad \underline{\text{or}} \qquad R^2 &= 6m^2/\lambda \\
R = +\sqrt{\frac{6m^2}{\lambda}} \qquad (R \geq 0) \quad &\text{how ?}
\end{aligned}$$

This tells us that field theory looks bad, it will have propagator with a wrong sign, it will have classical solution of the form $p^2 = -m^2$, (instead of $p^2 = m^2$); but they will be solved if we take

$$R(x) = \widetilde{R}(x) + \sqrt{\frac{6m^2}{\lambda}}$$

We will be expanding around $\widetilde{R}(x) = 0$; i.e. $R(x) = \sqrt{\frac{6m^2}{\lambda}}$

$$\widetilde{R}(x) = R(x) - \sqrt{\frac{6m^2}{\lambda}}$$

$$\mathcal{L} = \frac{1}{2}\,\partial_\mu \widetilde{R}\,\partial^\mu \widetilde{R} + \frac{1}{2}\left(\widetilde{R} + \sqrt{\frac{6m^2}{\lambda}}\right)^2 \partial_\mu \theta\,\partial^\mu \theta - \left(-\frac{m^2}{2}\left(\widetilde{R} + \sqrt{\frac{6m^2}{\lambda}}\right)^2 + \frac{\lambda}{4!}\left(\widetilde{R} + \sqrt{\frac{6m^2}{\lambda}}\right)^4\right)$$

$$\mathcal{L} = \frac{1}{2}\,\partial_\mu \widetilde{R}\,\partial^\mu \widetilde{R} + \frac{1}{2}\,\widetilde{R}^2\,\partial_\mu \theta\,\partial^\mu \theta - \left(-\frac{m^2\,\widetilde{R}^2}{2} + \frac{\lambda}{4!}\,\widetilde{R}^4\right)$$

$$
\begin{aligned}
&+\frac{1}{2}\,\frac{6m^2}{\lambda}\,\partial_\mu\theta\,\partial^\mu\theta+\widetilde{R}\sqrt{\frac{6m^2}{\lambda}}\,\partial_\mu\theta\,\partial^\mu\theta\\
&+\widetilde{R}\,m^2\sqrt{\frac{6m^2}{\lambda}}+\frac{1}{2}\,m^2\,\frac{6m^2}{\lambda}\\
&-\frac{\lambda}{4!}\left(\underbrace{\cancel{{}^4c_0\left(\sqrt{\frac{6m^2}{\lambda}}\right)^4}}_{\nearrow\,\text{Const}}+{}^4c_1\,\widetilde{R}\left(\sqrt{\frac{6m^2}{\lambda}}\right)^3+{}^4c_2\,\widetilde{R}^2\left(\sqrt{\frac{6m^2}{\lambda}}\right)^2+{}^4c_3\,\widetilde{R}^3\sqrt{\frac{6m^2}{\lambda}}\right)
\end{aligned}
$$

$$
\begin{aligned}
=\frac{1}{2}\,\partial_\mu\widetilde{R}\,\partial^\mu\widetilde{R}+\frac{\widetilde{R}^2}{2}\,\partial_\mu\theta\,\partial^\mu\theta+\frac{3m^2}{\lambda}\,\partial_\mu\theta\,\partial^\mu\theta+\widetilde{R}\sqrt{\frac{6m^2}{\lambda}}\,\partial_\mu\theta\,\partial^\mu\theta\\
+\left(\frac{m^2\,\widetilde{R}^2}{2}-\frac{\cancel{4!}}{2!\,2!}\,\frac{\lambda}{\cancel{4!}}\,\frac{\widetilde{R}^2\,6m^2}{\lambda}\right)-\frac{\lambda}{4!}\,\widetilde{R}^4\\
+\widetilde{R}\left(\cancel{m^2\sqrt{\frac{6m^2}{\lambda}}}-\frac{\lambda}{4!}\,4\,\cancel{\frac{6m^2}{\lambda}\sqrt{\frac{6m^2}{\lambda}}}\right)\\
-\frac{\lambda}{4!}\,\widetilde{R}^3\sqrt{\frac{6m^2}{\lambda}}
\end{aligned}
$$

$$
\begin{aligned}
\mathcal{L}=\frac{1}{2}\,\partial_\mu\widetilde{R}\,\partial^\mu\widetilde{R}+\left(\frac{\widetilde{R}^2}{2}+\frac{3m^2}{\lambda}+\widetilde{R}\sqrt{\frac{6m^2}{\lambda}}\right)\partial_\mu\theta\,\partial^\mu\theta\;-\;m^2\,\widetilde{R}^2-\frac{\lambda}{4!}\,\widetilde{R}^4\\
-\frac{\lambda}{4!}\,\widetilde{R}^3\sqrt{\frac{6m^2}{\lambda}}
\end{aligned}
$$

$$
\Rightarrow\qquad \text{mass of }\widetilde{R}\;=\;\frac{1}{2}\,m_{\widetilde{R}}^2\,\widetilde{R}^2\;=\;m^2\,\widetilde{R}^2
$$

$$
\Rightarrow\qquad \boxed{m_{\widetilde{R}}=\sqrt{2}\,m}
$$

Observations/features —

$\theta(x)$ now has well defined kinetic term at $\widetilde{R}(x)=0$.

$$
\text{Mass of }\widetilde{\theta}(x)\;=\;0.
$$

$$
\text{potential of }\theta(x)\;=\;0\qquad(\text{No potential at all for }\theta(x))\,.
$$

here $V\left(\theta(x)\right)=0$ because of $U(1)$ symmetry

(or) i.e. $\mathcal{L}$ is invariant under $\theta(x)\to\theta(x)+\alpha$

(No functions of $\theta(x)$ can be invariant under $\theta\to\theta+\alpha$; only functions of derivatives of $\theta(x)$ can be invariant).

$$
\partial_\mu\theta'=\partial_\mu\theta\qquad(\partial_\mu\alpha=0)\,.
$$

So we don't have potential term.

This implies Axionic fields have only derivative couplings. (Goldstone theorem)

$\theta(x)$: Goldstone boson (spinless) $\quad\hookrightarrow\ \theta(x)$ does not have any index

$$
\begin{cases}
A(x)=\text{boson (spinless)}.\\
A_\mu(x)=\text{vector boson (spin 1 as }\underline{1}\text{ index }\mu)\\
A_{\mu\nu}(x)=\text{tensor boson. (spin 2)}.
\end{cases}
$$

Goldstone Theorem :

Whenever a (global) continuous symmetry is spontaneously broken, there is a massless particle in the theory.

(Here $\theta(x)$ is massless).

Spontaneously broken means; If we try to minimize the energy (to find ground state of classical system). Any choice we make for ground state will fail to respect the symmetry.

i.e. $\mathcal{L}$ is still symmetric/invariant under $\theta(x) \to \theta(x) + \alpha$ but ground state is not.

Ground state is $\left(\widetilde{R}, \theta\right)$ at which $V(R, \theta)$ is minimum.

$V\left(\widetilde{R}\right)$ is minimum at $\widetilde{R} = 0$ $\left(\text{i.e. } R = \sqrt{\frac{6m^2}{\lambda}}\right)$

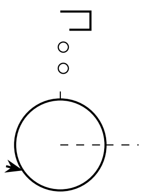
but θ can be anything.

say $(0, \theta')$ be ground state of fields.

$$\theta' \neq \theta' + \alpha \qquad \text{but } \theta' \text{ (any value) is not equal to } \theta' + \alpha.$$

So Ground state is not symmetric under $U(1)$. This is called spontaneous symmetry breaking

Analogy: Consider a water drop falls on perfect sphere it will choose any one path on sphere but that any path on surface of sphere will not respect the rotational symmetry of sphere and we say that symmetry is spontaneously broken.



But if we make a path through center then that path will respect the rotational symmetry of sphere.

Imposing local gauge Invariance :

let us impose local gauge invariance

$$\alpha \to \alpha(x)$$
$$\varphi \to e^{i\alpha(x)}\,\varphi \qquad \varphi = R\, e^{i\theta}$$
$$\varphi^* \to e^{-i\alpha(x)}\,\varphi^*$$

To have gauge invariance

$$\mathcal{L}' = \mathcal{L}$$
$$\mathcal{L}' = \mathcal{L}'\left(\widetilde{R}, \theta, \partial_\mu \widetilde{R}, \partial_\mu \theta\right) \downarrow$$

$$\mathcal{L}' = \frac{1}{2}\,\partial_\mu \widetilde{R}\,\partial^\mu \widetilde{R} + \left(\frac{\widetilde{R}^2}{2} + \frac{3m^2}{\lambda} + \widetilde{R}\sqrt{\frac{6m^2}{\lambda}}\right) \partial_\mu \theta\, \partial^\mu \theta - m^2\, \widetilde{R}^2 - \frac{\lambda}{4!}\,\widetilde{R}^4 - \frac{\lambda}{4!}\,\widetilde{R}^3\sqrt{\frac{6m^2}{\lambda}}$$

under $\quad \theta(x) \longrightarrow \theta(x) + \alpha(x)$

but $\quad \widetilde{R} \longrightarrow \widetilde{R} \qquad (R \text{ is independent of } \theta)$

We check if : $\partial_\mu \theta'\, \partial^\mu \theta' = \partial_\mu \theta\, \partial^\mu \theta$ under local gauge transf.

$$\theta' = \theta(x) + \alpha(x)$$

i.e. LHS

$$\begin{aligned}
&\partial_\mu\left(\theta(x)+\alpha(x)\right)\partial^\mu\left(\theta(x)+\alpha(x)\right)\\
&=\left(\partial_\mu\theta+\partial_\mu\alpha\right)\left(\partial^\mu\theta+\partial^\mu\alpha\right)\\
&=\partial_\mu\theta\,\partial^\mu\theta+\partial_\mu\alpha\,\partial^\mu\theta+\partial_\mu\theta\,\partial^\mu\alpha+\partial_\mu\alpha\,\partial^\mu\alpha
\end{aligned}$$

to be equal to $\partial_\mu\theta\,\partial^\mu\theta$

$$\Rightarrow \qquad \partial_\mu\alpha\,\partial^\mu\theta+\partial_\mu\theta\,\partial^\mu\alpha+\partial_\mu\alpha\,\partial^\mu\alpha \;=\; 0$$

So, we need to discard $\partial_\mu\theta\,\partial^\mu\theta$ term for gauge transformed θ;

we need to make $\mathcal{L}$, which is invariant under $\theta(x)\to\theta(x)+\alpha(x)$.

under gauge transf:

$$\begin{aligned}
\partial_\mu\theta(x)\;&\longrightarrow\;\partial_\mu\left(\theta+\alpha(x)\right)\\
&\longrightarrow\;\partial_\mu\theta+\partial_\mu\alpha(x)\;\neq\;\partial_\mu\theta(x)
\end{aligned}$$

but we can make $D_\mu\theta(x)$ such that $D_\mu\theta(x)\longrightarrow D_\mu\theta(x)$ under gauge transformation.

redefine:

$$D_\mu\theta(x)=\partial_\mu\theta(x)-A_\mu(x)$$

$$\text{Where}\quad A'_\mu(x)=A_\mu+\partial_\mu\alpha(x)\qquad \{\theta\in\mathbb{R}\Rightarrow A_\mu\in\mathbb{R}\}\;\; A_\mu(x)\text{ is real.}$$

$$\begin{aligned}
(D_\mu\theta)'&=\partial_\mu\theta'(x)-A'_\mu(x)\\
&=\partial_\mu\left(\theta+\alpha(x)\right)-\left(A_\mu+\partial_\mu\alpha(x)\right)\\
&=\partial_\mu\theta-A_\mu\\
&=D_\mu\theta(x)\qquad\text{which is gauge invariant!}
\end{aligned}$$

So local Gauge invariant $\mathcal{L}$ becomes : $\{\partial_\mu\theta\to D_\mu\theta=\partial_\mu\theta-A_\mu\}$

$$\begin{aligned}
\mathcal{L}=\frac{1}{2}\,\partial_\mu\widetilde{R}\,\partial^\mu\widetilde{R}+\left(\frac{\widetilde{R}^2}{2}+\frac{3m^2}{\lambda}+\widetilde{R}\sqrt{\frac{6m^2}{\lambda}}\right)&\left(\partial_\mu\theta-A_\mu\right)^2-m^2\,\widetilde{R}^2-\frac{\lambda}{4!}\,\widetilde{R}^4\\
-\frac{\lambda}{4!}\,\widetilde{R}^3\sqrt{\frac{6m^2}{\lambda}}\qquad&\underbrace{-\frac{1}{4}\,F_{\mu\nu}\,F^{\mu\nu}}_{\text{To allow gauge field to propagate.}}
\end{aligned}$$

Note that $\left(\partial_\mu\theta-A_\mu\right)^2=\left(\partial_\mu\theta\right)^2+A_\mu^2-2\,A_\mu\,\partial_\mu\theta$ is not diagonal, we can diagonalise it by field redefination of A_μ.

$$A_\mu\;\longrightarrow\;A_\mu+\partial_\mu\theta$$

So that,

$$\begin{aligned}
\partial_\mu\theta-A_\mu&=\partial_\mu\theta-A_\mu-\partial_\mu\theta\\
&=-A_\mu\qquad(\theta(x)\text{ has disappeared!})
\end{aligned}$$

and,

$$\begin{aligned}
F_{\mu\nu}=\partial_\mu A_\nu-\partial_\nu A_\mu&=\partial_\mu\left(A_\nu+\partial_\nu\theta\right)-\partial_\nu\left(A_\mu+\partial_\mu\theta\right)\\
(F_{\mu\nu}\text{ remains same})\qquad&=\partial_\mu A_\nu+\cancel{\partial_\mu\partial_\nu\theta}-\partial_\nu A_\mu-\cancel{\partial_\nu\partial_\mu\theta}\\
&=\partial_\mu A_\nu-\partial_\nu A_\mu\\
&=F_{\mu\nu}
\end{aligned}$$

$$\mathcal{L} = \frac{1}{2}\,\partial_\mu \widetilde{R}\,\partial^\mu \widetilde{R} + \left(\frac{\widetilde{R}^2}{2} + \frac{3m^2}{\lambda} + \widetilde{R}\sqrt{\frac{6m^2}{\lambda}}\right)(-A_\mu)^2 - m^2\,\widetilde{R}^2 - \frac{\lambda}{4!}\,\widetilde{R}^4$$
$$- \frac{\lambda}{4!}\,\widetilde{R}^3\sqrt{\frac{6m^2}{\lambda}} \;-\; \frac{1}{4}\,F_{\mu\nu}F^{\mu\nu}$$

$$\mathcal{L} = \frac{1}{2}\,\partial_\mu \widetilde{R}\,\partial^\mu \widetilde{R} + \frac{3m^2}{\lambda}\,A_\mu A^\mu + \left(\frac{\widetilde{R}^2}{2} + \widetilde{R}\sqrt{\frac{6m^2}{\lambda}}\right)A_\mu A^\mu - m^2\,\widetilde{R}^2 - \frac{\lambda}{4!}\,\widetilde{R}^4$$
$$- \frac{\lambda}{4!}\,\widetilde{R}^3\sqrt{\frac{6m^2}{\lambda}} \;-\; \frac{1}{4}\,F_{\mu\nu}F^{\mu\nu}$$

$\swarrow$ mass term for A_μ

This is what higgs mechanism does, A_μ becomes massive even in gauge theory. (Also $\theta(x)$ has disappeared!)

With local gauge invariance, Goldstone theorem no longer true, (as we could not find massless field); instead if we try to spontaneously break a local symmetry the gauge field $A_\mu(x)$ becomes massive.

We know that in relativistic theory a massless vector particle has only two polarisations. For massive we have $2J+1$ for spin J. (For A_μ, $J=1$ so we have 3 polarisations)

So the disappearance of $\theta(x)$ turns as third polarisation of massive $A_\mu(x)$. It is complete rearrangement of d.o.f in lorentz covariant way.

$\widetilde{R}(x)$ is known as Higgs field (it's not the field which participates in higgs mechanism; it's the one which does not participates (therefore it survives); $\theta(x)$ is called "would be Goldstone boson (which it was in global gauge invariance)"; since we coupled theory to a gauge field A_μ (via imposing local gauge invariance); The A_μ (gauge field) eat $\theta(x)$ & it went away.

End result is Higgs field $^{\widetilde{R}(x)}$ & gauge field $^{(A_\mu)}$ has mass.

Instead of just A_μ we have 3 massive gauge fields in real world called as W^+, W^-, Z, For that case there could be 1 or many higgs field $\left(\widetilde{R}(x)\right)$.

End of Higgs Mechanism

Non Abelian Gauge Symmetries

Suppose our scalar fields are complex with index I.

$$\varphi = \varphi_I(x) \qquad I = 1,2,3\ldots N.$$
$$\varphi^* = \varphi_I^*(x)$$
$$\varphi = \begin{pmatrix}\varphi_1\\ \varphi_2\\ \vdots\\ \varphi_N\end{pmatrix}$$

analogue of $e^{i\theta}\varphi$ here is taking $U\varphi$, where 'U' is unitary matrix $\left(UU^\dagger = I\right)$

$$\varphi_i = U_{ij}\,\varphi_j$$

$$\mathcal{L} = \partial_\mu\varphi_i^*\,\partial^\mu\varphi_i \;-\; m^2\,\varphi_i^*\varphi_i \;-\; \frac{\lambda}{6}\,(\varphi_i^*\varphi_i)^2$$

i.e. under

$$\varphi_i \;\to\; U\varphi_i$$

$$\varphi_i^* \;\to\; \varphi_i^{*T}\, U^\dagger$$

$$\text{thus,}\quad \varphi_i^{*T}\varphi_i \;\longrightarrow\; \varphi_i^{*T}\,\underline{U^\dagger U}\,\varphi_i = \varphi_i^{*T}\,\varphi_i$$

We could use $U = \begin{pmatrix} e^{i\alpha} & & & \\ & e^{i\alpha} & & \\ & & \ddots & \\ & & & e^{i\alpha} \end{pmatrix}$ to do global phase rotation of each $\varphi_i(x)$

(or) $U = \begin{pmatrix} e^{i\alpha_1} & & & \\ & e^{i\alpha_2} & & \\ & & \ddots & \\ & & & e^{i\alpha_n} \end{pmatrix}$ to do independent global phase rotation of φ_i

But fun comes when we use general unitary matrix which mixes all φ_i and corresponding to that global symmetry we can find conserved current & so on.

Try promoting $U = U(x)$ local gauge invariance of each field $\varphi_i(x)$

our $\mathcal{L} = \partial_\mu\varphi_i\,\partial^\mu\varphi_i^* - m^2\varphi_i^*\varphi_i - \frac{\lambda}{6}\,(\varphi_i^*\varphi_i)$ is not gauge invariant under

$$\varphi_i'(x) = U(x)_{ij}\,\varphi_i(x)$$
$$\varphi_i'^*(x) = \varphi_j^*\,U_{ji}^*(x)$$

$$\Big(\text{under dagger op.}\quad U \longrightarrow U^\dagger\;;\quad U_{ij} \longrightarrow U_{ji}^*\Big)$$

$$\begin{aligned}\partial_\mu\varphi_i' &= \partial_\mu\left(U_{ij}(x)\,\varphi_i(x)\right)\\ &= \varphi_i\,\partial_\mu U_{ij}(x) + U_{ij}(x)\,\partial_\mu\varphi_i(x)\\ &\neq \partial_\mu\varphi_i\end{aligned}$$

We thus look for $D_\mu'\varphi' = U\,D_\mu\varphi$ under $\varphi(x) \to U(x)\,\varphi(x)$.

so that $D'^\mu\varphi'^*\,D_\mu'\varphi' = D^\mu\varphi^*\,D_\mu\varphi$

We look for D_μ of type $D_\mu = \partial_\mu + i\,A_\mu(x)$

$$D_\mu'\,U\varphi = U\,D_\mu\varphi$$
$$D_\mu'\,U = U\,D_\mu$$
$$D_\mu' = U\,D_\mu\,U^\dagger$$

$$\cancel{\partial_\mu} + i\,A_\mu'(x) = U\left(\partial_\mu\,\mathbb{I} + i\,A_\mu(x)\right)U^\dagger = U\,\partial_\mu U^\dagger + U\,U^\dagger\,\cancel{\partial_\mu\mathbb{I}} + i\,U\,A_\mu(x)\,U^\dagger$$

$$\boxed{A_\mu'(x) \;=\; U(x)\,A_\mu(x)\,U^\dagger(x) \;-\; i\,U\,\partial_\mu U^\dagger} \quad \uparrow$$

or $UU^\dagger = I$:

$$(\partial_\mu U)\,U^\dagger + U\,\partial_\mu U^\dagger = 0 \quad\Rightarrow\quad U\,\partial_\mu U^\dagger = -\,(\partial_\mu U)\,U^\dagger$$

$$\mathcal{L} = (D_\mu\varphi)^\dagger\,D^\mu\varphi - m^2\,\varphi^\dagger\varphi - \frac{\lambda}{6}\left(\varphi^\dagger\varphi\right)^2$$

$$\text{(or)}\qquad \mathcal{L} = D_\mu\varphi_i\,D^\mu\varphi_i^* - m^2\,\varphi_i^*\varphi_i - \frac{\lambda}{6}\,(\varphi_i^*\varphi_i)^2$$

is invariant under

$$\varphi_i \;\to\; U_{ij}(x)\,\varphi_j(x)$$
$$\varphi_i^* \;\to\; \varphi_j^*(x)\,U_{ji}^*(x)$$

This can be applied to single complex scalar field $\mathcal{L}$. (i.e.) $N = 1$

$$
\begin{aligned}
A_\mu \;\to\; U A_\mu U^\dagger - i\, U\, \partial_\mu U^\dagger \qquad\qquad & U = e^{i\alpha(x)} \\
A'_\mu = A_\mu - i\, e^{i\alpha(x)}\, e^{-i\alpha(x)}\, (i\, \partial_\mu \alpha(x)) & \\
A'_\mu = A_\mu + \partial_\mu \alpha(x) &
\end{aligned}
$$

Which is what we get in abelian $A_\mu(x)$

Any unitary matrix can be written as

$$U(x) = \exp\left(i\, \Lambda(x)\right)$$

$$U(x)\, U^\dagger(x) = e^{i\Lambda(x)}\, e^{-i\Lambda^\dagger(x)} = I. \qquad \Big(\text{provided } \Lambda^\dagger(x) = \Lambda(x) \;\; \text{(hermitian)}\Big)$$

General hermitian matrix can be written as

$$\Lambda(x) = \begin{pmatrix} \text{Real numbers (diagonal)} & \text{Complex no's} \\ \text{Complex Conjugate of upper triangle} & \end{pmatrix}$$

So,

$$
\begin{aligned}
A'_\mu(x) &= U A_\mu U^\dagger + i\, (\partial_\mu U)\, U^\dagger \\
&= e^{i\Lambda(x)}\, A_\mu(x)\, e^{-i\Lambda(x)} + i\, \Big(\partial_\mu\, e^{i\Lambda(x)}\Big)\, e^{-i\Lambda(x)}
\end{aligned}
$$

For 1$^{\text{st}}$ order

$$
\begin{aligned}
&= (1 + i\Lambda(x))\, A_\mu(x)\, (1 - i\Lambda(x)) + i\, (\partial_\mu\, (1 - i\Lambda(x)))\, (1 - i\Lambda(x)) \\
&= (A_\mu + i\Lambda(x)\, A_\mu(x))\, (1 - i\Lambda(x)) \;+\; \partial_\mu \Lambda(x) \\
&= A_\mu - i\, A_\mu(x)\, \Lambda(x) + i\Lambda(x)\, A_\mu(x) + \partial_\mu \Lambda(x) \qquad (\Lambda(x) \text{ is small})
\end{aligned}
$$

$$\boxed{A'_\mu \;=\; A_\mu + i\, [\Lambda(x),\, A_\mu(x)] + \partial_\mu \Lambda(x)}$$

(or)

$$\delta A_\mu = \partial_\mu \Lambda(x) + i\, [\Lambda(x),\, A_\mu(x)]$$

So,

$$\mathcal{L} = (D_\mu \varphi)^\dagger\, (D^\mu \varphi) - V\left(\varphi^\dagger \varphi\right) \qquad \text{is invariant under}$$

$$
\begin{aligned}
\varphi \;&\to\; U\varphi \\
\text{(Matrix valued)} \hookleftarrow \quad A_\mu \;&\to\; U A_\mu U^\dagger + i\, (\partial_\mu U)\, U^\dagger
\end{aligned}
$$

To allow propagation of $A_\mu(x)$; we need analogue of $F_{\mu\nu}$.

In scalar QED:

$$
\begin{aligned}
[D_\mu,\, D_\nu]\, \varphi &= [\partial_\mu + iA_\mu\,,\; \partial_\nu + iA_\nu)\, \varphi \\
&= \Big(\cancelto{0}{[\partial_\mu, \partial_\nu]} + i\, [\partial_\mu,\, A_\nu] + i\, [A_\mu,\, \partial_\nu] + \cancelto{0}{i^2\, [A_\mu,\, A_\nu]}\; \text{(Abelian)}\Big)\, \varphi \\
&= i\, (\partial_\mu A_\nu - A_\nu \partial_\mu + A_\mu \partial_\nu - \partial_\nu A_\mu)\, \varphi \\
&= i\, (\partial_\mu A_\nu \varphi - A_\nu \partial_\mu \varphi + A_\mu \partial_\nu \varphi - \partial_\nu A_\mu \varphi) \\
&= i\, ((\partial_\mu A_\nu)\, \varphi + \cancel{A_\nu\, \partial_\mu \varphi} - \cancel{A_\nu\, \partial_\mu \varphi} + A_\mu \partial_\nu \varphi - (\partial_\nu A_\mu)\, \varphi - \cancel{A_\mu\, (\partial_\nu \varphi)})
\end{aligned}
$$

$$[D_\mu,\, D_\nu]\, \varphi = i\, F_{\mu\nu}\, \varphi$$

$$\text{So} \qquad F_{\mu\nu} = -i\,[D_\mu,\, D_\nu]$$

$$\downarrow$$

Here $F_{\mu\nu}$ is gauge invariant.

if we impose $F_{\mu\nu} = -i\,[D_\mu,\, D_\nu]$ on non abelian case

$$\begin{aligned} F_{\mu\nu}\,\varphi &= -i\,[D_\mu,\, D_\nu]\,\varphi \\ &= -i\,[\partial_\mu + iA_\mu\,,\ \partial_\nu + iA_\nu)\,\varphi \\ &= -i\left(\cancelto{0}{[\partial_\mu,\, \partial_\nu]}\varphi + i\,[A_\mu,\, \partial_\nu]\,\varphi + i\,[\partial_\mu,\, A_\nu]\,\varphi - [A_\mu,\, A_\nu]\,\varphi\right) \end{aligned}$$

$$\begin{aligned} F_{\mu\nu}\,\varphi &= (\partial_\mu A_\nu(x) - \partial_\nu A_\mu(x) - i\,[A_\mu(x),\, A_\nu(x)])\,\varphi \\ F_{\mu\nu} &= \partial_\mu \underset{\sim}{A_\nu}(x) - \partial_\nu \underset{\sim}{A_\mu}(x) - i\left[\underset{\sim}{A_\mu}(x),\, \underset{\sim}{A_\nu}(x)\right] \end{aligned}$$

Here $F_{\mu\nu}$ is not gauge invariant. $\qquad \downarrow$ $A_\mu(x)$ is matrix valued & matrices don't commute in general.

$$\underset{\sim}{F'_{\mu\nu}}(x) = U\,\underset{\sim}{F_{\mu\nu}}(x)\,\underset{\sim}{U^\dagger} \qquad \text{under gauge transformation} \quad \begin{cases} \varphi \to U\varphi \\ A_\mu \to U A_\mu U^\dagger + i\,(\partial_\mu U)\,U^\dagger \end{cases}$$

To add kinetic term for $A_\mu(x)$ we try

$$-\frac{1}{4}\,F_{\mu\nu}\,F^{\mu\nu} \qquad \text{But this is not gauge invariant.}$$

$$\downarrow \qquad -\frac{1}{4}\,UF_{\mu\nu}\,\underline{U^\dagger\,U}\,F^{\mu\nu}\,U^\dagger \;=\; -\frac{1}{4}\,U\,F_{\mu\nu}\,F^{\mu\nu}\,U^\dagger$$

However,

$$\mathrm{Tr}\left(-\frac{1}{4}\,F_{\mu\nu}\,F^{\mu\nu}\right) = -\frac{1}{4}\,\mathrm{Tr}\,(F_{\mu\nu}\,F^{\mu\nu}) \qquad \text{is gauge invariant}$$

$$\begin{aligned} \mathrm{Tr}\,(F_{\mu\nu}\,F^{\mu\nu}) \;\longrightarrow\; \mathrm{Tr}\left(UF_{\mu\nu}\,U^\dagger U\,F^{\mu\nu}\,U^\dagger\right) &= \mathrm{Tr}\left(U\,F_{\mu\nu}\,F^{\mu\nu}\,U^\dagger\right) \\ &= \mathrm{Tr}\left(F_{\mu\nu}\,F^{\mu\nu}\,U^\dagger\,U\right) \\ &= \mathrm{Tr}\,(F_{\mu\nu}\,F^{\mu\nu}) \end{aligned}$$

So,

$$\mathcal{L} = (D_\mu\varphi)^\dagger\,D^\mu\varphi \;-\; V\left(\varphi^\dagger\varphi\right) \;-\; \boxed{\frac{1}{4}\,\mathrm{Tr}\,(F_{\mu\nu}\,F^{\mu\nu})}$$

$\uparrow$ Yang-Mills lagrangian.

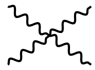
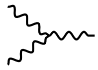
is invariant under local non abelian gauge transformation.

Note that $-\frac{1}{4}\,\mathrm{Tr}\,(F_{\mu\nu}\,F^{\mu\nu})$ is an interacting theory.

Unlike abelian case $F^{\mu\nu}F_{\mu\nu}$ is free term for propagation of A_μ, here $A_\mu(x)$ is non abelian & it turns out here $A_\mu(x)$ is interacting with itself.

$$\begin{aligned} F^{\mu\nu}(x)\,F_{\mu\nu}(x) &= (\partial_\mu A_\nu - \partial_\nu A_\mu - i\,[A_\mu,\, A_\nu])^2 \\ &= (\partial_\mu A_\nu - \partial_\nu A_\mu)^2 + [A_\mu,\, A_\nu]\,[A^\mu,\, A^\nu] \\ &\quad + (-i)\,(\partial_\mu A_\nu - \partial_\nu A_\mu)\,[A^\mu,\, A^\nu] \\ &\quad + (-i)\,[A_\mu,\, A_\nu]\,(\partial^\mu A^\nu - \partial^\nu A^\mu) \end{aligned}$$

$$-\frac{1}{4}\,\mathrm{Tr}\,(F_{\mu\nu}\,F^{\mu\nu}) = -\frac{1}{4}\,\underset{①}{\mathrm{Tr}(\partial_\mu A_\nu - \partial_\nu A_\mu)^2} \;+\; -\frac{1}{4}\,\underset{②}{\mathrm{Tr}([A_\mu,\,A_\nu]\,[A^\mu,\,A^\nu])}$$
$$+\frac{i}{4}\,\underset{③}{\mathrm{Tr}((\partial_\mu A_\nu - \partial_\nu A_\mu)\,[A^\mu,\,A^\nu] + [A_\mu,\,A_\nu]\,(\partial^\mu A^\nu - \partial^\nu A^\mu))}$$

① — Free (quadratic)
② — interacting (quartic fields)  
③ — interacting (cubic fields)

This theory describes strong interaction and weak interaction in 2 different ways.

For strong interactions

we have Quarks : Ψ_I $\quad I = 1, 2, 3$ ↙ color index.

$$\text{Gauge symmetry} \qquad \Psi_I \to U_{IJ}\,\Psi_J \qquad \left(UU^\dagger = I\right)$$
$$U \in SU(3)$$

$\mathcal{L}$ for single flavour ↓

$$\mathcal{L} = i\,\overline{\Psi}\,\gamma^\mu\,(\partial_\mu + i\,A_\mu(x))\,\Psi_J \;-\; \frac{1}{4}\,\mathrm{Tr}\,(F^{\mu\nu}\,F_{\mu\nu})$$
$$= i\,\overline{\Psi}_i\,\gamma^\mu\left(\partial_\mu\,\delta_{ij} + i\,(A_\mu)_{ij}\right)\Psi_j \;-\; \frac{1}{4}\,\mathrm{Tr}\,(F^{\mu\nu}\,F_{\mu\nu})$$
$$-\,m\,\overline{\Psi}_i\,\Psi_i$$

$A_\mu(x) \in SU(3)$ i.e. independent parameters are $3^2 - 1 = 8$

so there are 8 gauge fields called gluons.

$$\mathcal{L} = \sum_{f=1}^{6} i\,\overline{\Psi}^{(f)}_{\alpha a}\left((\gamma^\mu)_{\alpha\beta}\,(\partial_\mu\,\delta_{ab} + i\,(A_\mu)_{ab}) - m\,\delta_{ab}\,\delta_{\alpha\beta}\right)\Psi^{(f)}_{\beta b}$$
$$-\frac{1}{4}\,\mathrm{Tr}\,(F^{\mu\nu}\,F_{\mu\nu})$$

↪ (Total $\mathcal{L}$ for QCD)

↓

Since there is no mass term for $(A_\mu(x))_{ab}$. If we add mass term $m\,A_\mu A^\mu$ then it would not be gauge invariant. But question of weather $A_\mu(x)$/gluons are massless or not does not make sense as $A_\mu(x)$ is always confined (by strong Force; They never propagate freely unless at high Temp & density)

So we don't know mass of $(A_\mu(x))_{ab}$; despite theory predicts that they are massless. (despite theory is very short range)

For weak interaction —

We use same $\mathcal{L}$.

$$\mathcal{L} = -\frac{1}{4}\,\mathrm{Tr}\,(F^{\mu\nu}\,F_{\mu\nu})$$

(This time it couples to weakly interacting particles. i.e. all fermions quarks & leptons).

$\mathcal{L}$ invariant under

$$\varphi \to U\varphi$$

$$A_\mu \to U A_\mu U^\dagger + i\,(\partial_\mu U)\,U^\dagger$$

Here $U(x) \in SU(2) \times U(1) \longrightarrow$ Total $3 + 1 = 4$ generators. i.e. 4 gauge fields/particles.

Again we have massless theory (due to gauge invariance!) however since theory is short range we can look for mass via higgs mechanism.

Out of 4 $A_\mu(x)$ we get $\underline{1}$ massless particle (photon) & other 3 $A_\mu(x)$ $\left(W^+, W^-, Z^0\right)$ gets mass.

Here we have "sort of" unified EM & weak interaction; as we used Yang Mills Lagrangian based on $SU(2) \times U(1)$; but unified means have one coupling const. but in $SU(2) \times U(1)$ coupling const of two groups can be independent of each other (because $SU(2) \times U(1)$ is not simple group; its a group made up of 2 factors; True unification require simple group!)